ON THE HODGE CONJECTURE FOR FERMAT VARIETIES

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Abstract. We prove the Hodge conjecture for the Fermat varieties of degrees 44, 51, 52 and 114 in every dimension, and more generally of every degree $2^a 11^b$, every degree $2^a 13^b$ and every degree $2^a 3^b 19^c$ or $2^a 3^b 17^c$ with $a \leq 1$, together with their products and the abelian varieties of Fermat type of these degrees. The new cycles come from families of curves on Fermat threefolds, through a residue formula for the derivative of their Abel–Jacobi maps; six of the seven residues are certified by interval arithmetic. We also prove the Hodge conjecture for Fermat fourfolds of every odd degree, and determine which combinations of the standard conjectures around the Hodge conjecture imply it.

1. Introduction

Let X be a smooth complex projective variety. A class in $H^{2p}(X, \mathbb{Q})$ is a *Hodge class* if its image in $H^{2p}(X, \mathbb{C})$ lies in $H^{p,p}(X)$, and it is *algebraic* if it is a rational linear combination of the classes of subvarieties of codimension p. Algebraic classes are Hodge classes. Hodge asked for the converse [Hod52].

Conjecture 1.1 (Hodge)**.** *For every smooth complex projective variety X and every $p \geq 0$, every Hodge class in $H^{2p}(X, \mathbb{Q})$ is algebraic.*

We write (HC) for this statement. It holds for $p \leq 1$ and $p \geq \dim X - 1$, so for every variety of dimension at most three, and it is open in general. We do not prove it, and no statement below should be read as a claim that it holds.[1]

Fermat varieties. Let $X_m^n \subseteq \mathbb{P}^{n+1}$ be the Fermat variety $x_0^m + \cdots + x_{n+1}^m = 0$. Fermat varieties are the varieties on which the conjecture has been tested most closely in every dimension [Ran80, Shi79]. Their Hodge classes are explicit: the group μ_m^{n+2} acts on X_m^n, the primitive cohomology splits into lines indexed by characters of this group, and the Hodge classes are spanned by the lines whose characters satisfy a balancing condition (Proposition 3.6). The algebraic cycles that one can write down are fewer: linear subspaces, the cycles that the inductive structure of Shioda and Katsura [SK79] builds from them, and the cycles of Aoki [Aok87]. The conjecture for X_m^n was proved by Ran and Shioda when m is prime or $m = 4$ [Ran80, Shi79], by Shioda when $m \leq 20$ [Shi79], and by Aoki when m is a power of a prime [Aok87]. Aoki then reduced the conjecture for all Fermat varieties of a given degree m to finitely many classes, one for each divisor of m that is a product of an even number of distinct elements of $\{4, 3, 5, 7, 11, \dots\}$, and showed that the classes he could reach suffice for the degrees $2^a 3^b 5^c 7^d$

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Source, figures and verification code: `https://github.com/creelie/e84hc`, archived on Zenodo under the concept DOI 10.5281/zenodo.23234362. This version is release v1.7.0.

[1]Here and below a statement is called *proved* when it is proved in this paper or in refereed work. Results that are available only as preprints are marked as such where they are used. The one preprint on which a theorem of this paper depends, [Ope26b], enters only as an explicit hypothesis.

with $c = 0$ or $d = 0$ and for the degrees p^e and $2p^e$, p prime [Aok83, Aok00]. Peterson has recently treated the divisor 35 [Pet26]. Our main theorem treats the divisors 57, 44, 52 and 51.

Theorem A. *Let $m = 2^a3^b19^c$ or $m = 2^a3^b17^c$ with $a \in \{0, 1\}$ and $b, c \geq 0$, or $m = 2^a11^b$ or $m = 2^a13^b$ with $a, b \geq 0$. Then the Hodge conjecture holds for the Fermat variety X_m^n of every dimension n, for every product of Fermat varieties of degree m, and for every abelian variety of Fermat type of degree m.*

An abelian variety is of Fermat type of degree m if it is isogenous to a product of Jacobians of curves dominated by the Fermat curve of degree m [Aok00]. In particular the Hodge conjecture holds for the Fermat varieties of degrees 44, 51, 52 and 114 in every dimension. When none of 57, 44, 52 and 51 divides m, Theorem A is due to Aoki. When $57 \mid m$, the only earlier argument is that of the preprint [MMRV26]. It states the theorem for $m = 57$, and through Aoki's reduction it would give the other degrees as well, but it reduces the remaining class to a Hodge class on the Fermat fourfold of degree 114 and concludes with a statement of Kang [Kan16] whose proof has a gap (Remark 3.21). Our proof does not use that statement, so for $57 \mid m$ it is the first complete proof; for the degrees 114, 171, 342, 513, ... the theorem has not been stated before. For fourfolds, a computer census of the Fermat fourfolds of even degree at most 250 [Jum26a], combined with [Pet26], leaves the degrees 110, 114, 220 and 228; Theorem A settles 114 and 228 (Remark 4.35). When $44 \mid m$, $52 \mid m$ or $51 \mid m$ there is no earlier proof: [MMRV26] treats every degree below 65 except 44, 51 and 52, and the degrees 44, 88, 176, 352, ..., 52, 104, 208, 416, ... and 51, 102, 153, 306, ... appear in Theorem A for the first time (Remarks 4.36 to 4.38). The unrefereed preprint [Ope26b] claims the Hodge conjecture for abelian varieties of CM type, which would give it for every Fermat variety (Theorem E); Theorem A does not depend on that claim.

The idea of the proof. For $m = 114$ Aoki's reduction leaves a single class, and it suffices to show that one balanced multiset β of ten residues modulo 114 is represented by an algebraic cycle on the Fermat eightfold (Corollary 4.5). It is invisible to linear subspaces and to Aoki's cycles, which never produce an odd number of residues of order 19, while β contains one (Lemma 4.3). We write β as the join of two characters α_1, α_2 of the Fermat threefold X_{114}^3 whose rational Hodge structures $M_{\alpha_1}, M_{\alpha_2} \subseteq H^3$ have level one and complementary types (Lemma 4.6). If each of them is carried, by an algebraic correspondence, by the first cohomology of a curve, then the Hodge classes in $M_{\alpha_1} \otimes M_{\alpha_2}$ come from a product of two curves, where they are algebraic by the Lefschetz theorem on $(1, 1)$-classes, and the inductive structure takes them to the eightfold (Proposition 4.17).

So the main step is a coniveau statement for two pieces of H^3 of a threefold. We prove it by deforming curves. A family of binary forms $u_0, \ldots, u_4$ with $\sum_k u_k = 0$ and a suitable factorization gives, through $x_k^m = u_k$, a family of curves on X_m^3; the derivative of its Abel–Jacobi map, paired with the $(2, 1)$-form of α, is a sum of residues of an explicit rational 1-form on $\mathbb{P}^1$, and if it does not vanish, M_α is carried by curves. Peterson found this argument for one family on the Fermat threefold of degree 35 [Pet26, §3]. We set it up for every family of this shape, which we call block families (Definition 4.7, Proposition 4.14 and Theorem 4.16), and write down two of them in degree 114. For $13\alpha_2$ there is a family of cubics whose residue is a rational function of the parameter, computed by hand (Proposition 4.19). For α_1 the smallest family we know consists of curves of degree 24, and the nonvanishing of its residue at one point is proved by a certified computation in interval arithmetic (Proposition 4.20, Appendix B).

For $m = 44$ the remaining class is a multiset of eight residues with a single element of order 11, and the smallest balanced multisets of this kind have eight elements. We write it as the join of a character α of the threefold X_{44}^3 and a character b of the Fermat curve with

complementary types (Lemma 4.22). Only M_α has to be carried by curves, since H^1 of the Fermat curve already is, and the Hodge classes in $M_\alpha \otimes H^1$ then come from a product of two curves (Proposition 4.18). For α there is a family of curves of degree 11, and the nonvanishing of its residue is again certified in interval arithmetic (Proposition 4.23).

For $m = 52$ the remaining class is again a multiset of ten residues, with a single element of order 13, and we write it as the join of two characters of the threefold X^3_{52} with complementary types (Lemma 4.25). Both pieces of H^3 need families of curves; we use families of degrees 14 and 5, whose residues are certified in interval arithmetic (Propositions 4.26 and 4.28).

For $m = 51$ we work at the level 102, where the remaining class is again a multiset of ten residues, now with a single element of order 17, and the join of two characters of X^3_{102} with complementary types (Lemma 4.30). The families have degrees 6 and 18, and their residues are certified in the same way (Propositions 4.31 and 4.33). These six certificates are the only steps of the paper that use a computer.

Fermat fourfolds of odd degree. The second unconditional result concerns fourfolds.

Theorem B. *Let m be odd. Then the Hodge conjecture holds for the Fermat fourfold X^4_m.*

By Proposition 3.6 the Hodge classes of X^4_m are spanned by lines indexed by sextuples of nonzero residues modulo m. When m is prime to 3, we show that each such sextuple contains two entries a and $-a$, or is, up to order, $(x, x + m/5, x + 2m/5, x + 3m/5, x + 4m/5, -5x)$ (Theorem 3.9). The classes of the first kind are algebraic by the inductive structure of Shioda and Ran, and those of the second kind by the cycles of Aoki [Aok87], pulled back along a covering of Fermat fourfolds. The proof of the classification uses the Dirichlet characters modulo the largest order of an entry, and its only analytic input is the classical fact that $B_{1,\chi} \neq 0$ for odd primitive χ. When $3 \mid m$ the classification fails, and we argue by induction instead (Section 3.7): a sextuple without such a pair either has a *move*, which trades it, by means of one cycle of Aoki, for a quadruple or for a sextuple of smaller orders, or it lies in one of three explicit orbits, in degrees 21, 33 and 39, whose classes we obtain on the Fermat fourfolds of degrees 21, 66 and 78. When m is prime to 15, Theorem B follows from Aoki's theorem that every Hodge character then splits into pairs [Aok83]; it is also known when m is a power of a prime [Aok87] and when $m = 3^a5^b7^c$ [Aok00, Pet26]. In the remaining degrees it is new for $m > 199$, apart from the statement of Kang mentioned above; smaller degrees were covered by computer searches (Remark 3.7). The classification for m prime to 6, together with the fact that $B_{1,\chi} \neq 0$, has been checked in Lean 4 with Mathlib (Appendix A).

The routes to the conjecture. The rest of the paper asks a different question: *from which of the statements that the literature has isolated does* (HC) *follow?* We answer it, prove the conditional theorems that the answer allows, and stop at the first statement on each route that we cannot prove. The statements are the following; they are made precise in Definition 2.7.

- (F2) The Hodge conjecture for abelian varieties.
- (F3′) The Hodge conjecture modulo abelian varieties: every Hodge class on every X is a sum of classes $\Gamma_*\gamma$, where γ is a Hodge class on an abelian variety B and Γ is an algebraic cycle on $B \times X$.
- (L) The Lefschetz standard conjecture $B(X)$ of Grothendieck [Gro69], for every X.
- (M) Every Hodge class is motivated in the sense of André [And96].
- (V) The variational Hodge conjecture: in a smooth projective family over a smooth connected base, a flat family of rational classes that is algebraic on one fibre is algebraic on every fibre.
- (CM) The Hodge conjecture for abelian varieties of CM type.

(IP) Propagation from CM points: (V) for polarized abelian schemes and classes that are algebraic on a fibre of CM type.

The statement (CM) is the main theorem claimed in the preprint [Ope26b], which has not been refereed. We have not verified its proof, and Section 5.1 says which parts of it we have checked.

Theorem C. *The following are equivalent:*

(i) (HC)*;*
(ii) (F2) *and* (F3′)*;*
(iii) (L) *and* (F3′)*;*
(iv) (V) *and* (F3′)*;*
(v) (CM)*,* (IP) *and* (F3′)*;*
(vi) (L) *and* (M)*.*

The implications from (ii)–(vi) to (i) combine known theorems of André, Deligne, Abdulali and Milne with two observations of this paper: (F3′) implies (M) (Proposition 2.9), and (CM) together with (IP) implies (F2) (Proposition 5.4). The theorem is proved in Section 5.3. The next theorem says that these are all the ways in which the conjecture follows from the statements above, together with the other statements the paper meets: the semiregularity of Schoen's subvariety (Question 6.14), and the Hodge conjecture for Fermat varieties and for the Weil classes of the split Weil families over $\mathbb{Q}(\sqrt{-3})$.

Theorem D (The routes)**.** *Let S be a set of the statements of Table 1 that does not contain* (HC)*, and suppose that* (HC) *follows from S by the implications of Table 1. Then*

(a) *either S contains* (F3′) *together with one of* (F2)*,* (L)*,* (V)*, or both of* (CM) *and* (IP)*;*
(b) *or S contains both* (L) *and* (M)*.*

Each of the five minimal sets so described implies (HC)*, and each implies every statement of Table 1 except the semiregularity of Schoen's subvariety.*

So there are two ways in. One passes through abelian varieties and needs (F3′) together with a statement that implies (F2); the other avoids abelian varieties and assumes (L) and (M), which by Theorem C is the Hodge conjecture in another form. In particular (F2) and (F3′) can be avoided as hypotheses, but never as consequences, and Schoen's subvariety lies on no route at all (Corollary 7.2). The routes are drawn in Figure 1.

The remaining results are conditional on one statement, or concern one family. The first returns to Fermat varieties.

Theorem E. *Assume* (CM)*. Then the Hodge conjecture holds for every Fermat variety X_m^n, for every product of Fermat varieties of any degrees and dimensions, and for every smooth projective variety onto which such a product maps surjectively. Without any hypothesis,* (F3′) *holds for all of these varieties.*

Unconditional proofs are known only for special degrees (Remark 3.7), among them those of Theorems A and B. The proof of Theorem E uses the inductive structure of Shioda and Katsura [SK79] to dominate every Fermat variety by a product of Fermat curves, blown up along smooth centres, and the fact that Fermat curves have Jacobians of CM type.

Abelian varieties of Weil type are where (F2) is hardest. On the split Weil family of an imaginary quadratic field over its Shimura variety S (Section 5.4), write $\Sigma \subseteq S$ for the set of members on which the Weil classes are algebraic, and $\Sigma_d \subseteq \Sigma$ for the set where they have algebraic representatives of degree at most d (Definition 5.5).

Theorem F. *Each Σ_d is Zariski closed in S, and $\Sigma = \bigcup_d \Sigma_d$. The following are equivalent:*

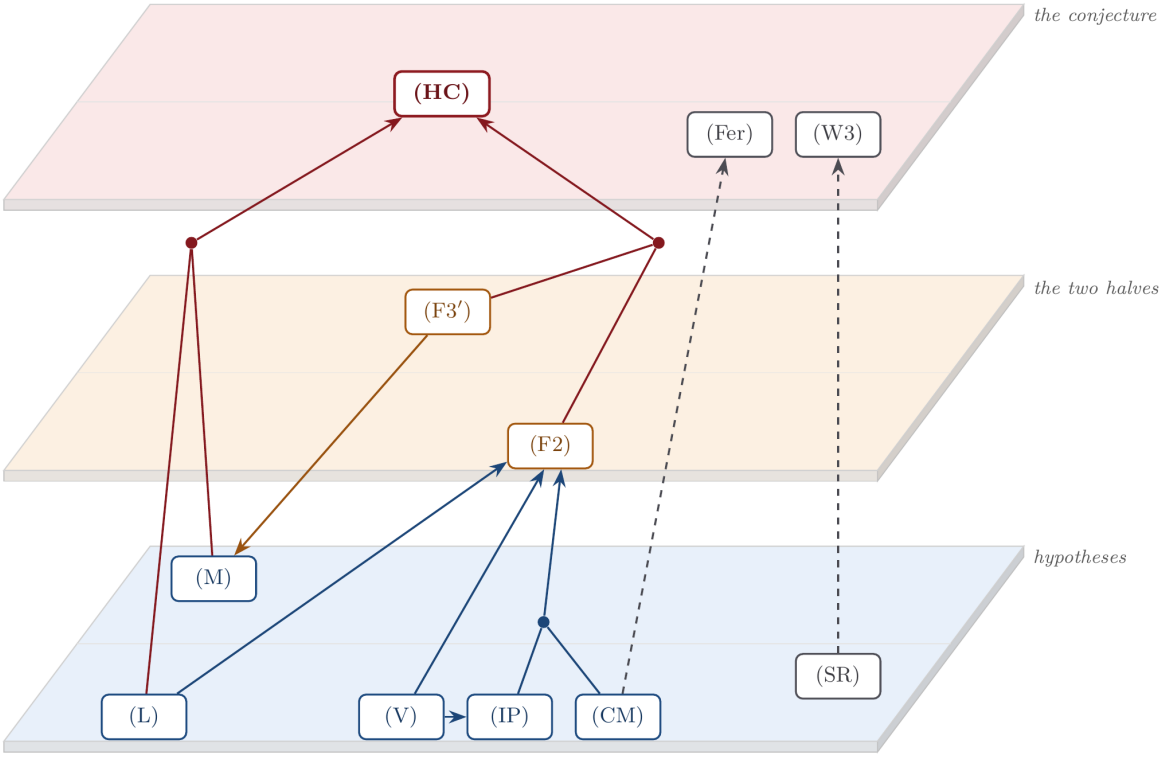


FIGURE 1. The routes of Theorem D. Hypotheses lie on the bottom layer, the two halves (F2) and (F3′) in the middle, and (HC) on top. Arrows are implications of Table 1. At each dot two lines combine into a conjunction: (CM) ∧ (IP) ⇒ (F2), (L) ∧ (M) ⇒ (HC) and (F2) ∧ (F3′) ⇒ (HC). The arrows from (HC) and (F2) to the statements they contain are not drawn. The dashed arrows on the right lead to the special cases (Fer) and (W3) of Section 7, and not to (HC).

(i) $\Sigma = S$;
(ii) $\Sigma_d = S$ *for some* d;
(iii) *for some* d, *the CM points in* Σ_d *are not contained in any finite union of proper special subvarieties of* S.

Under (CM) *every CM point of* S *lies in* Σ.

So under (CM) the Weil classes are algebraic on the whole family exactly when, at the CM points, they have algebraic representatives of uniformly bounded degree on a set that is not trapped in finitely many proper Shimura subvarieties. The equivalence of (ii) and (iii) is where the André–Oort conjecture, proved by Pila, Shankar and Tsimerman [PST21], enters.

The last result concerns the only cycles known to represent Weil classes at members with maximal Hodge group in every dimension, those constructed by Schoen [Sch88]. Let X be a curve of genus $n+1 \geq 3$, let $C \to X$ be the étale cyclic triple cover given by a line bundle of order three, and let B be its Prym variety, an abelian variety of dimension $2n$ of Weil type for $\mathbb{Q}(\sqrt{-3})$ with polarization class η.

Theorem G. *There is an irreducible subvariety* $Y \subseteq B$ *of dimension* n, *built from one of the three components into which Schoen's cover splits over the canonical system of* X, *whose class*

pairs nontrivially with the Weil classes of B. If the Hodge group of B is maximal, which holds for very general (X, C), then

$$\mathrm{cl}(Y) = c\,\eta^n + w, \qquad c \in \mathbb{Q}_{>0}, \quad 0 \neq w \text{ a Weil class.}$$

If Y is semiregular for one such B, the Weil classes are algebraic on every member of the split Weil family of $\mathbb{Q}(\sqrt{-3})$ in dimension $2n$.

For $n \leq 3$ this family is already covered by Schoen's work [Sch88, Sch98]. For $n \geq 4$ it is not: the Prym locus has dimension $3n$ and the family has dimension n^2, so Y would have to deform in $n^2 - 3n$ further directions. The proof of the pairing in Theorem G is a short computation with twisted coefficients on the symmetric product of X, and it also gives a proof of Schoen's theorem that B carries algebraic Weil classes (Corollary 6.7). We give the evidence on both sides of Question 6.14 in Section 6.6, including a precise analogy with the Abel–Jacobi curve, which is not semiregular, a description of Y as a Brill–Noether locus of the Prym variety, smooth of dimension n at its general point, whose analogue for double covers is Mumford's Prym theta divisor, and a question of Markman about a weak form of the semiregularity theorem.

What is not proved. None of the statements (F2), (F3′), (L), (M), (V), (IP) is proved here, and (CM) is used only as a hypothesis. The Hodge conjecture is open, and by Corollary 7.2 every set of these statements from which it follows also implies (F2) and (F3′). Section 8 lists what is proved and what is not, and names the first open cases: for (F2), the Weil classes of nonsplit abelian sixfolds and of abelian eightfolds, and the exceptional classes on the square of a Mumford fourfold; for (F3′), the square of a K3 surface with real multiplication.

For Fermat varieties, the method of Theorem B does not reach fourfolds of even degree: $m/2$ is its own negative, the subgroups K_2 of Section 3.6 are too small, and Hodge characters of further kinds occur [Aok83]. Theorem A reaches the even degrees $2 \cdot 3^b 19^c$, $2^a 11^b$ and $2^a 13^b$, but not, for instance, the fourfolds of degree 110, where the class that remains belongs to the divisor 55 of Aoki's reduction and we have found no family of curves of the kind used here (Remark 4.35). Of the divisors divisible by 4 we treat only $4 \cdot 11$ and $4 \cdot 13$; for others, such as $4 \cdot 19$, we have no cycles.

Organization. Section 2 proves the reductions that hold for every variety and defines the statements. Section 3 describes the Hodge classes of Fermat varieties and proves Theorem B and Theorem E, and Section 4 proves Theorem A. Section 5 treats (CM) and (IP), proves Theorem C, and proves Theorem F. Section 6 proves Theorem G and the transfer theorem for Weil classes. Section 7 proves Theorem D. Section 8 collects what is proved and what is not. Appendix A describes the machine checks of the finite steps, and Appendix B gives the certified computations behind Propositions 4.20, 4.23, 4.26, 4.28, 4.31 and 4.33. Apart from these six propositions, no proof in this paper depends on a computer.

Conventions. Varieties are complex, smooth, projective and, unless said otherwise, connected. Cohomology is singular cohomology with rational coefficients. For a variety X we write $\mathrm{Hdg}^p(X) \subseteq H^{2p}(X, \mathbb{Q})$ for the Hodge classes and $\mathrm{Alg}^p(X) \subseteq \mathrm{Hdg}^p(X)$ for the algebraic classes. Our Tate twist is $V(r)^{a,b} = V^{a+r,b+r}$, so that $V(r)$ has weight $\mathrm{wt}(V) - 2r$.

2. Reductions that hold for every variety

2.1. **Correspondences.** Let B and X be varieties, possibly disconnected, with B of pure dimension b. For an algebraic cycle Γ of codimension c on $B \times X$ with rational coefficients, the map

$$\Gamma_* \colon H^k(B, \mathbb{Q}) \to H^{k+2c-2b}(X, \mathbb{Q}), \qquad y \mapsto \mathrm{pr}_{X*}(\mathrm{cl}(\Gamma) \cup \mathrm{pr}_B^* y),$$

is a morphism of Hodge structures $H^k(B)(b-c) \to H^{k+2c-2b}(X)$. It carries algebraic classes to algebraic classes, the composite of two such maps is the map of the composite correspondence, and for a morphism $f\colon B \to X$ the graph Γ_f and its transpose give f_* and f^* [Ful98, Chapter 16]. For a cycle with components of several codimensions we add the maps of the components.

A pure rational Hodge structure V of weight w is *polarizable* if it carries a bilinear form Q such that $Q(Cx, \bar{x}) > 0$ for $x \neq 0$, where C acts on $V^{a,b}$ by i^{a-b}. The cohomology of a variety and its Tate twists are polarizable, by the Hodge–Riemann relations [Voi02, Chapter 6]. Polarizable Hodge structures are semisimple.[2]

Lemma 2.1. *Let $\varphi\colon V \to W$ be a surjective morphism of polarizable Hodge structures of weight $2p$. Then φ maps the Hodge classes of V, the rational classes of type (p,p), onto those of W.*

Proof. By semisimplicity the kernel of φ has a complement V' that is a sub-Hodge structure, and φ restricts to an isomorphism of Hodge structures $V' \to W$. Its inverse carries a Hodge class of W to a Hodge class of $V' \subseteq V$. □

Lemma 2.2 (Dominated cohomology)**.** *Let X be a variety, and let $B_1, \dots, B_r$ be varieties with algebraic cycles Γ_j on $B_j \times X$, such that the images of the maps Γ_{j*} span $H^{2p}(X, \mathbb{Q})$. Then every Hodge class of degree $2p$ on X is a sum $\sum_j \Gamma_{j*}\gamma_j$ with γ_j Hodge classes on B_j. In particular, if the Hodge conjecture holds for every B_j, it holds for X in degree $2p$.*

Proof. Write $\Gamma_j = \sum_c \Gamma_{j,c}$ with $\Gamma_{j,c}$ of pure codimension c, and $b_j = \dim B_j$ (if B_j is not equidimensional, treat its components separately). The component $\Gamma_{j,c}$ maps into degree $2p$ exactly from $H^k(B_j)$ with $k = 2p + 2b_j - 2c$, which is even. The sum of the maps

$$\bigoplus_{j,c} H^{2p+2b_j-2c}(B_j, \mathbb{Q})(b_j - c) \longrightarrow H^{2p}(X, \mathbb{Q})$$

is a surjective morphism of polarizable Hodge structures of weight $2p$. A Hodge class in $H^{2p+2b_j-2c}(B_j)(b_j - c)$ is a Hodge class of B_j. Apply Lemma 2.1. Algebraic classes go to algebraic classes under correspondences, which gives the last assertion. □

2.2. **Degree two and hard Lefschetz.**

Theorem 2.3 (Lefschetz [Lef24], Kodaira–Spencer [KS53])**.** *Every Hodge class of degree two is algebraic.*

Proof. The exponential sequence gives an exact sequence

$$H^1(X, \mathcal{O}_X^\times) \to H^2(X, \mathbb{Z}) \to H^2(X, \mathcal{O}_X),$$

whose second map is the composite of $H^2(X, \mathbb{Z}) \to H^2(X, \mathbb{C})$ with the projection to $H^{0,2}(X)$. An integral class of type $(1,1)$ therefore is the first Chern class of a holomorphic line bundle, which is algebraic by GAGA, and its first Chern class is the class of a divisor. A rational Hodge class has an integral multiple. □

Let h be the class of a hyperplane section of X, of dimension n, and L the cup product with h. By the hard Lefschetz theorem $L^{n-k}\colon H^k(X, \mathbb{Q}) \to H^{2n-k}(X, \mathbb{Q})$ is an isomorphism for $k \leq n$ [Voi02, Theorem 6.25].

Proposition 2.4. *For $2p < n$, if every Hodge class of degree $2p$ on X is algebraic, so is every Hodge class of degree $2n - 2p$. Every Hodge class of degree 0, 2, $2n-2$ or $2n$ is algebraic, and* (HC) *holds for every variety of dimension at most three.*

[2] If $W \subseteq V$ is a sub-Hodge structure, the restriction of Q to W is again a polarization, hence nondegenerate, so $V = W \oplus W^\perp$. The orthogonal $W^\perp$ is a sub-Hodge structure, because the decomposition $V_{\mathbb{C}} = \bigoplus V^{a,b}$ is orthogonal for the hermitian form $Q(Cx, \bar{y})$ and $W_{\mathbb{C}}$ is the sum of its intersections with the $V^{a,b}$.

Proof. L^{n-2p} maps $H^{p,p}$ onto $H^{n-p,n-p}$ and rational classes to rational classes, so it maps $\mathrm{Hdg}^p(X)$ onto $\mathrm{Hdg}^{n-p}(X)$; and it maps the class of a subvariety Z to the class of $Z \cap H_1 \cap \cdots \cap H_{n-2p}$ for general hyperplanes H_i. Hence $\mathrm{Hdg}^{n-p}(X) = L^{n-2p}\,\mathrm{Alg}^p(X) \subseteq \mathrm{Alg}^{n-p}(X)$. Degrees 0 and $2n$ are spanned by the fundamental class and the class of a point; degree 2 is Theorem 2.3, and degree $2n-2$ follows from it. For $n \leq 3$ every even degree is one of these. □

2.3. **Surjective morphisms and blow-ups.**

Proposition 2.5. *Let $f\colon Z \to X$ be a surjective morphism, with X connected, and let $r = \dim Z - \dim X$. Let h_Z be an ample class on Z, and $d > 0$ the degree of h_Z^r on a general fibre. Then $\alpha = d^{-1} f_*(h_Z^r \cup f^*\alpha)$ for every $\alpha \in H^*(X, \mathbb{Q})$. In particular the correspondence $\Gamma = d^{-1}(h_Z^r \times 1) \cdot \Gamma_f$ from Z to X satisfies $\Gamma_* f^* = \mathrm{id}$, so Γ_* is surjective, and* (HC) *for Z implies* (HC) *for X.*

Proof. By the projection formula $f_*(h_Z^r \cup f^*\alpha) = f_*(h_Z^r) \cup \alpha$, and $f_*(h_Z^r) \in H^0(X, \mathbb{Q}) = \mathbb{Q}$ is the degree d, which is positive because h_Z is ample on the fibres. The last claim follows from Lemma 2.2. □

Proposition 2.6. *Let $Y \subseteq X$ be a smooth closed subvariety, possibly disconnected, of codimension $c \geq 2$ in every component, let $\pi\colon \tilde{X} \to X$ be the blow-up along Y, let $j\colon E \to \tilde{X}$ be the exceptional divisor and $q\colon E \to Y$ the projective bundle, and let $\xi = c_1(\mathcal{O}_E(1))$. Then*

$$H^k(\tilde{X}, \mathbb{Q}) = \pi^* H^k(X, \mathbb{Q}) \,\oplus\, \bigoplus_{i=0}^{c-2} j_*(\xi^i \cup q^* H^{k-2i-2}(Y, \mathbb{Q})),$$

and each summand is the image of a correspondence: π^* from X, and $y \mapsto j_*(\xi^i \cup q^* y)$ from Y.

Proof. The decomposition is the blow-up formula [Voi02, Theorem 7.31]. The map $y \mapsto j_*(\xi^i \cup q^* y)$ is the correspondence given by the image in $Y \times \tilde{X}$ of a cycle on E representing ξ^i, pushed forward along $(q, j)\colon E \to Y \times \tilde{X}$. □

2.4. **The statements.**

Definition 2.7. Let X range over all varieties and p over all integers.

(1) (F2): the Hodge conjecture for every complex abelian variety.[3]
(2) (F3′): $\mathrm{Hdg}^p(X) = \mathrm{Ab}^p(X)$, where $\mathrm{Ab}^p(X)$ is the span of the classes $\Gamma_*\gamma$ of degree $2p$ with B an abelian variety (the point included), γ a Hodge class on B and Γ an algebraic cycle on $B \times X$ [BMB26, Definition 20.34].
(3) (L): for every X and every ample class on X, the Lefschetz involution $*_L$ defined below is induced by an algebraic cycle on $X \times X$. This is Grothendieck's standard conjecture $B(X)$ for every X [Gro69]: by [Kle68, 1.4.4] and [And96, Proposition 1.2], the $\mathbb{Q}$-algebra of operators generated by L and $*_L$ is the one generated by L and the operator Λ of $B(X)$.
(4) (M): every Hodge class is motivated in the sense of [And96, §1].
(5) (V): for every smooth projective morphism $f\colon \mathcal{X} \to S$ over a smooth connected quasi-projective variety, and every global section ξ of $R^{2p} f_* \mathbb{Q}$ over S^{an}, if ξ_s is algebraic for one $s \in S$ then ξ_t is algebraic for every $t \in S$.
(6) (CM): the Hodge conjecture for every abelian variety of CM type.
(7) (IP): the case of (V) in which f is a polarized abelian scheme and $\mathcal{X}_s$ is of CM type.

[3] In [BMB26] the name (F2) denotes a statement about the Hodge classes not generated by divisors and Weil classes, and it is proved there to be equivalent to the Hodge conjecture for abelian varieties. We use the simpler form.

An abelian variety A is *of CM type* if $\mathrm{End}(A)\otimes\mathbb{Q}$ contains a commutative semisimple $\mathbb{Q}$-algebra of dimension $2\dim A$; the point is of CM type, and products of abelian varieties of CM type are of CM type.

Each $\Gamma_*\gamma$ in (F3′) is a Hodge class, so $\mathrm{Ab}^p(X)\subseteq\mathrm{Hdg}^p(X)$; taking B a point shows that $\mathrm{Ab}^p(X)$ contains $\mathrm{Alg}^p(X)$. Thus (F3′) asks only that Hodge classes be algebraic *modulo* what abelian varieties supply. By Proposition 2.5, applied to $A\times\mathbb{P}^1\to A$, the conjecture for varieties that are not abelian already contains the conjecture for abelian varieties; this is why the half of the conjecture that concerns other varieties has to be stated modulo abelian varieties to be a separate problem.

2.5. **The Lefschetz standard conjecture and motivated classes.** Let X have dimension n and ample class h. Every class in $H^k(X)$ is uniquely $\sum_j L^j x_j$ with x_j primitive of degree $k-2j$, and the *Lefschetz involution* is defined by $*_L(L^j x_j)=L^{n-k+j}x_j$ [And96, §1.1]. On H^k with $k\le n$ it is L^{n-k}, and on H^{2n-k} it is the inverse of L^{n-k}. A class $\alpha\in H^{2p}(X,\mathbb{Q})$ is *motivated* [And96, Définition 1] if there are a variety Y, ample classes h_X on X and h_Y on Y, and algebraic classes β,γ on $X\times Y$ with

$$\alpha=\mathrm{pr}_{X*}(\beta\cup *_L\gamma),$$

where $*_L$ is the Lefschetz involution of $X\times Y$ for the ample class $\mathrm{pr}_X^*h_X+\mathrm{pr}_Y^*h_Y$. We use three facts from [And96]. (a) Motivated classes are Hodge classes, they contain the algebraic classes, and they form a graded $\mathbb{Q}$-algebra that is stable under pull-back and push-forward along projections [And96, Proposition 2.1 and §2.5]; hence a correspondence given by an algebraic cycle maps motivated classes to motivated classes.[4] (b) Every Hodge class on an abelian variety is motivated [And96, Théorème 0.6.2]. (c) If the Lefschetz involution of every variety, for every ample class, is induced by an algebraic cycle, then every motivated class is algebraic [And96, §2, after Définition 1]. Fact (c) is immediate from the definition: $*_L\gamma$ is then the image of an algebraic class under an algebraic correspondence.

Proposition 2.8. (L) *and* (M) *together imply* (HC).

Proof. By (M) every Hodge class is motivated, and by (L) and fact (c) every motivated class is algebraic. □

Proposition 2.9. (F3′) *implies* (M).

Proof. Under (F3′) every Hodge class is a sum of classes $\Gamma_*\gamma$ with γ a Hodge class on an abelian variety. Such a γ is motivated by fact (b), and $\Gamma_*\gamma$ is motivated by fact (a). □

2.6. **Families.**

Lemma 2.10 (Theorem of the fixed part)**.** *Let $f\colon\mathcal{X}\to S$ be a smooth projective morphism over a smooth connected quasi-projective variety, and ξ a global section of $R^kf_*\mathbb{Q}$. If ξ_s is a Hodge class for one $s\in S$, then ξ_t is a Hodge class for every $t\in S$.*

Proof. By Deligne's theorem of the fixed part [Del71, Corollaire 4.1.2] the space $H^0(S,R^kf_*\mathbb{Q})$ carries a Hodge structure for which each restriction $r_t\colon H^0(S,R^kf_*\mathbb{Q})\to H^k(\mathcal{X}_t,\mathbb{Q})$ is an injective morphism of Hodge structures.[5] An injective morphism of Hodge structures that sends

[4] For a correspondence, $\Gamma_*\gamma=\mathrm{pr}_{X*}(\mathrm{cl}(\Gamma)\cup\mathrm{pr}_B^*\gamma)$ is built from γ by a pull-back along a projection, a product with an algebraic class and a push-forward along a projection.

[5] Injective because a section of a local system over a connected base is determined by its value at one point. The Hodge structure is the image of $H^k(\bar{\mathcal{X}},\mathbb{Q})\to H^k(\mathcal{X}_t,\mathbb{Q})$ for a smooth compactification $\bar{\mathcal{X}}$ of $\mathcal{X}$, which by [Del71, 4.1.1] is the whole invariant part. A global section that is of type (p,q) at one point is of type (p,q) everywhere [Del71, 4.1.3.1].

a class x to a class of type (p,p) forces x to be of type (p,p), because it respects the Hodge decompositions. So ξ is a Hodge class of $H^0(S, R^k f_*\mathbb{Q})$, and $\xi_t = r_t(\xi)$ is a Hodge class for every t. □

Lemma 2.11. (HC) *implies each of* (F2), (F3′), (L), (M), (V), (CM) *and* (IP).

Proof. (F2) and (CM) are special cases, and (F3′) holds because $\mathrm{Ab}^p(X)$ contains $\mathrm{Alg}^p(X)$. For (L): on each $H^k(X)$ the involution $*_L$ is a morphism of Hodge structures $H^k(X) \to H^{2n-k}(X)(n-k)$, so under the Künneth decomposition and Poincaré duality each of its components is a Hodge class on $X \times X$,[6] and (HC) makes it algebraic. Algebraic classes are motivated, which gives (M). For (V): if ξ_s is algebraic, it is a Hodge class, so ξ_t is a Hodge class for every t by Lemma 2.10, and (HC) makes it algebraic. (IP) is a case of (V). □

3. Fermat varieties

3.1. Varieties dominated by CM abelian varieties.

Definition 3.1. Let $\mathcal{D}$ be the class of varieties X, possibly disconnected, such that $H^*(X,\mathbb{Q})$ is spanned by the images of maps $\Gamma_*\colon H^*(B,\mathbb{Q}) \to H^*(X,\mathbb{Q})$, where B runs over the abelian varieties of CM type, the point included, and Γ over the algebraic cycles on $B \times X$.

Proposition 3.2. (i) *Every $X \in \mathcal{D}$ satisfies* (F3′). *If* (CM) *holds, every $X \in \mathcal{D}$ satisfies* (HC).

(ii) *$\mathcal{D}$ contains every abelian variety of CM type and every finite set of points, and every curve whose Jacobian is of CM type.*

(iii) *$\mathcal{D}$ is closed under finite disjoint unions, under taking connected components, and under products.*

(iv) *If $Z \in \mathcal{D}$ and $f\colon Z \to X$ is a surjective morphism, then $X \in \mathcal{D}$.*

(v) *If $X \in \mathcal{D}$ and $Y \subseteq X$ is a smooth closed subvariety of codimension at least two in every component, with $Y \in \mathcal{D}$, then the blow-up of X along Y lies in $\mathcal{D}$.*

Proof. (i) By Lemma 2.2, applied in each degree to finitely many Γ_* whose images span, every Hodge class on X is a sum of classes $\Gamma_*\gamma$ with γ a Hodge class on an abelian variety of CM type. Such a class lies in $\mathrm{Ab}^p(X)$, and under (CM) it is algebraic.

(ii) For B of CM type take Γ the diagonal. For a finite set of points take B a point and Γ the points of X one at a time. For a curve C with Jacobian J of CM type: $H^0(C)$ and $H^2(C)$ are spanned by the fundamental class and the class of a point, which are images from the point, and $H^1(C) = \alpha^* H^1(J)$ for an Abel–Jacobi map $\alpha\colon C \to J$, and α^* is the correspondence given by the transpose of the graph of α.

(iii) A disjoint union: the cohomology is the direct sum, and a cycle on $B \times X_i$ is a cycle on $B \times \bigsqcup_i X_i$. A connected component X' of X: restriction $H^*(X) \to H^*(X')$ is surjective and is the correspondence given by the transpose of the graph of the inclusion, so composing gives correspondences from the B's onto $H^*(X')$. A product: by the Künneth formula $H^*(X \times X')$ is spanned by the classes $\mathrm{pr}^* x \cup \mathrm{pr}'^* x'$, and if $x = \Gamma_* y$ and $x' = \Gamma'_* y'$ then $\mathrm{pr}^* x \cup \mathrm{pr}'^* x' = \pm(\Gamma \times \Gamma')_*(\mathrm{pr}_B^* y \cup \mathrm{pr}_{B'}^* y')$, where $\Gamma \times \Gamma'$ is a cycle on $(B \times B') \times (X \times X')$ and $B \times B'$ is of CM type.

(iv) By (iii) we may assume X connected and replace Z by a component that maps onto X. By Proposition 2.5 there is a correspondence Γ from Z to X with Γ_* surjective; composing it with the correspondences that dominate $H^*(Z)$ gives correspondences that dominate $H^*(X)$.

[6] A $\mathbb{Q}$-linear map $u\colon H^a(X) \to H^b(X)$ corresponds to an element of $H^a(X)^\vee \otimes H^b(X) \cong H^{2n-a}(X)(n) \otimes H^b(X)$, a summand of $H^{2n-a+b}(X \times X)(n)$, and the correspondence it defines acts as u. The map u is a morphism of Hodge structures $H^a(X) \to H^b(X)(r)$ exactly when this element is a Hodge class of the appropriate weight.

(v) By Proposition 2.6 the cohomology of the blow-up is spanned by the images of correspondences from X and from Y; compose them with the correspondences that dominate $H^*(X)$ and $H^*(Y)$. □

3.2. **Fermat curves.**

Lemma 3.3. *For every $m \geq 1$ the Jacobian of the Fermat curve X^1_m is of CM type.*

Proof. For $m \leq 2$ the curve is rational and the Jacobian is a point. Let $m \geq 3$, let $g = (m-1)(m-2)/2$ be the genus, and use the affine model $u^m + v^m = 1$, which is isomorphic to X^1_m over $\mathbb{C}$. The group $G = \mu_m \times \mu_m$ acts by $(\zeta, \zeta') \cdot (u, v) = (\zeta u, \zeta' v)$, and this action extends to the smooth projective curve. The forms

$$\omega_{r,s} = u^{r-1} v^{s-m}\, du, \qquad r, s \geq 1, \quad r + s \leq m - 1,$$

are holomorphic[7], and there are $(m-1)(m-2)/2 = g$ of them, so they form a basis of $H^{1,0}$. The form $\omega_{r,s}$ is an eigenvector of G with character $(\zeta, \zeta') \mapsto \zeta^r \zeta'^s$, and distinct pairs (r, s) give distinct characters, since $1 \leq r, s \leq m-2$. The complex conjugates span $H^{0,1}$ and have the characters $(-r, -s) = (m-r, m-s)$; since $(m-r) + (m-s) \geq m+1$, none of these occurs in $H^{1,0}$. So $H^1(X^1_m, \mathbb{C})$ is the direct sum of $2g$ lines on which G acts by $2g$ distinct characters.

Let $E \subseteq \mathrm{End}(H^1(X^1_m, \mathbb{Q}))$ be the image of the group algebra $\mathbb{Q}[G]$. It is commutative and semisimple, as a quotient of a commutative semisimple algebra, and $E \otimes \mathbb{C}$ is the image of $\mathbb{C}[G]$, which acts on each of the $2g$ lines through its own character; since the characters are distinct, $E \otimes \mathbb{C} \cong \mathbb{C}^{2g}$ and $\dim_{\mathbb{Q}} E = 2g$. The elements of G act on the Jacobian, and H^1 of the Jacobian is H^1 of the curve, so $E \subseteq \mathrm{End}(J) \otimes \mathbb{Q}$. Thus J is of CM type. □

The characters can be grouped into orbits of $(\mathbb{Z}/m)^\times$, and each orbit spans the cohomology of an isogeny factor of J with complex multiplication by a quotient of $\mathbb{Q}(\mu_m)$ [KR78]; this is the classical computation of Weil [Wei52].

3.3. **The inductive structure.** Fix $m \geq 1$ and write $X^n = X^n_m$, the Fermat variety $x_0^m + \cdots + x_{n+1}^m = 0$ in $\mathbb{P}^{n+1}$, for $n \geq 0$; thus X^0 is a set of m points. Fix ε with $\varepsilon^m = -1$.

Proposition 3.4 (Shioda–Katsura [SK79])**.** *Let $r, s \geq 1$, let $Y \subseteq X^r \times X^s$ be the subvariety $x_{r+1} = y_{s+1} = 0$, which is isomorphic to $X^{r-1} \times X^{s-1}$ and has codimension two, and let Z be the blow-up of $X^r \times X^s$ along Y. The rational map $\varphi\colon X^r \times X^s \dashrightarrow X^{r+s}$,*

$$\varphi(x, y) = \big[x_0 y_{s+1} : \cdots : x_r y_{s+1} :\ \varepsilon x_{r+1} y_0 : \cdots : \varepsilon x_{r+1} y_s\big],$$

is defined exactly off Y, and it extends to a surjective morphism $Z \to X^{r+s}$.

Proof. The image lies in X^{r+s}, because $y_{s+1}^m \sum_{i \leq r} x_i^m + \varepsilon^m x_{r+1}^m \sum_{j \leq s} y_j^m = -y_{s+1}^m x_{r+1}^m + x_{r+1}^m y_{s+1}^m = 0$. All coordinates vanish exactly on Y: if $y_{s+1} \neq 0$ they force $x_0 = \cdots = x_r = 0$ and then $x_{r+1}^m = -\sum_{i \leq r} x_i^m = 0$, which is impossible; so $y_{s+1} = 0$, and in the same way $x_{r+1} = 0$. On the blow-up, on the chart where $y_{s+1} = t\, x_{r+1}$, the map is $[t x_0 : \cdots : t x_r : \varepsilon y_0 : \cdots : \varepsilon y_s]$; here $y_0, \ldots, y_s$ do not all vanish, because otherwise $y_{s+1}^m = 0$ as well. The other chart is symmetric. So φ extends to a morphism on Z [SK79, Lemma 1.2]. It is dominant, since from a general point of X^{r+s} one recovers $[x_0 : \cdots : x_r]$ and $[y_0 : \cdots : y_s]$, and then x_{r+1} and y_{s+1}

[7] On the affine curve $v^{s-m} du = -v^{s-1} u^{1-m} dv$ shows that $\omega_{r,s}$ has no poles where u or v is finite. At the m points at infinity, u and v have simple poles and du has a double pole, so $\omega_{r,s}$ has order $-(r-1) - (s-m) - 2 = m - 1 - r - s \geq 0$ there.

up to finitely many choices; its image is closed and X^{r+s} is irreducible for $r+s \geq 2$, so it is surjective.[8] □

3.4. **Proof of Theorem E.**

Theorem 3.5. *Every Fermat variety X^n_m lies in $\mathcal{D}$. So do all products of Fermat varieties of any degrees and dimensions, and every variety onto which such a product maps surjectively.*

Proof. Fix m. We show $X^n \in \mathcal{D}$ by induction on n. The set of points X^0 lies in $\mathcal{D}$, and so does the curve X^1, by Proposition 3.2(ii) and Lemma 3.3. Let $n \geq 2$ and suppose $X^k \in \mathcal{D}$ for $k < n$. Apply Proposition 3.4 with $r = 1$ and $s = n-1$. The product $X^1 \times X^{n-1}$ and the centre $Y \cong X^0 \times X^{n-2}$ lie in $\mathcal{D}$ by Proposition 3.2(iii); the blow-up Z lies in $\mathcal{D}$ by Proposition 3.2(v); and X^n, the image of Z, lies in $\mathcal{D}$ by Proposition 3.2(iv). The remaining assertions are Proposition 3.2(iii) and (iv). □

Proof of Theorem E. Combine Theorem 3.5 with Proposition 3.2(i). □

The proof shows more than Theorem E states: every Hodge class on a Fermat variety is the image, under an explicit algebraic correspondence, of a Hodge class on a power of the Jacobians of Fermat curves of the same degree. Theorem E is therefore exactly as strong as (CM) for these powers.

3.5. **The Hodge classes of a Fermat variety.** The group $G_n = \mu_m^{n+2}/\mu_m$ acts on X^n_m by multiplying the coordinates, and its characters are the vectors $a = (a_0, \dots, a_{n+1}) \in (\mathbb{Z}/m)^{n+2}$ with $\sum a_i = 0$. Let $\mathfrak{A}^n_m$ be the set of characters with every $a_i \neq 0$. For $a \in \mathfrak{A}^n_m$ write $\langle a \rangle = \sum_i \bar{a}_i/m$, where $\bar{a}_i \in \{1, \dots, m-1\}$ represents a_i; it is an integer between 1 and $n+1$.

Proposition 3.6. (i) *The primitive cohomology $H^n_{\mathrm{prim}}(X^n_m, \mathbb{C})$ is the direct sum of lines $V(a)$, one for each $a \in \mathfrak{A}^n_m$, on which G_n acts by pull-back through the character a; and $V(a) \subseteq H^{n-q,q}$ with $q = \langle a \rangle - 1$. Hence*

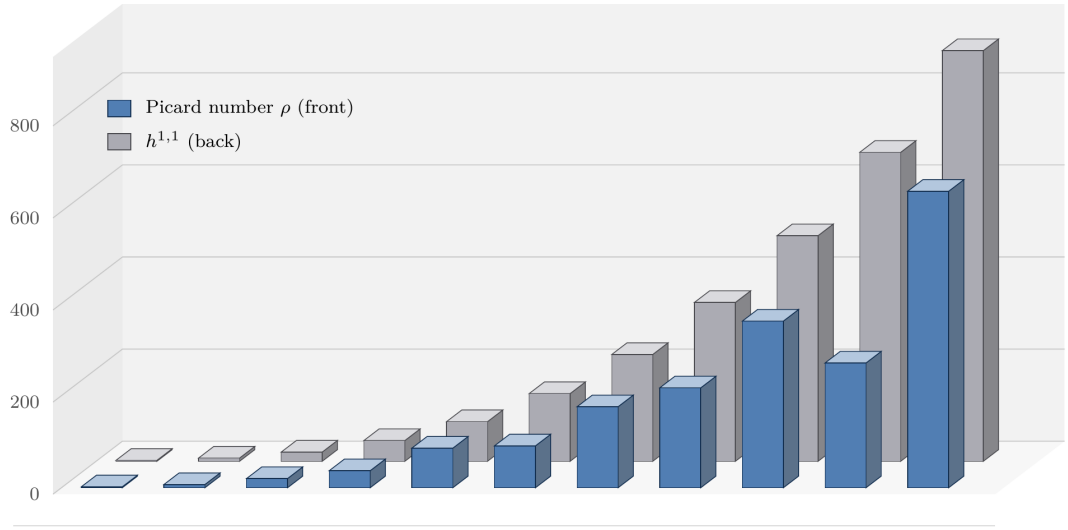
$$\dim H^n_{\mathrm{prim}} = |\mathfrak{A}^n_m| = \frac{(m-1)^{n+2} + (-1)^n(m-1)}{m}, \qquad h^{n-q,q}_{\mathrm{prim}} = \#\{a : \langle a \rangle = q+1\}.$$

(ii) *Let n be even. The space of primitive Hodge classes, tensored with $\mathbb{C}$, is the sum of the lines $V(a)$ for which $\langle ta \rangle = n/2+1$ for every $t \in (\mathbb{Z}/m)^\times$.*

Proof. (i) By Griffiths' description of the cohomology of a hypersurface by residues [Gri69, §8], $H^{n-q,q}_{\mathrm{prim}}(X^n_m)$ is isomorphic to the graded piece of degree $(q+1)m - (n+2)$ of the Jacobian ring $\mathbb{C}[x_0, \dots, x_{n+1}]/(x_0^{m-1}, \dots, x_{n+1}^{m-1})$, the class of a monomial x^b corresponding to the residue of $x^b\Omega/F^{q+1}$, where $F = \sum x_i^m$ and $\Omega = \sum (-1)^i x_i \, dx_0 \wedge \cdots \widehat{dx_i} \cdots \wedge dx_{n+1}$. That residue is an eigenvector of G_n with character $a = b + (1, \dots, 1)$, because Ω has character $(1, \dots, 1)$ and F is invariant. As b runs over the exponents with $0 \leq b_i \leq m-2$, the vector a runs once over the characters with $1 \leq \bar{a}_i \leq m-1$; the degree condition is $\sum \bar{a}_i = (q+1)m$, that is, $\langle a \rangle = q+1$, and it forces $\sum a_i = 0$ in $\mathbb{Z}/m$. The count of $\mathfrak{A}^n_m$ follows from the recursion $N_k = (m-1)^{k-1} - N_{k-1}$, $N_1 = 0$, for the number N_k of k-tuples of nonzero residues with sum zero.

(ii) The eigenspace decomposition is defined over $\mathbb{Q}(\mu_m)$, and the automorphism $\zeta \mapsto \zeta^t$ of $\mathbb{Q}(\mu_m)$ carries $V(a)$ to $V(ta)$. So for each orbit O of $(\mathbb{Z}/m)^\times$ on $\mathfrak{A}^n_m$ the sum $\bigoplus_{a \in O} V(a)$ is defined over $\mathbb{Q}$, these rational subspaces decompose $H^n_{\mathrm{prim}}(X^n_m, \mathbb{Q})$, and the one for O consists of Hodge classes exactly when every $V(a)$, $a \in O$, has type $(n/2, n/2)$, that is, when $\langle a \rangle = n/2+1$

[8] Shioda and Katsura show more: the morphism $Z \to X^{r+s}$ is the composite of the quotient by a cyclic group of order m and a birational morphism that contracts two disjoint subvarieties of the quotient [SK79, Theorem 1.7]. We use only that it is a surjective morphism.



m	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**
ρ	2	7	20	37	86	91	176	217	362	271	644
$h^{1,1}$	2	7	20	45	86	147	232	345	490	671	892

FIGURE 2. The Fermat surfaces X_m^2, $2 \le m \le 12$: the Picard number ρ (front row) against $h^{1,1}$ (back row), both computed from Proposition 3.6. On a surface every Hodge class is algebraic (Theorem 2.3); the figure shows how the share of $H^{1,1}$ that is defined over $\mathbb{Q}$ varies with the arithmetic of m. The values of $h^{1,1}$ are recomputed from the Jacobian ring in Macaulay2 (Appendix A).

on O. A rational class with a component on some $V(a)$ has components on the whole orbit of a, so the Hodge classes are the sum of the rational subspaces of the orbits of this kind. □

For a Fermat surface the Picard number is $\rho(X_m^2) = 1 + \#\{a \in \mathfrak{A}_m^2 : \langle ta \rangle = 2 \text{ for all } t \in (\mathbb{Z}/m)^\times\}$, and $h^{1,1} = (2m^3 - 6m^2 + 7m)/3$. Figure 2 compares the two for $m \le 12$. For m prime to 6, $\rho = 3(m-1)(m-2)+1$, and the Néron–Severi group is spanned by the $3m^2$ lines on the surface [AS83]; the quartic, quintic and sextic give $\rho = 20$, 37 and 86.

Remark 3.7. The Hodge conjecture for X_m^n is known without hypothesis when m is prime or $m = 4$ [Ran80, Shi79], when $m \le 20$ [Shi79], and when m is a power of a prime [Aok87, Corollary 2-3]; see [dSJ21, §2]. Aoki proves it for every abelian variety of Fermat type of degree m when $m = 2^a 3^b 5^c 7^d$ with $c = 0$ or $d = 0$, and when m is p^e or $2p^e$ [Aok00, Theorem 0.1]; as noted after the proof of Theorem E, this gives it for X_m^n. The preprint [Pet26] adds the degrees $2^a 3^b 5^c 7^d$ not divisible by 420, and Theorem A the degrees $2^a 3^b 19^c$ with $a \le 1$ that are divisible by 57, the degrees $2^a 11^b$ that are divisible by 44, the degrees $2^a 13^b$ that are divisible by 52 and the degrees $2^a 3^b 17^c$ with $a \le 1$ that are divisible by 51 (Remarks 4.35 to 4.38). When every prime factor of m exceeds $n+2$, every Hodge character is a union of pairs [Aok83, Theorem A], and the classes are spanned by linear subspaces [Ran80, Shi79]; for fourfolds this covers the degrees whose prime factors are all at least 7. For fourfolds of degree $m \le 100$ prime to 6 the conjecture is proved in [dSJ21, Proposition 3.1] by a computer search, and the preprint [Jum26b] gives a computer-assisted proof for fourfolds of odd degree at most 199. Kang states it for every Fermat fourfold [Kan16, Corollary 3.2], and the preprint [MMRV26], which treats all dimensions in degree less than 65 except 44, 51 and 52, uses that statement; its proof has

the gap described in Remark 3.21. So Theorem B is new for the odd degrees $m > 199$ that are divisible by 3 or by 5 and have a prime factor larger than 7, the first of which are 201, 205, 207, 213 and 215, and its proof, unlike those of [dSJ21, Jum26b], does not use a computer. In all these proofs the Hodge classes are written as combinations of the classes of linear subspaces and of cycles built from them, characters a that split into pairs $a_i + a_j = 0$ corresponding to linear subspaces. Theorem E needs no such decomposition: it moves the whole question to the Jacobians of Fermat curves. The preprint [Ope26a] announces the Hodge conjecture on every self-power of a very general member of the family of diagonal hypersurfaces; the Fermat varieties are special members of that family.

3.6. **Fermat fourfolds of degree prime to six.** In this subsection m is prime to 6, and we prove Theorem B for such m; Section 3.7 treats the odd m divisible by 3. For $a \in \mathbb{Z}/m$ we write $\bar{a} \in \{0, \dots, m-1\}$ for its representative, extending the notation of Section 3.5.

Definition 3.8. Let A be a multiset of nonzero elements of $\mathbb{Z}/m$ whose sum is 0. It is *balanced* if the integer $\sum_{a \in A} \overline{ta}$ is the same for every $t \in (\mathbb{Z}/m)^\times$; we call this integer $h(A)$. It *contains a pair* if $a \in A$ and $-a \in A$ for some a; since m is odd, $a \neq -a$, so these are two different members of A. When $5 \mid m$, the *fiber* of $x \in \mathbb{Z}/m$ is the set

$$\Phi(x) = \{x + jm/5 : \ 0 \le j \le 4\}$$

of the five solutions y of $5y = 5x$, and A is 5-*standard* if $A = \Phi(x) + \{-5x\}$ for some x, as multisets.

For a balanced multiset of n elements, comparing t with $-t$ gives $h(A) = nm - h(A)$, so $h(A) = nm/2$. By Proposition 3.6(ii), a character $\alpha \in \mathfrak{A}_m^4$ spans a line of Hodge classes exactly when the multiset of its entries is balanced, and the same holds for the characters $\beta \in \mathfrak{A}_m^2$ of the Fermat surface. Aoki writes $\sigma_{5,x} = (x, x+d, x+2d, x+3d, x+4d, -5x)$, $d = m/5$, for the 5-standard characters [Aok87].

Theorem 3.9. *Let m be prime to 6.*

(i) *Every balanced multiset of four nonzero elements of $\mathbb{Z}/m$ contains a pair.*
(ii) *Every balanced multiset of six nonzero elements of $\mathbb{Z}/m$ contains a pair or is 5-standard.*

Part (i) is Aoki's theorem that the Hodge classes of a Fermat surface of degree prime to 6 are spanned by lines [Aok83, Theorem A]; we prove it again because the proof of (ii) uses it and runs on the same lines. When $5 \nmid m$, part (ii) says that every Hodge character of X_m^4 splits into three pairs, which is also contained in [Aok83, Theorem A], since then every prime factor of m exceeds 6. The case $5 \mid m$ is new. The theorem has been checked in Lean (Appendix A).

Proof of Theorem B for m prime to 6, given Theorem 3.9. Write $G = G_4$ for the group acting on $X = X_m^4$. By Proposition 3.6(ii), and since $H^4(X, \mathbb{Q})$ is the sum of the primitive part and the line spanned by the square of the hyperplane class, it is enough to show that every line $V(\alpha)$ with α balanced lies in $\mathrm{Alg}^2(X) \otimes \mathbb{C}$. Then the Hodge classes and the algebraic classes span the same complex subspace, and since $\mathrm{Alg}^2(X) \subseteq \mathrm{Hdg}^2(X)$ are rational subspaces they are equal. A subvariety $Z \subseteq X$ *represents* α [Aok87] if the α-component

$$\frac{1}{|G|} \sum_{g \in G} \alpha(g)^{-1} g^*[Z] = \frac{1}{|G|} \sum_{g \in G} \alpha(g)^{-1} [g^{-1} Z]$$

of $[Z]$ is not zero; it lies in $V(\alpha)$ and in $\mathrm{Alg}^2(X) \otimes \mathbb{C}$, so then $V(\alpha) \subseteq \mathrm{Alg}^2(X) \otimes \mathbb{C}$. Permuting the coordinates is an automorphism of X that permutes the lines $V(\alpha)$ accordingly, so the order of the entries of α does not matter. By Theorem 3.9(ii) there are two cases.

α contains a pair. Order the entries as $\alpha = (\beta_0, \beta_1, \beta_2, \beta_3, a, -a)$. Since $\overline{ta} + \overline{-ta} = m$ for every unit t, the multiset of entries of $\beta = (\beta_0, \dots, \beta_3)$ is balanced, so β is a Hodge character of the Fermat surface X^2_m, and $V(\beta)$ consists of algebraic classes by Theorem 2.3. The character $(a, -a)$ of the set X^0_m of m points spans a line of $H^0(X^0_m, \mathbb{C})$, which is spanned by classes of points. By the inductive structure, in the form of Shioda and Ran stated in [Aok87, Theorem 1-4(i)], $V(\alpha)$ is then spanned by algebraic classes.

α is 5-standard. Then $5 \mid m$; write $d = m/5$ and $\alpha = (x, x+d, \dots, x+4d, -5x)$. Aoki constructs a subvariety of codimension two representing α when $\gcd(\bar{x}, d) = 1$ [Aok87, Theorem 2-1]. In general let $g = \gcd(\bar{x}, d)$, $m' = m/g$ and $d' = d/g = m'/5$. Every entry of α is a multiple of g, so $\alpha = g\alpha'$ with

$$\alpha' = (x', x'+d', \dots, x'+4d', -5x') \in (\mathbb{Z}/m')^6, \qquad x' = \bar{x}/g, \quad \gcd(x', d') = 1,$$

where $g\alpha'$ means the image of α' under the injective homomorphism $\mathbb{Z}/m' \to \mathbb{Z}/m$, $y \mapsto gy$. The entries of α' are nonzero, because those of α are. Every unit t' modulo m' lifts to a unit t modulo m, and then $\overline{t'\alpha'_i}/m' = \overline{t\alpha_i}/m$ for each i; so α' is balanced, and it is a Hodge character of $X' = X^4_{m'}$. In particular $d' \geq 3$, since for $d' = 1$ one of $x', x'+1, \dots, x'+4$ would vanish modulo 5. By [Aok87, Theorem 2-1] there is a subvariety $Y' \subseteq X'$ of codimension two that represents α'. Let

$$\pi \colon X \to X', \qquad [x_0 : \cdots : x_5] \mapsto [x_0^g : \cdots : x_5^g].$$

It is a finite surjective morphism, and it is equivariant for the homomorphism $G \to G'$, $\zeta \mapsto \zeta^g$, to the group G' acting on X'. For $\xi \in V(\alpha')$ and $\zeta \in G$ we get $\zeta^*\pi^*\xi = \pi^*(\zeta^g)^*\xi = \alpha'(\zeta^g)\pi^*\xi = \alpha(\zeta)\pi^*\xi$, so π^* maps $V(\beta')$ into $V(g\beta')$ for every character β' of G', and distinct β' go to distinct $g\beta'$. As $\pi_*\pi^*$ is multiplication by $\deg \pi$, the map π^* is injective on cohomology. Hence the α-component of $\pi^*[Y']$, the class of the cycle π^*Y', is the image under π^* of the α'-component of $[Y']$, which is not zero; so the cycle π^*Y' represents α, and again $V(\alpha) \subseteq \mathrm{Alg}^2(X) \otimes \mathbb{C}$. □

The rest of this subsection proves Theorem 3.9. We fix a balanced multiset A of $n \in \{4, 6\}$ nonzero elements of $\mathbb{Z}/m$ that contains no pair, and show that $n = 6$ and A is 5-standard.

Step 1: the counting function. Let e be the largest additive order of an element of A. It divides m, so it is prime to 6, and $e > 1$. The map $\kappa \colon \mathbb{Z}/e \to \mathbb{Z}/m$, $y \mapsto (m/e)y$, is an injective homomorphism, and the elements of $\mathbb{Z}/m$ of order exactly e are the $\kappa(z)$ with $z \in U = (\mathbb{Z}/e)^\times$. For $z \in U$ put

$$F(z) = \mathrm{mult}_A(\kappa z) - \mathrm{mult}_A(-\kappa z),$$

the difference of the multiplicities in A. Then $F(-z) = -F(z)$. If $a_0 = \kappa(z_0) \in A$ has order e, then $-a_0 \notin A$, so $F(z_0) \geq 1$. We call $\|F\| = \sum_{z \in U} |F(z)|$ the *mass* of F.

Lemma 3.10. *$\|F\| \leq 2n$. If $\|F\| = 2n$, then every element of A has order e and $\sum_{z \in U} F(z)\, z = 0$ in $\mathbb{Z}/e$.*

Proof. $\|F\| \leq \sum_z (\mathrm{mult}_A(\kappa z) + \mathrm{mult}_A(-\kappa z)) = 2N$, where $N \leq n$ is the number of elements of A of order e. If $\|F\| = 2n$, then $N = n$, and writing $a = \kappa(z_a)$ for $a \in A$,

$$\sum_z F(z) z = \sum_{a \in A} z_a - \sum_{a \in A} (-z_a) = 2 \sum_{a \in A} z_a,$$

while $\kappa(\sum_a z_a) = \sum_a a = 0$; as κ is injective, $\sum_a z_a = 0$. □

Step 2: the characters. For a Dirichlet character χ modulo e let $B_{1,\chi} = e^{-1}\sum_{s=1}^{e}\chi(s)s$. If χ is odd and primitive, then $B_{1,\chi} \neq 0$. This is classical: by the relative class number formula [Was97, Theorem 4.17] the product of the numbers $-B_{1,\chi}/2$, over the odd characters of the Galois group of $\mathbb{Q}(\mu_e)$, is a nonzero rational multiple of the relative class number, a positive integer; equivalently, $B_{1,\chi}$ is a nonzero multiple of $L(1,\bar{\chi})$ [Was97, Theorem 4.9]. This is the only analytic input of the proof.

Lemma 3.11. $\sum_{z\in U} F(z)\chi(z)^{-1} = 0$ *for every primitive Dirichlet character* χ *modulo* e.

Proof. If χ is even, the sum vanishes because F is odd. Let χ be odd, and regard it as a character of $(\mathbb{Z}/m)^\times$ through reduction modulo e; it is not trivial. Since A is balanced,

$$\sum_{a\in A}\sum_{t\in(\mathbb{Z}/m)^\times}\chi(t)\,\overline{ta} = h(A)\sum_{t}\chi(t) = 0.$$

Let $a \in A$ have order f. If $f \neq e$, then $e \nmid f$, since $f \leq e$. For a divisor f of m let $W_f \subseteq (\mathbb{Z}/m)^\times$ be the units congruent to 1 modulo f. If χ were trivial on W_f, it would be trivial on $W_fW_e = W_{\gcd(f,e)}$ (by the Chinese remainder theorem), so it would be induced from a character modulo the proper divisor $\gcd(f,e)$ of e, against primitivity. So χ is not trivial on W_f; as $\overline{ta}$ depends only on t modulo f, the inner sum vanishes. If $f = e$, write $a = \kappa(z)$. For a unit t with image $s \in U$ we have $ta = \kappa(sz)$ and $\overline{ta} = (m/e)\,\overline{sz}$, where now $\overline{sz} \in \{0,\dots,e-1\}$. Reduction $(\mathbb{Z}/m)^\times \to U$ is onto, with fibres of size $\varphi(m)/\varphi(e)$, so

$$\sum_{t}\chi(t)\,\overline{ta} = \frac{m\,\varphi(m)}{e\,\varphi(e)}\sum_{s\in U}\chi(s)\,\overline{sz} = \frac{m\,\varphi(m)}{\varphi(e)}\,\chi(z)^{-1}B_{1,\chi},$$

after the substitution $s \mapsto sz^{-1}$. Adding over A and dividing by $m\varphi(m)B_{1,\chi}/\varphi(e) \neq 0$ gives $\sum_z \mathrm{mult}_A(\kappa z)\chi(z)^{-1} = 0$. The substitution $z \mapsto -z$ turns $\sum_z \mathrm{mult}_A(-\kappa z)\chi(z)^{-1}$ into $\chi(-1)^{-1}$ times the same sum, so it vanishes too, and so does their difference. □

Step 3: splitting along the subgroups K_p. For a prime $p \mid e$ let $K_p \subseteq U$ be the subgroup of units congruent to 1 modulo e/p, the kernel of $U \to (\mathbb{Z}/(e/p))^\times$.

Lemma 3.12. (a) $|K_p| = p$ *if* $p^2 \mid e$, *and* $|K_p| = p-1$ *if* $p \parallel e$. *So* $|K_p| \geq 4$, *with equality only for* $p = 5 \parallel e$.

(b) *Let* $p^v \parallel e$. *The product of the* K_q *with* $q \neq p$ *consists of units congruent to* 1 *modulo* p^v. *Hence* K_p *meets it trivially, the product* H *of all the* K_q *is direct, and* -1 *does not lie in the product of the* K_q *with* $q \neq p$.

(c) *A Dirichlet character modulo* e *that is not primitive is trivial on some* K_p.

Proof. (a) $|K_p| = \varphi(e)/\varphi(e/p)$. (b) For $q \neq p$, p^v divides e/q. An element of K_p congruent to 1 modulo p^v is congruent to 1 modulo $\mathrm{lcm}(e/p, p^v) = e$, and $-1 \not\equiv 1 \pmod p$ as p is odd. (c) A character induced from a proper divisor f of e is trivial on the units congruent to 1 modulo f, and f divides e/p for some $p \mid e$. □

Definition 3.13. Let $K_1,\dots,K_r$ be subgroups of a finite abelian group, H their product and C a coset of H. A function $f\colon C \to \mathbb{Z}$ is *split along* $(K_1,\dots,K_r)$ if $f = f_1 + \cdots + f_r$ with functions $f_i\colon C \to \mathbb{C}$ such that $f_i(zk) = f_i(z)$ for $z \in C$ and $k \in K_i$; for $r = 0$ this means $f = 0$. A *line* along K_i is a coset wK_i.

Lemma 3.14. *On every coset of* H *in* U, *the function* F *is split along the family* $(K_p)_{p\mid e}$.

Proof. By Fourier inversion on U, $F = \varphi(e)^{-1}\sum_\chi \hat{F}(\chi)\chi$ with $\hat{F}(\chi) = \sum_z F(z)\chi(z)^{-1}$. By Lemma 3.11 only characters that are not primitive contribute, and by Lemma 3.12(c) each of them is trivial on some K_p. Collect the terms according to such a p. □

We use four simple properties of split functions, for f split along $(K_1,\dots,K_r)$ on C and $H' = K_2\cdots K_r$.

(S1) For $k \in K_1$ and every coset $S \subseteq C$ of H', the function $z \mapsto f(z) - f(zk)$ on S is split along $(K_2,\dots,K_r)$: the terms f_1 cancel.
(S2) If f vanishes on one coset $S_0 \subseteq C$ of H', then f is split along $(K_2,\dots,K_r)$ on every coset $S \subseteq C$ of H': choose $k \in K_1$ with $Sk = S_0$ and apply (S1).
(S3) For $r = 2$ and $C = xK_1K_2$: $f(xij) = f(xi) + f(xj) - f(x)$ for $i \in K_1$ and $j \in K_2$.
(S4) For $r = 1$, f is invariant under K_1.

Step 4: three lemmas on split functions. In the next three lemmas, $K_1,\dots,K_r$ (or $K, K_2,\dots,K_r$) are subgroups of a finite abelian group whose product H is direct, and C is a coset of H. Fix $x \in C$ and put $H' = K_2\cdots K_r$. For c in the first subgroup, the *slices* of C are the cosets $S_c = xcH'$; they are pairwise distinct, and for a function f on C we write $\mu(c) = \sum_{z\in S_c} |f(z)|$.

Lemma 3.15 (grid)**.** *Suppose $|K_i| \geq 4$ for every i, with equality for at most one i. Let $f\colon C \to \mathbb{Z}$ be split along $(K_1,\dots,K_r)$, not identically zero, with $\sum_C |f| \leq 6$. Then for some i with $|K_i| \leq 6$ there are a line $L = wK_i \subseteq C$ and $\sigma \in \{\pm 1\}$ such that $f = \sigma$ on L and $f = 0$ on $C \setminus L$.*

Proof. Induction on r. For $r = 0$ the function vanishes, which is excluded. Let $r \geq 1$ and $K = K_1$. First an observation.

(P) *Let $c, c' \in K$ with $\mu(c) + \mu(c') \leq 6$, and suppose $g(z) = f(z) - f(zc^{-1}c')$ is not identically zero on S_c. Then $\mu(c) + \mu(c') \geq |K_i|$ for some $i \geq 2$.* Indeed, g is split along $(K_2,\dots,K_r)$ on S_c by (S1), and $\sum_{S_c} |g| \leq \mu(c) + \mu(c')$ because $z \mapsto zc^{-1}c'$ maps S_c onto $S_{c'}$. By induction $|g|$ is 1 on a line along some K_i, $i \geq 2$, and 0 elsewhere, so $\sum_{S_c} |g| = |K_i|$.

Case A: $f(zk) = f(z)$ for all $z \in C$, $k \in K$. Let $f(z_0) \neq 0$. Then f equals $f(z_0)$ on the $|K| \geq 4$ points of z_0K, so $|f(z_0)| = 1$ and $|K| \leq 6$, and a nonzero value off z_0K would add at least $|K| \geq 4$ more. So $f = f(z_0)$ on z_0K and 0 elsewhere.

Case B1: $\mu(c_0) = 0$ for some c_0. By (S2), f is split along $(K_2,\dots,K_r)$ on every slice. On a slice where it does not vanish, by induction it is ± 1 on a line along some K_i with $i \geq 2$ and $|K_i| \leq 6$ and 0 elsewhere, which uses mass $|K_i| \geq 4$; a second such slice would bring the mass to 8. This is the claim.

Case B2: f is not K-invariant, and $\mu \geq 1$ on every slice. As $\sum \mu \leq 6 < 2|K|$, some c_s has $\mu(c_s) = 1$. Choose $c, k \in K$ such that $f(z) \neq f(zk)$ for some $z \in S_c$. By (P), applied to c and ck, or because $\mu(c) + \mu(ck) > 6$, one of them, c_1, has $\mu(c_1) \geq 2$. Then $f(z) - f(zc_1^{-1}c_s)$ does not vanish identically on S_{c_1}, since translation carries S_{c_1} onto S_{c_s} and $\mu(c_1) \neq \mu(c_s)$. By (P), $\mu(c_1) + 1 \geq |K_i| \geq 4$ for some $i \geq 2$, so

$$6 \geq \sum_{c\in K} \mu(c) \geq \mu(c_1) + (|K| - 1) \geq 3 + 3.$$

All these are equalities: $|K| = 4$ and $|K_i| = \mu(c_1) + 1 = 4$, two subgroups of order four, which is excluded. □

Lemma 3.16 (a large subgroup carrying the sign)**.** *Let $|K| \geq 10$ and $|K_i| \geq 4$ for $i \geq 2$, with equality for at most one i. Let $f\colon C \to \mathbb{Z}$ be split along $(K, K_2,\dots,K_r)$, and suppose that for some $\varepsilon \in K \setminus \{1\}$ and $\varepsilon' \in H'$ we have $f(z\varepsilon\varepsilon') = -f(z)$ for all $z \in C$. If $\sum_C |f| \leq 12$ and $f(x) \geq 1$, then $f = 1$ on a line along some K_i, $i \geq 2$, with $|K_i| \leq 6$.*

Proof. Multiplication by $\varepsilon\varepsilon'$ carries S_c onto $S_{c\varepsilon}$ and preserves $|f|$, so $\mu(c\varepsilon) = \mu(c)$. If some slice has $\mu = 0$, then f is split along $(K_2,\dots,K_r)$ on $S_1 \ni x$ by (S2), and $2\mu(1) = \mu(1) + \mu(\varepsilon) \leq 12$; by Lemma 3.15 f is σ on a line along some K_i in S_1 and 0 elsewhere on S_1, and $\sigma = 1$ because

the line passes through x. Otherwise $\mu \geq 1$ everywhere, and as $\sum \mu \leq 12 < 2|K|$, some c has $\mu(c) = 1$, hence $\mu(c\varepsilon) = 1$. The function $f(z) - f(z\varepsilon)$ on S_c is split along $(K_2, \dots, K_r)$ by (S1) and has mass at most 2, so it vanishes by Lemma 3.15. Let z_0 be the point of S_c with $f(z_0) = \pm 1$. The points $z_0\varepsilon$ and $z_0\varepsilon\varepsilon'$ of $S_{c\varepsilon}$ carry the values $f(z_0)$ and $-f(z_0)$, both nonzero, so they coincide because $\mu(c\varepsilon) = 1$; then $f(z_0) = -f(z_0)$, a contradiction. □

Lemma 3.17 (two subgroups)**.** *Let $K \cap L = 1$, $|K| = 4$, $|L| = 6$, and let $\varepsilon \in K$ and $\varepsilon' \in L$ be different from 1, with $\varepsilon^2 = 1$. Let $f\colon xKL \to \mathbb{Z}$ be split along (K, L), with $f(z\varepsilon\varepsilon') = -f(z)$, $\sum |f| \leq 12$ and $f(x) \geq 1$. Then*

(a) *$f(xi) = 1$ for all $i \in K$; or*
(b) *$f(xj) = 1$ for all $j \in L$; or*
(c) *$\sum |f| = 12$, and there are $u\colon K \to \{\pm 1\}$ and $v\colon L \to \{\pm 1\}$ with $u(i\varepsilon) = -u(i)$, $v(j\varepsilon') = -v(j)$ and $2f(xij) = u(i) + v(j)$.*

Proof. By (S3), $f(xij) = f(xi) + f(xj) - f(x)$. Put $\delta = f(x) - f(x\varepsilon')$, $u(i) = 2f(xi) - \delta$ and $v(j) = 2f(xj) - 2f(x) + \delta$, so that $2f(xij) = u(i) + v(j)$. Comparing $f(xi\varepsilon \cdot \varepsilon') = -f(xi)$ with (S3) gives $f(xi\varepsilon) + f(xi) = \delta$, that is $u(i\varepsilon) = -u(i)$; for $i = 1$ it gives $f(x\varepsilon) = -f(x\varepsilon')$, and then the same comparison for $x\varepsilon \cdot j\varepsilon'$ gives $v(j\varepsilon') = -v(j)$. Since $j \mapsto j\varepsilon'$ permutes L and changes the sign of v,

$$2\sum_j |u(i) + v(j)| = \sum_j \big(|u(i) + v(j)| + |u(i) - v(j)|\big) = 2\sum_j \max(|u(i)|, |v(j)|),$$

so $T = \sum_{i,j} \max(|u(i)|, |v(j)|) = 2\sum |f| \leq 24$. All $u(i)$ and $v(j)$ have the same parity, because $u(i) + v(j) = 2f(xij)$. Write $K = \{1, \varepsilon, a, a\varepsilon\}$, $V = \sum_j |v(j)|$ and $M(s) = \sum_j \max(s, |v(j)|)$; then $T = 2M(|u(1)|) + 2M(|u(a)|)$, $M(s) \geq \max(6s, V)$, and $V \geq 2|v(1)|$ because $|v(\varepsilon')| = |v(1)|$.

If the parity is odd, all 24 terms of T are at least 1, so all equal 1, and we are in case (c). If it is even and $u(1) \geq 1$, then $u(1) \geq 2$ and $24 \geq T \geq 24 + 2V$, so $u(1) = 2$, $v = 0$, and $f(xj) = (u(1) + v(j))/2 = 1$: case (b). If it is even and $u(1) \leq 0$, then $v(1) = 2f(x) - u(1) \geq 2$. If $u(1) \neq 0$, then $T \geq 2 \cdot 12 + 2V > 24$; so $u(1) = 0$, and in the same way $u(a) = 0$. Then $u = 0$ on K, $T = 4V \leq 24$, so $2v(1) \leq V \leq 6$ and $v(1) = 2$, and $f(xi) = (u(i) + v(1))/2 = 1$: case (a). □

Step 5: the core dichotomy.

Proposition 3.18. *Let $e > 1$ be prime to 6, $U = (\mathbb{Z}/e)^\times$, and let $F\colon U \to \mathbb{Z}$ satisfy $F(-z) = -F(z)$, $\|F\| \leq 12$ and $F(x) \geq 1$ for some x, and be split along $(K_p)_{p|e}$ on every coset of their product H. Then*

(1) *$F \geq 1$ on a line wK_p for $p = 5$ or $p = 7$; or*
(2) *the primes dividing e are 5 and 7, $|K_5| = 4$, $|K_7| = 6$, and there are $y \in U$, $\varepsilon \in K_5$ and $\varepsilon' \in K_7$ with $\varepsilon\varepsilon' = -1$, and functions $u\colon K_5 \to \{\pm 1\}$ and $v\colon K_7 \to \{\pm 1\}$ with $u(i\varepsilon) = -u(i)$, $v(j\varepsilon') = -v(j)$ and $2F(yij) = u(i) + v(j)$ for $i \in K_5$, $j \in K_7$; the mass of F on yK_5K_7 is 12.*

Proof. By Lemma 3.12(a) every $|K_p|$ is at least 4, only K_5 can have order 4, and $|K_p| \leq 9$ forces $p \in \{5, 7\}$.

Case (a): $-1 \notin H$. The cosets xH and $-xH$ are distinct and carry the same mass, so $\sum_{xH} |F| \leq 6$. By Lemma 3.15, $F = \sigma$ on a line along some K_p with $|K_p| \leq 6$, so $p \in \{5, 7\}$, and $\sigma = 1$ because the line passes through x.

Case (b): $-1 \in H$ and $|K_p| \geq 10$ for some p. Write $-1 = \varepsilon\varepsilon'$ with $\varepsilon \in K_p$ and ε' in the product of the other K_q; $\varepsilon \neq 1$ by Lemma 3.12(b). Lemma 3.16 gives (1).

Case (c): $-1 \in H$ and $|K_p| \le 9$ for all p. Then every prime factor of e is 5 or 7. If only one prime p divides e, then $-1 \in K_p$ and F is K_p-invariant on xK_p by (S4), so $F(x) = F(-x) = -F(x)$, which is impossible. So $H = K_5K_7$, and $-1 = \varepsilon\varepsilon'$ with $\varepsilon \in K_5$, $\varepsilon' \in K_7$, both different from 1 by Lemma 3.12(b). From $\varepsilon^2\varepsilon'^2 = 1$ and $K_5 \cap K_7 = 1$ we get $\varepsilon^2 = \varepsilon'^2 = 1$, so $|K_5|$ and $|K_7|$ are even: $|K_5| = 4$ and $|K_7| = 6$. Lemma 3.17 on xK_5K_7 gives (1) or (2). □

Step 6: the sum of the entries.

Lemma 3.19. *Let $p \mid e$ with $|K_p| \in \{4, 6\}$, and let $F\colon U \to \mathbb{Z}$ be odd with $F \ge 1$ on a line wK_p and $\|F\| \le 2|K_p|$. Then $\|F\| = 2|K_p|$ and $\sum_z F(z)z \ne 0$ in $\mathbb{Z}/e$.*

Proof. If $e = p$ then $K_p = U \ni -1$, and $F(w) \ge 1$, $F(-w) \ge 1$ contradict oddness. So $e/p \ge 5$, $-1 \notin K_p$, and the lines wK_p and $-wK_p$ are disjoint. As $F \ge 1$ on the first and $F \le -1$ on the second, $\|F\| = 2|K_p|$, $F = 1$ on wK_p, $F = -1$ on $-wK_p$, and $F = 0$ elsewhere. Then $\sum_z F(z)z = 2w\sum_{k\in K_p} k \equiv 2|K_p|\,w \pmod{e/p}$. If it vanished, e/p would divide $2|K_p| \in \{8, 12\}$, which is impossible for $e/p > 1$ prime to 6. □

Lemma 3.20. *In case (2) of Proposition 3.18, if $\|F\| = 12$, then $\sum_z F(z)z \ne 0$ in $\mathbb{Z}/e$.*

Proof. The coset yK_5K_7 carries the whole mass, so $F = 0$ off it. Let $\psi\colon \mathbb{Z}/e \to \mathbb{Z}/5$ be reduction. Since $5 \mid e/7$, every $j \in K_7$ has $\psi(j) = 1$. Since $|K_5| = 4$, $5 \parallel e$, and ψ is injective on K_5: an element of K_5 congruent to 1 modulo 5 is congruent to 1 modulo $\operatorname{lcm}(5, e/5) = e$. From $\varepsilon\varepsilon' = -1$, $\psi(\varepsilon) = -1$. Write $K_5 = \{1, \varepsilon, a, a\varepsilon\}$; then $\psi(a) \ne \pm 1$. As $v(j\varepsilon') = -v(j)$, $\sum_j v(j) = 0$. Hence

$$\begin{aligned}\psi\Big(2\sum_z F(z)z\Big) &= \sum_{i\in K_5}\sum_{j\in K_7} \big(u(i) + v(j)\big)\,\psi(y)\psi(i)\\ &= 6\,\psi(y)\sum_i u(i)\psi(i) = 12\,\psi(y)\big(u(1) + u(a)\psi(a)\big),\end{aligned}$$

using $u(\varepsilon) = -u(1)$ and $u(a\varepsilon) = -u(a)$. Here 12 and $\psi(y)$ are units of $\mathbb{Z}/5$. If $\sum_z F(z)z$ vanished, then $u(1) + u(a)\psi(a) = 0$, and $\psi(a) = \pm 1$, a contradiction. □

Step 7: end of the proof. By Lemma 3.10, Lemma 3.14 and the remarks in Step 1, F satisfies the hypotheses of Proposition 3.18 with $\|F\| \le 2n \le 12$. In case (2) the mass is 12, so $n = 6$ and $\|F\| = 12 = 2n$, and Lemma 3.10 contradicts Lemma 3.20. So $F \ge 1$ on a line wK_p with $p \in \{5, 7\}$. Since F is odd, $-1 \notin K_p$, so wK_p and $-wK_p$ are disjoint and $\|F\| \ge 2|K_p|$.

If $p = 7$, then $|K_7| = 7$ is impossible as $14 > 12$, and $|K_7| = 6$ forces $n = 6$ and $\|F\| = 12 = 2n$, which Lemma 3.10 and Lemma 3.19 exclude. So $p = 5$. If $n = 4$, then $|K_5| = 5$ is impossible as $10 > 8$, and $|K_5| = 4$ forces $\|F\| = 8 = 2n$, excluded in the same way. This proves (i).

Let $n = 6$ and $p = 5$, and put $x = \kappa(w)$. For $k \in K_5$ we have $5k = 5$ in $\mathbb{Z}/e$, so $5\kappa(wk) = \kappa(5w) = 5x$; and $F(wk) \ge 1$ means $\kappa(wk) \in A$. So A contains the $|K_5|$ distinct elements of $L = \kappa(wK_5) \subseteq \Phi(x)$. Recall that $\sum_{y\in\Phi(x)} y = 5x$ and, writing $\bar{x} = qm/5 + r$ with $0 \le r < m/5$,

$$\sum_{y\in\Phi(x)} \bar{y} = \sum_{j=0}^{4}\Big(\big((q + j) \bmod 5\big)\frac{m}{5} + r\Big) = 2m + 5r = 2m + \overline{5x}.$$

Since $t\Phi(x) = \Phi(tx)$, also $\sum_{y\in\Phi(x)} \overline{ty} = 2m + \overline{5tx}$ for every unit t.

If $|K_5| = 5$, then $L = \Phi(x)$, and the sixth element b of A satisfies $5x + b = 0$; so $A = \Phi(x) + \{-5x\}$ is 5-standard.

If $|K_5| = 4$, then $L = \Phi(x) \setminus \{y_0\}$ for one y_0, and $A = L + \{b, c\}$. Here $e \neq 5$, since otherwise $K_5 = U$ would contain -1; so $5x = \kappa(5w) \neq 0$. Hence $y_0 \neq 0$ (as $5y_0 = 5x$), and $y_0 \neq 5x$ (otherwise $20x = 0$, so $5x = 0$, as m is odd). The sum of A gives $5x - y_0 + b + c = 0$, and for every unit t,

$$h(A) = 2m + \overline{5tx} - \overline{ty_0} + \overline{tb} + \overline{tc}.$$

So the multiset $Q = \{5x, b, c, -y_0\}$ of nonzero elements has sum 0, and $\sum_{q \in Q} \overline{tq} = \overline{5tx} + \overline{tb} + \overline{tc} + m - \overline{ty_0} = h(A) - m$ for every t: it is balanced. By (i) it contains a pair, formed by two of its four members. The pair $\{5x, -y_0\}$ is excluded by $y_0 \neq 5x$, and $\{b, c\}$ is excluded because A contains no pair. Each of the remaining four, $b = -5x$, $c = -5x$, $b = y_0$ or $c = y_0$, together with $5x - y_0 + b + c = 0$, gives $\{b, c\} = \{y_0, -5x\}$, so $A = \Phi(x) + \{-5x\}$ is 5-standard. This proves (ii). □

Remark 3.21. Kang states the Hodge conjecture for every Fermat fourfold [Kan16, Corollary 3.2], and [MMRV26] rely on that statement. The proof bounds a refined motivic dimension of $X_m^1 \times X_m^{n-1}$ by [Kan15, Proposition 2.2(iv)], whose proof writes the level filtration of a product as $F^m(H \otimes H') = \bigoplus_{t+s=m} F^t H \otimes F^s H'$, where F^p is the largest rational sub-Hodge structure contained in the p-th step of the Hodge filtration. This decomposition fails already for $H = H' = H^1(E, \mathbb{Q})$, E an elliptic curve, and $m = 1$: the Künneth component of the diagonal of $E \times E$ in $H^1(E) \otimes H^1(E)$ is a nonzero rational class of type $(1, 1)$, so it lies in $F^1(H \otimes H')$, while $F^1 H^1(E) = 0$ and the right-hand side vanishes. We do not know whether the inequality of [Kan15, Proposition 2.2(iv)], or [Kan16, Corollary 3.2], holds; Theorem B does not use them.

3.7. **Fermat fourfolds of odd degree divisible by three.** In this subsection m is odd and divisible by 3, and we prove Theorem B for such m. Here Theorem 3.9 fails: the set $\{1, 4, 9, 15, 16, 18\}$ of residues modulo 21 is balanced, contains no pair and is not 5-standard. So we replace the classification by an induction. A balanced sextuple either has a *move*, which reduces it to a smaller one by means of one cycle of Aoki, or lies in one of three explicit orbits (Theorem 3.27), whose classes we write down directly (Lemma 3.26). We keep the notation of Section 3.6.

Algebraic multisets. In this paragraph $M \geq 2$ is any integer; the even levels $M = 66$ and $M = 78$ occur in Lemma 3.26. We use Definition 3.8 at level M; when M is even, a *pair* is a submultiset $\{a, -a\}$, and $\{M/2, M/2\}$ is one. Let A be a balanced multiset of an even number $n \geq 2$ of nonzero elements of $\mathbb{Z}/M$. Ordering its elements gives a character $\alpha \in \mathfrak{A}_M^{n-2}$, and by Proposition 3.6(ii) the line $V(\alpha)$ consists of Hodge classes; for $n = 2$, X_M^0 is a set of M points. Permuting the coordinates of X_M^{n-2} permutes these lines, so the following notion does not depend on the ordering: A is *algebraic* if $V(\alpha) \subseteq \mathrm{Alg}^{(n-2)/2}(X_M^{n-2}) \otimes \mathbb{C}$. For an odd prime $\ell \mid M$ and $x \in \mathbb{Z}/M$ we put

$$\Phi_\ell(x) = \{x + jM/\ell : \ 0 \leq j < \ell\}, \qquad \sigma_{\ell,x} = \Phi_\ell(x) + \{-\ell x\},$$

so that $\Phi_5(x)$ is the fiber $\Phi(x)$ of Definition 3.8.

Lemma 3.22. *Let A and B be balanced multisets of an even number of nonzero elements of $\mathbb{Z}/M$.*

(a) *If $|A| \leq 4$, then A is algebraic.*
(b) *If A and B are algebraic, so is $A + B$. If P is a union of pairs and $A + P$ is algebraic, then A is algebraic.*
(c) *Let $t \in (\mathbb{Z}/M)^\times$. Then tA is balanced, and it is algebraic if and only if A is.*

(d) *Let $g \mid M$, $M' = M/g$, let A' be a multiset of nonzero elements of $\mathbb{Z}/M'$ with sum 0, and let $gA' \subseteq \mathbb{Z}/M$ be its image under $y \mapsto gy$. Then gA' is balanced if and only if A' is, and in that case gA' is algebraic if and only if A' is.*

(e) *Let M be odd, ℓ an odd prime dividing M, and $x \in \mathbb{Z}/M$ with $\ell x \neq 0$. Then $\sigma_{\ell,x}$ is a balanced multiset of $\ell+1$ nonzero elements, and it is algebraic.*

Proof. (a) For $|A| = 2$, $A = \{a, -a\}$, and $H^0(X_M^0)$ is spanned by the classes of points. For $|A| = 4$, $V(\alpha)$ consists of Hodge classes on the surface X_M^2, which are algebraic by Theorem 2.3.

(b) This is the inductive structure of Shioda and Ran, [Aok87, Theorem 1-4]: a union of pairs is a character of the kind denoted there by δ, whose line is spanned by the class of a linear subspace [Aok87, Theorem 1-1].

(c) The first assertion is clear. As in the proof of Proposition 3.6(ii), the automorphism $\zeta \mapsto \zeta^t$ of $\mathbb{Q}(\mu_M)$, acting on $H^{n-2}(X_M^{n-2}, \mathbb{Q}) \otimes \mathbb{Q}(\mu_M)$ through the second factor, carries the line $V(\alpha)$, which is defined over $\mathbb{Q}(\mu_M)$, onto $V(t\alpha)$, and it preserves $\mathrm{Alg}^{(n-2)/2} \otimes \mathbb{Q}(\mu_M)$.

(d) Every unit modulo M' lifts to a unit modulo M, and for a unit t modulo M with image t' we have $\overline{tgy} = g\,\overline{t'y}$ for $y \in \mathbb{Z}/M'$, the bars denoting representatives modulo M and M'; this gives the first assertion. Let $\pi\colon X_M^{n-2} \to X_{M'}^{n-2}$ be the morphism $[x_i] \mapsto [x_i^g]$ and α' an ordering of A'. In the proof of Theorem B for m prime to 6 we saw that π^* is injective and maps $V(\alpha')$ into $V(g\alpha')$; both are lines, so $\pi^* V(\alpha') = V(g\alpha')$. If A' is algebraic, then so is gA', because π^* preserves algebraic classes. Conversely $V(\alpha') = (\deg \pi)^{-1} \pi_* \pi^* V(\alpha') = \pi_* V(g\alpha')$, and π_* preserves algebraic classes.

(e) Since $\ell x \neq 0$, $0 \notin \Phi_\ell(x)$. The sum of $\sigma_{\ell,x}$ is $\ell x + \binom{\ell}{2} M/\ell - \ell x = 0$, as ℓ is odd. For a unit t we have $t\Phi_\ell(x) = \Phi_\ell(tx)$, and writing $\overline{tx} = qM/\ell + r$ with $0 \le r < M/\ell$,

$$\sum_{y \in \Phi_\ell(tx)} \bar{y} = \sum_{j=0}^{\ell-1} \Big(((q+j) \bmod \ell) \frac{M}{\ell} + r \Big) = \frac{(\ell-1)M}{2} + \overline{\ell t x};$$

adding $\overline{-\ell t x} = M - \overline{\ell t x}$ gives $(\ell+1)M/2$. So $\sigma_{\ell,x}$ is balanced. For $\ell = 3$ it is algebraic by (a). For $\ell \ge 5$ the argument in the proof of Theorem B for m prime to 6, case α 5-standard, applies with 5 replaced by ℓ: with $g = \gcd(\bar{x}, M/\ell)$ we have $\sigma_{\ell,x} = g\,\sigma_{\ell,x'}$, where $\sigma_{\ell,x'}$ lives at the odd level $M' = M/g$ and $\gcd(x', M'/\ell) = 1$; moreover $M'/\ell \ge 3$, since for $M' = \ell$ the set $\Phi_\ell(x')$ would contain 0. Aoki constructs a subvariety of $X_{M'}^{\ell-1}$ of codimension $(\ell-1)/2$ that represents $\sigma_{\ell,x'}$ [Aok87, Theorem 2-1], so $\sigma_{\ell,x'}$ is algebraic, and so is $\sigma_{\ell,x}$ by (d). □

Moves. From now on m is odd and divisible by 3. A *sextuple* is a balanced multiset of six nonzero elements of $\mathbb{Z}/m$. The *profile* $o(A)$ of a sextuple A is the sequence of the additive orders of its elements in nonincreasing order, and profiles are compared lexicographically.

Definition 3.23. Let A be a sextuple, ℓ an odd prime dividing m, and $x \in \mathbb{Z}/m$ with $\ell x \neq 0$. Let $S \subseteq A$ consist of one copy of each element of $\Phi_\ell(x)$ that lies in A, let $k = |S|$, and put

$$A'' = (A - S) + (-(\Phi_\ell(x) \setminus S)) + \{\ell x\},$$

a multiset of $7 + \ell - 2k$ elements. The pair (ℓ, x) is a *direct move* for A if $|A''| \le 4$, and a *lowering move* if $|A''| = 6$ and $o(A'') < o(A)$.

Lemma 3.24. *In Definition 3.23, A'' is a balanced multiset of nonzero elements, and $A'' + \sigma_{\ell,x} = A + P$ for a union of pairs P. If A'' is algebraic, then so is A. In particular A is algebraic if it has a direct move.*

Proof. The elements of A'' are nonzero, because $0 \notin \Phi_\ell(x)$ and $\ell x \neq 0$. The identity holds with $P = (\Phi_\ell(x) \setminus S) + (-(\Phi_\ell(x) \setminus S)) + \{\ell x, -\ell x\}$. The sum and the function $t \mapsto \sum_a \overline{ta}$ are

additive in the multiset, and A, P and $\sigma_{\ell,x}$ are balanced (Lemma 3.22(e)); so A'' is balanced. If A'' is algebraic, then $A'' + \sigma_{\ell,x} = A + P$ is algebraic by Lemma 3.22(e),(b), and so is A, by Lemma 3.22(b). After a direct move, A'' has at most four elements, so it is algebraic by Lemma 3.22(a). □

Lemma 3.25. *Let A be a sextuple, $x \in \mathbb{Z}/m$ an element of order e, and $\ell \neq e$ an odd prime dividing e. If A contains at least $(\ell+1)/2$ distinct elements of $\Phi_\ell(x)$ of order e, then (ℓ, x) is a direct or a lowering move for A.*

Proof. Here $\ell x \neq 0$, as $e \nmid \ell$. Since $\ell \mid e$, the element $m/\ell = (m/e)(e/\ell)$ lies in the subgroup of order e, and hence so does $\Phi_\ell(x)$; and ℓx has order e/ℓ. If $k \geq (\ell+3)/2$, then $|A''| \leq 4$. Otherwise $k = (\ell+1)/2$, $|A''| = 6$, and S consists of elements of order e. Passing from A to A'' removes these k elements and adds $\ell + 1 - k = k$ elements of order at most e, one of which, ℓx, has order e/ℓ. So A and A'' have the same elements of order larger than e, and A'' has fewer elements of order e; hence $o(A'') < o(A)$. □

Three exceptional orbits. Let

$$O_{21} = \{1, 4, 9, 15, 16, 18\}, \qquad O_{33} = \{1, 4, 16, 22, 25, 31\}, \qquad O_{39} = \{1, 7, 16, 22, 34, 37\},$$

subsets of $\mathbb{Z}/21$, $\mathbb{Z}/33$ and $\mathbb{Z}/39$. In terms of residues: O_{21} consists of the three elements congruent to 1 modulo 3 whose residue modulo 7 is a nonzero square, and the three elements $3v$ with v a nonsquare modulo 7; O_{33} consists of the five elements congruent to 1 modulo 3 whose residue modulo 11 is a nonzero square, and 22; and O_{39} consists of the six elements congruent to 1 modulo 3 whose residue modulo 13 lies in $\{1, 3, 9\} \cup \{7, 8, 11\}$, the union of two cosets of the subgroup generated by 3 in $(\mathbb{Z}/13)^\times$.

Lemma 3.26. *The sets O_{21}, O_{33} and O_{39} are balanced and algebraic.*

Proof. That they are balanced, that is, $\sum_a \overline{ta} = 3m$ for every unit t, is a direct check of 6, 10 and 12 sums, as t and $-t$ give $3m$ together. For an even level $M = 2M'$ and $x \in \mathbb{Z}/M$ with $2x \neq 0$, the multiset $T(x) = \{x, x + M', -2x, M'\}$ of nonzero elements has sum 0, and it is balanced: for a unit t, which is odd, put $y = \overline{tx} \neq M'$; then $tM' = M'$, and the representatives of the entries of $tT(x)$ are $y, y + M', M - 2y, M'$ if $y < M'$, and $y, y - M', 2M - 2y, M'$ if $y > M'$, adding up to $2M$ in both cases. These are Aoki's 2-standard elements [Aok87, §1].

O_{21}. With $\sigma_{3,2} = \{2, 9, 16\} + \{15\}$ and $Q = \{1, 4, 18, 19\}$ we have

$$O_{21} + \{2, 19\} = \sigma_{3,2} + Q.$$

As O_{21}, the pair and $\sigma_{3,2}$ are balanced, so is Q. By Lemma 3.22(a) $\sigma_{3,2}$ and Q are algebraic, so $O_{21} + \{2, 19\}$ is algebraic by Lemma 3.22(b), and so is O_{21}.

O_{33}. By Lemma 3.22(d) it suffices to show that $a = 2O_{33} = \{2, 8, 32, 44, 50, 62\} \subseteq \mathbb{Z}/66$ is algebraic; it is balanced because O_{33} is. In $\mathbb{Z}/66$,

$$a + \{1, 65\} + \{25, 41\} + \{33, 33\} = Q + T(32) + T(8),$$

with $Q = \{1, 25, 44, 62\}$, $T(32) = \{2, 32, 33, 65\}$ and $T(8) = \{8, 33, 41, 50\}$. As before Q is balanced, all three quadruples on the right are algebraic, so is their sum, and a is algebraic by Lemma 3.22(b).

O_{39}. In the same way, $a = 2O_{39} = \{2, 14, 32, 44, 68, 74\} \subseteq \mathbb{Z}/78$ satisfies

$$a + \{5, 73\} + \{7, 71\} + \{39, 39\} = Q + T(32) + T(5),$$

with $Q = \{2, 7, 73, 74\}$, $T(32) = \{14, 32, 39, 71\}$ and $T(5) = \{5, 39, 44, 68\}$, and we conclude as for O_{33}. □

None of the three sets has a direct or a lowering move; this is a finite check, which the programs of Appendix A also carry out. For $m = 21$ Lemma 3.26 is contained in [dSJ21], and for $m = 33$ and $m = 39$ in [Jum26b]: the identity used for O_{33} is [Jum26b, Theorem 1.4], while [Jum26b] treats O_{39} through the correspondence of Shioda and Katsura [Jum26b, Theorem 1.2]. We have not found the identity at level 78 in the literature.

The reduction.

Theorem 3.27. *Let m be odd and divisible by 3, and let A be a sextuple that contains no pair and generates $\mathbb{Z}/m$. Then A has a direct or a lowering move, or $m \in \{21, 33, 39\}$ and $tA = O_m$ for some $t \in (\mathbb{Z}/m)^\times$.*

Proof of Theorem B for odd m divisible by 3, given Theorem 3.27. As in the proof for m prime to 6, it suffices to show that every sextuple A is algebraic. We argue by induction on m and, for fixed m, on $o(A)$; at the odd levels prime to 6 every sextuple is algebraic by the proof in Section 3.6. If A contains a pair $\{a, -a\}$, then $A - \{a, -a\}$ is balanced, so A is algebraic by Lemma 3.22(a),(b). If A lies in the subgroup $g\mathbb{Z}/m$ for some $g > 1$ dividing m, then $A = gA'$ for a sextuple A' at the odd level $m/g < m$, which is algebraic, and so is A, by Lemma 3.22(d). Otherwise Theorem 3.27 applies. If A has a direct move, it is algebraic by Lemma 3.24. If it has a lowering move, then A'' is a sextuple with $o(A'') < o(A)$, hence algebraic, and A is algebraic by Lemma 3.24. In the remaining case tA is algebraic by Lemma 3.26, and so is A, by Lemma 3.22(c). □

The rest of this subsection proves Theorem 3.27. We fix A as there and assume that A has no direct and no lowering move.

Step 1: maximal orders and the character relation. An order e of an element of A is *maximal* if no element of A has an order that is a proper multiple of e. As A generates $\mathbb{Z}/m$, m is the least common multiple of the maximal orders. For a maximal order e we use the objects of Step 1 of Section 3.6: the embedding $\kappa\colon \mathbb{Z}/e \to \mathbb{Z}/m$, $U = (\mathbb{Z}/e)^\times$, and the odd function F on U; we let N_e be the number of elements of A of order e. As A contains no pair, $|F(z)| = \mathrm{mult}_A(\kappa z) + \mathrm{mult}_A(-\kappa z)$, so $\|F\| = 2N_e \le 12$. Lemma 3.10 holds with the same proof. For a prime $p \mid e$ the subgroup $K_p \subseteq U$ and the product H are defined as in Step 3 there; Lemma 3.12(b),(c) hold with the same proofs, and by Lemma 3.12(a), $|K_p| \ge 4$ for $p \ge 5$, with equality only for $p = 5 \parallel e$, while $|K_3| = 3$ if $9 \mid e$ and $|K_3| = 2$ if $3 \parallel e$.

Lemma 3.28. *Let B be a balanced multiset of nonzero elements of $\mathbb{Z}/m$. For $a \in B$ let f_a be its order, and write $a = (m/f_a)z_a$ with $z_a \in (\mathbb{Z}/f_a)^\times$. Let χ be an odd primitive Dirichlet character of conductor $c \mid m$. Then*

$$\sum_{a \in B,\ c \mid f_a} \frac{\bar\chi(z_a)}{\varphi(f_a)} \prod_{\ell \mid f_a,\ \ell \nmid c} (1 - \chi(\ell)) = 0, \tag{1}$$

where ℓ runs over primes and $\bar\chi(z_a)$ is the value at the residue of z_a modulo c.

Proof. First, for a multiple f of c,

$$\sum_{\substack{1 \le s \le f \\ (s,f)=1}} \chi(s)\, s = f\, B_{1,\chi} \prod_{\ell \mid f,\ \ell \nmid c} (1 - \chi(\ell)). \tag{2}$$

Indeed, for N divisible by c, $\sum_{s=1}^{N} \chi(s)s = \sum_{r=1}^{c} \chi(r) \sum_{j=0}^{N/c-1} (r + jc) = NB_{1,\chi}$, since $\sum_r \chi(r) = 0$. Let f_0 be the product of the primes dividing f but not c. As $\chi(s) = 0$ unless $(s, c) = 1$, and

then $(s,f)=1$ if and only if $(s,f_0)=1$, Möbius inversion gives

$$\sum_{(s,f)=1}\chi(s)s=\sum_{d|f_0}\mu(d)\chi(d)\,d\sum_{s'=1}^{f/d}\chi(s')s'=fB_{1,\chi}\sum_{d|f_0}\mu(d)\chi(d),$$

which is (2). Now regard χ as a character of $(\mathbb{Z}/m)^\times$ through reduction modulo c. Since B is balanced, $\sum_{a\in B}\sum_t\chi(t)\,\overline{ta}=h(B)\sum_t\chi(t)=0$. If $c\nmid f_a$, the inner sum vanishes, as in the proof of Lemma 3.11: χ is not trivial on the units congruent to 1 modulo f_a, since otherwise it would be induced from a character modulo the proper divisor $\gcd(f_a,c)$ of c. If $c\mid f_a$, then $\overline{ta}=(m/f_a)\,\overline{tz_a}$, with $\overline{tz_a}\in\{0,\dots,f_a-1\}$, and $\chi(t)$ depends only on t modulo f_a; so, substituting $s\mapsto sz_a^{-1}$ and using (2),

$$\sum_t\chi(t)\,\overline{ta}=\frac{m\,\varphi(m)}{f_a\varphi(f_a)}\,\bar\chi(z_a)\sum_{s\in(\mathbb{Z}/f_a)^\times}\chi(s)\,\bar s=\frac{m\,\varphi(m)B_{1,\chi}}{\varphi(f_a)}\,\bar\chi(z_a)\prod_{\ell|f_a,\ \ell\nmid c}(1-\chi(\ell)).$$

Adding over a and dividing by $m\varphi(m)B_{1,\chi}\neq0$ gives (1). □

For a maximal order e and $c=e$, the only elements in (1) are those of order e, and (1) becomes the relation $\sum_z \mathrm{mult}_A(\kappa z)\chi(z)^{-1}=0$ of the proof of Lemma 3.11. So Lemma 3.11 holds for every maximal order e, and with it Lemma 3.14: F is split along $(K_p)_{p|e}$ on every coset of H.

Lemma 3.29. *Let e be a maximal order, p a prime dividing e and $w\in U$. Then $\kappa(wK_p)\subseteq\Phi_p(\kappa w)$, with equality if $p^2\mid e$. If $e\neq p$, $|K_p|\geq(p+1)/2$ and $F\geq1$ on wK_p, then A has a direct or a lowering move.*

Proof. An element $k\in K_p$ is congruent to 1 modulo e/p, so $\kappa(wk)-\kappa(w)=(m/e)\,w(k-1)$ lies in $(m/e)(e/p)\mathbb{Z}=(m/p)\mathbb{Z}$. If $p^2\mid e$, both sides have p elements. If $F\geq1$ on wK_p, then the $|K_p|$ elements $\kappa(wk)$ of order e lie in A, and Lemma 3.25 applies. □

Step 2: every maximal order is divisible by 3. Let e be a maximal order prime to 3; then e is prime to 6. The function F is odd, $\|F\|\leq12$, $F(z_0)\geq1$ when $\kappa(z_0)\in A$ has order e, and F is split, so Proposition 3.18 applies. In its case (2), $\|F\|=12$, so all six elements of A have order e, and Lemma 3.10 gives $\sum_z F(z)z=0$, against Lemma 3.20. In case (1), $F\geq1$ on a line wK_p with $p\in\{5,7\}$. As F is odd, $-1\notin K_p$, so $e\neq p$, and $|K_p|\geq4\geq(p+1)/2$; by Lemma 3.29 A has a move, which we excluded.

Step 3: no maximal order is divisible by 9. Let e be a maximal order with $9\mid e$. Each K_p consists of units congruent to 1 modulo e/p, and $3\mid e/p$; so H consists of units congruent to 1 modulo 3, and $-1\notin H$. Let $F(x)\geq1$ and $C=xH$. The cosets C and $-C$ are distinct and carry the same mass, so $\sum_C|F|\leq6$. Write $K_3=\{1,k,k^2\}$ and $H'=\prod_{p\neq3}K_p$, and let $S_c=xcH'$, for $c\in K_3$, be the slices of C, with $\mu(c)=\sum_{S_c}|F|$, as in Step 4 of Section 3.6.

(i) *No line zK_3 contains two points $z_1\neq z_2$ with $F(z_1)F(z_2)>0$.* Otherwise, replacing z_1,z_2 by $-z_1,-z_2$ if necessary, $F(z_1),F(z_2)\geq1$, and $\kappa(z_1),\kappa(z_2)$ are two elements of A of order e in $\Phi_3(\kappa z_1)$ (Lemma 3.29); as $e\neq3$, Lemma 3.25 gives a move.

(ii) *$\mu(c)\geq1$ for every c.* Otherwise, by (S2), F is split along $(K_p)_{p\neq3}$ on $S_1\ni x$, where its mass is at most 6. These subgroups have orders at least 4, with equality for at most one of them, so by Lemma 3.15 $F=1$ on a line xK_p with $p\in\{5,7\}$ and $|K_p|\leq6$, and Lemma 3.29 gives a move.

(iii) For $c\neq c'$ in K_3 the function $G(z)=F(z)-F(zc^{-1}c')$ on S_c is split along $(K_p)_{p\neq3}$ by (S1), and $\sum_{S_c}|G|\leq\mu(c)+\mu(c')\leq5$ by (ii). It is not zero: by (ii) some $z\in S_c$ has $F(z)\neq0$, and $F(zc^{-1}c')=F(z)$ would contradict (i), as z and $zc^{-1}c'$ lie on the line zK_3. By

Lemma 3.15, $|G| = 1$ on a line along some K_p with $|K_p| \le 5$ and $G = 0$ elsewhere, so $p = 5$ and $\mu(c) + \mu(c') \ge |K_5| \ge 4$. Adding over the three pairs $\{c, c'\}$ gives $2\sum_c \mu(c) \ge 12$. As $\sum_c \mu(c) \le 6$, we get $|K_5| = 4$ and $\mu(c) = 2$ for every c.

Now take $c = 1$ and $c' = k$: $G = \sigma$ on a line $L \subseteq S_1$ along K_5 and $G = 0$ elsewhere, for a sign σ. By (i), $F(z)F(zk) \le 0$ for all z. Off L, $F(z) = F(zk)$, so both vanish; on L, $F(z) - F(zk) = \sigma$, so $(F(z), F(zk))$ is $(\sigma, 0)$ or $(0, -\sigma)$. Since $\mu(1) = 2$, $F = \sigma$ at two points of L, and $F = 0$ at the other two points z of L, where $F(zk) = -\sigma$. The same argument with $c' = k^2$ gives a line $L' \subseteq S_1$ and a sign σ' such that F takes values in $\{0, \sigma'\}$ on S_1, with support in L'; so $L' = L$ and $\sigma' = \sigma$, and $F(zk^2) = -\sigma$ at the two points $z \in L$ with $F(z) = 0$. For such a z we get $F(zk) = F(zk^2) = -\sigma$, against (i).

Step 4: the case $3 \parallel e$. By Steps 2 and 3, every maximal order e satisfies $3 \parallel e$; write $e = 3e^*$. Then $e^* > 1$: for $e = 3$, $U = K_3 = \{\pm 1\}$, and F would be K_3-invariant by (S4), so $F(1) = F(-1) = -F(1)$, while $F \ne 0$. Fix a maximal order e. By the Chinese remainder theorem $U = (\mathbb{Z}/3)^\times \times U^*$ with $U^* = (\mathbb{Z}/e^*)^\times$, and we write $z = (s, u)$ and $\kappa(s, u)$. Then $K_3 = (\mathbb{Z}/3)^\times \times \{1\}$, and for $p \mid e^*$, $K_p = \{1\} \times K_p^*$, where $K_p^* \subseteq U^*$ consists of the units congruent to 1 modulo e^*/p; so $H = K_3 \times H^*$ with $H^* = \prod_{p|e^*} K_p^*$.

For $u \in U^*$ the points $(1, u)$ and $(-1, u)$ form a line along K_3. If $F(1, u)F(-1, u) > 0$, then A has a move, as in (i) of Step 3. So $F(1, u)F(-1, u) \le 0$ for all u, and we define $D\colon U^* \to \mathbb{Z}$ by $D(u) = F(1, u) - F(-1, u)$. The elements of A of order e are its *tops* at e.

Lemma 3.30. (a) *D is even, $\|D\| = 2N_e$, and D is split along $(K_p^*)_{p|e^*}$ on every coset of H^*.*

(b) *For $u \in U^*$, the tops of the form $\kappa(s, \pm u)$ are $|D(u)|$ in number, and each of them has $s = \operatorname{sign} D(u)$.*

Proof. (a) Since F is odd, $D(-u) = F(1, -u) - F(-1, -u) = -F(-1, u) + F(1, u) = D(u)$. As $F(1, u)F(-1, u) \le 0$, $|D(u)| = |F(1, u)| + |F(-1, u)|$, and these add up to $\|F\| = 2N_e$. Applying (S1) with the element $(-1, 1)$ of K_3 to the cosets $(1, u)(\{1\} \times H^*)$ shows that D is split. (b) If $D(u) > 0$, then $F(1, u) \ge 0 \ge F(-1, u)$; as A contains no pair, it contains $\kappa(1, u)$ with multiplicity $F(1, u)$, $\kappa(1, -u) = -\kappa(-1, u)$ with multiplicity $-F(-1, u)$, and neither $\kappa(-1, u)$ nor $\kappa(-1, -u)$. The case $D(u) < 0$ is symmetric, and if $D(u) = 0$ there are no such tops. □

Step 5: even split functions.

Lemma 3.31. *Let $e^* > 1$ be prime to 6, and let $D\colon U^* \to \mathbb{Z}$ be even, not zero, with $\|D\| \le 12$, and split along $(K_p^*)_{p|e^*}$ on every coset of H^*. Then one of the following holds.*

(E1) *$D = \sigma(\mathbf{1}_L + \mathbf{1}_{-L})$ for a sign σ and a line L along K_5^* or K_7^* with $L \cap (-L) = \emptyset$.*

(E2) *e^* is a prime $q \le 13$ and D is constant.*

(E3) *$e^* = 35$, and $D(u) = g(u \bmod 5) + h(u \bmod 7)$ with even functions g on $(\mathbb{Z}/5)^\times$ and h on $(\mathbb{Z}/7)^\times$.*

Proof. By Lemma 3.12(a), $|K_p^*| \ge 4$, with equality only for $p = 5 \parallel e^*$, and $|K_p^*| \le 6$ only for $p \in \{5, 7\}$.

e^* is not squarefree, say $p^2 \mid e^*$. Every K_q^* consists of units congruent to 1 modulo e^*/q, hence modulo p; so $-1 \notin H^*$. For a coset C of H^*, the cosets C and $-C$ are distinct and carry the same mass, so $\sum_C |D| \le 6$. By Lemma 3.15, on each coset C where $D \ne 0$, $D = \sigma_C$ on a line L_C along K_5^* or K_7^* and $D = 0$ elsewhere on C; as D is even, D is then σ_C on $-L_C \subseteq -C$. Each such pair of cosets carries mass at least 8, so there is only one, and (E1) holds.

$e^* = q$ is prime. Then $U^* = K_q^*$, so D is constant by (S4), and $\|D\| = |D(1)|(q - 1) \le 12$ gives $q \le 13$: (E2).

e^* is squarefree with at least two prime factors, one of them $p \geq 11$. Now $H^* = U^*$, by the Chinese remainder theorem. Put $H'^* = \prod_{q \neq p} K_q^*$, and consider the slices $S_c = cH'^*$, $c \in K_p^*$, with masses $\mu(c)$. Let $\varepsilon_p \in K_p^*$ be congruent to -1 modulo p; then $-1 = \varepsilon_p \varepsilon'$ with $1 \neq \varepsilon' \in H'^*$, so $-1 \notin K_p^*$, and $-S_c = S_{c\varepsilon_p}$, so $\mu(c\varepsilon_p) = \mu(c)$, with $c\varepsilon_p \neq c$. Suppose that every slice carries mass. As $|K_p^*| \geq 10$ and $\sum \mu \leq 12$, at most one slice has $\mu \geq 3$, hence none, by the pairing; and some c_0 has $\mu(c_0) = 1$. For every c, the function $D(z) - D(zc_0^{-1}c)$ on S_{c_0} is split along $(K_q^*)_{q \neq p}$ by (S1) and has mass at most 3, so it vanishes by Lemma 3.15. Hence D is constant on $z_0 K_p^*$, where z_0 is the point of S_{c_0} with $D(z_0) \neq 0$, and vanishes elsewhere. As $D(-z_0) = D(z_0) \neq 0$, this gives $-1 \in K_p^*$, a contradiction. So some slice has $\mu = 0$, and by (S2) D is split along $(K_q^*)_{q \neq p}$ on every slice. As $\mu(c) = \mu(c\varepsilon_p)$, every slice has mass at most 6, and we conclude by Lemma 3.15 as in the first case, with slices in place of cosets: (E1).

Otherwise e^* is squarefree and its prime factors are 5 and 7, so $e^* = 35$ and $U^* = K_5^* K_7^*$. By (S3), $D(ij) = D(i) + D(j) - D(1)$ for $i \in K_5^*$, $j \in K_7^*$; this gives $D(u) = g(u \bmod 5) + h(u \bmod 7)$, with $g(i) = D(i) - D(1)$ and $h(j) = D(j)$, as i and j are determined by u mod 5 and u mod 7. Since D is even, $g(-i) - g(i) = h(j) - h(-j)$ for all i, j; both sides are a constant γ, and replacing i by $-i$ shows $\gamma = -\gamma$. So g and h are even: (E3). □

Step 6: two lemmas.

Lemma 3.32. *Let $M' > 1$ be prime to 6. There are primitive Dirichlet characters modulo M' of both parities. Among those of a given parity, the values at 3 are not all equal, unless $M' = 5$ and the parity is even.*

Proof. A character modulo M' is a product of characters ψ_p modulo the prime powers $p^a \parallel M'$, and it is primitive if and only if every ψ_p is. The characters modulo p^a are the powers θ^j of a generator θ, which is faithful; so $\theta(-1) = -1$, and $\theta(9) \neq 1$ because $9 \not\equiv 1 \pmod p$. The character θ^j has parity $(-1)^j$, and it is primitive if and only if $(p-1) \nmid j$ when $a = 1$, and $p \nmid j$ when $a \geq 2$. Both conditions hold for $j = 1, 2$, so both parities occur. Replacing $\psi_p = \theta^j$ by $\theta^{j \pm 2}$ keeps the parity and multiplies the value at 3 by $\theta(9)^{\pm 1} \neq 1$. One of θ^{j+2} and θ^{j-2} is primitive, since p cannot divide both $j+2$ and $j-2$, and $p-1$ divides both only if $p = 5$ and $j \equiv 2 \pmod 4$. So the values at 3 in a parity class are constant only if every ψ_p is the quadratic character modulo $p^a = 5$, that is, only if $M' = 5$ and the parity is even. □

Lemma 3.33. *Let $e = 3e^*$ be a maximal order and $b \in A$ an element whose order $f \neq e$ is a multiple of e^*. Then $f = e^*$.*

Proof. The order f divides some maximal order e', and $3 \mid e'$ by Step 2; so $e = 3e^*$ divides e', and $e' = e$ by the maximality of e. Hence $e^* \mid f \mid e$ and $f \neq e$, so $f = e^*$. □

Step 7: the cases (E1) and (E3) do not occur. Let $e = 3e^*$ be a maximal order. By Lemma 3.30(a), Lemma 3.31 applies to D.

Case (E1). Let $L = wK_p^*$ with $p \in \{5, 7\}$. By Lemma 3.30(b), $N_e = |L|$, and the tops are the elements $\kappa(\sigma, \varepsilon_u u)$, $u \in L$, one for each u, with signs ε_u. For a sign ε, the tops with $\varepsilon_u = \varepsilon$ lie in $\kappa\big((\sigma, \varepsilon w)K_p\big) \subseteq \Phi_p(\kappa(\sigma, \varepsilon w))$ by Lemma 3.29, and $e \neq p$; so by Lemma 3.25 fewer than $(p+1)/2$ of them have $\varepsilon_u = \varepsilon$. If $|L| = 5$, this is impossible. Since $2|L| \leq 12$, the remaining cases are $|L| = 4$, $p = 5$, with two tops of each sign, and $|L| = 6$, $p = 7$, with three of each sign.

Let $p = 5$ and $|L| = 4$, so $5 \parallel e^*$. Here $e^* \neq 5$, since $K_5^* = U^*$ would contain -1; so $e^* = 5e'$ with $e' > 1$ prime to 30, and reduction modulo 5 identifies K_5^* with $(\mathbb{Z}/5)^\times$. Put $\varepsilon'(k) = \varepsilon_{wk}$, a function on $(\mathbb{Z}/5)^\times$ with two values 1 and two values -1. Let $\chi = \chi_5 \chi'$ be an odd primitive character modulo e^*, with χ_5 modulo 5 and χ' modulo e'. By Lemma 3.33, the elements $a \in A$

with $e^* \mid f_a$ are the tops and the elements of order e^*, and (1) with $c = e^*$, multiplied by $\varphi(e) = 2\varphi(e^*)$, reads

$$(1 - \chi(3)) \sum_{u \in L} \bar\chi(\varepsilon_u u) + 2 \sum_{f_b = e^*} \bar\chi(z_b) = 0.$$

As χ is odd and $\chi(k) = \chi_5(k)$ for $k \in K_5^*$, the first sum is $\bar\chi(w)E(\chi_5)$ with $E(\chi_5) = \sum_k \varepsilon'(k)\bar\chi_5(k)$. Since $\sum_k \varepsilon'(k) = 0$, E vanishes at the trivial character, and, with λ_5 the quadratic character, $E(\lambda_5) = 2(\varepsilon'(1) + \varepsilon'(4))$. So either $\varepsilon' = \pm\lambda_5$, and then E vanishes at the two characters of order 4; or $E(\lambda_5) = 0$, and then E does not vanish at some character of order 4, as $\varepsilon' \neq 0$. If no element of A has order e^*, choose χ_5 with $E(\chi_5) \neq 0$, and by Lemma 3.32 (note $e' \neq 5$) a primitive χ' with $\chi'(-1) = -\chi_5(-1)$ and $\chi'(3) \neq \chi_5(3)^{-1}$; the relation then fails. If exactly one element b has order e^*, choose $\chi_5 \neq 1$ with $E(\chi_5) = 0$ and a primitive χ' with $\chi'(-1) = -\chi_5(-1)$; the relation becomes $2\bar\chi(z_b) = 0$, which is absurd. If two elements have order e^*, then all six elements of A lie in $\kappa(\mathbb{Z}/e)$; writing them as $\kappa(y)$, we get $\sum y = 0$ in $\mathbb{Z}/e$, while the residues of the y modulo 3 are σ for the four tops and 0 for the other two, so $4\sigma \equiv 0 \pmod 3$, which is false.

Let $p = 7$ and $|L| = 6$. Then $N_e = 6$, so A consists of tops, $m = e$, and $\sum_{u \in L} \varepsilon_u u = 0$ in $\mathbb{Z}/e^*$, because $\sum_{a \in A} a = 0$. Here $7 \parallel e^*$ and, as before, $e^* = 7e'$ with $e' > 1$ prime to 42. Put $\varepsilon'(k) = \varepsilon_{wk}$ for $k \in K_7^*$, identified with $(\mathbb{Z}/7)^\times$, and let P be the set where $\varepsilon' = 1$. Reducing modulo 7 gives $w \sum_k \varepsilon'(k)k \equiv 0$, and $\sum_k \varepsilon'(k)k = 2\sum_P k - 21$, so $\sum_P k \equiv 0 \pmod 7$. The only three-element subsets of $\{1, \dots, 6\}$ with this property are $\{1, 2, 4\}$ and $\{3, 5, 6\}$; so $\varepsilon' = \pm\lambda_7$, the Legendre symbol modulo 7, and $\varepsilon_u = \tau\lambda_7(u)$ for a sign τ. If $e' \neq 5$, let ψ be an even primitive character modulo e' with $\psi(3) \neq -1$ (Lemma 3.32) and $\chi = \lambda_7\psi$, which is odd and primitive modulo e^*. Then $\chi(3) = -\psi(3) \neq 1$, and $\bar\chi(\varepsilon_u u) = \varepsilon_u\lambda_7(u)\bar\psi(u) = \tau\bar\psi(w)$ for $u \in L$, so the left side of (1) with $c = e^*$ is a nonzero multiple of $6\tau\bar\psi(w) \neq 0$. If $e' = 5$, then $m = 105$; take $\chi = \lambda_7$, of conductor 7. As 3 and 5 are nonsquares modulo 7, (1), multiplied by $\varphi(105)$, reads $(1 - \lambda_7(3))(1 - \lambda_7(5)) \sum_u \lambda_7(\varepsilon_u u) = 4 \cdot 6\tau = 0$, which is false.

Case (E3), not (E1). Here $e^* = 35$. The function D is constant on the six classes $\{(\pm i, \pm j)\}$ of four elements each, $i \in (\mathbb{Z}/5)^\times$, $j \in (\mathbb{Z}/7)^\times$; write its values as a matrix $(M_{\alpha\beta})$ with $\alpha \in \{1, 2\}$ and $\beta \in \{1, 2, 3\}$, so that $M_{\alpha\beta} = g_\alpha + h_\beta$ and $4\sum |M_{\alpha\beta}| = \|D\| \leq 12$. The difference $\delta = M_{1\beta} - M_{2\beta}$ does not depend on β. If $\delta = 0$, the two rows agree, and $\sum |M_{\alpha\beta}| \leq 3$ leaves one column with both entries σ and zeros elsewhere: this is (E1), with a line along K_5^*. If $|\delta| \geq 2$, every column contributes at least 2, which is too much. If $|\delta| = 1$, every column is $(\delta, 0)$ or $(0, -\delta)$. If all three columns agree we are in (E1), with a line along K_7^*. Otherwise $\|D\| = 12$, so A consists of tops, and D takes one sign on eight elements of U^* and the other on four. By Lemma 3.30(b) four tops have residue ± 1 modulo 3 and two have the opposite residue, so the residues of the z_a modulo 3 add up to ± 2. But $\sum_a z_a = 0$ in $\mathbb{Z}/e$, because $\sum_a a = 0$ and κ is injective.

Step 8: the case (E2), and the end of the proof. By Step 7, every maximal order e is in case (E2): $e = 3q$ with a prime $q \in \{5, 7, 11, 13\}$, and $D = c_0$ is constant. By Lemma 3.30(b) all tops at e have the same residue $\sigma = \operatorname{sign} c_0$ modulo 3, there are $|c_0|$ of them over each pair $\pm u$ of residues modulo q, and $N_e = |c_0|(q-1)/2$. For an odd character ψ modulo q, $\psi(3) = 1$ only if ψ is trivial on the subgroup generated by 3.

$q = 13$. Then $|c_0| = 1$ and $N_e = 6$, so A consists of tops and $m = 39$. Let $P \subseteq (\mathbb{Z}/13)^\times$ be the set of residues modulo 13 of the elements of A, one from each pair $\pm u$, and $\pi = \mathbf{1}_P - \mathbf{1}_{-P}$. For odd ψ modulo 13, (1) with $c = 13$ gives $(1 - \psi(3)) \sum_P \bar\psi = 0$, and $\sum_v \pi(v)\bar\psi(v) = 2\sum_P \bar\psi$; for even ψ that sum vanishes because π is odd. So the Fourier transform of π is supported on the characters trivial on $\langle 3 \rangle = \{1, 3, 9\}$, and π is invariant under $\langle 3 \rangle$. The cosets of $\langle 3 \rangle$ are

$C_j = 2^j\langle 3\rangle$, $j \in \mathbb{Z}/4$, with $-C_j = C_{j+2}$; hence $P = C_j \cup C_{j+1}$ for some j. A unit t with $t \equiv \sigma \pmod 3$ and $t \equiv 2^{3-j} \pmod{13}$ gives $tA = O_{39}$, since $C_3 \cup C_0 = \{7, 8, 11\} \cup \{1, 3, 9\}$.

$q = 11$. Then $|c_0| = 1$ and $N_e = 5$. The sixth element $b = -\sum$ (tops) lies in $\kappa(\mathbb{Z}/33)$, so $m = 33$; its residue modulo 3 is $-5\sigma = \sigma \neq 0$, so $3 \mid f_b$, and $f_b = 3$ as $f_b \neq 33$; so $b \equiv 0 \pmod{11}$. With P and π as before, (1) with $c = 11$ involves only the tops, and shows that the Fourier transform of π is supported on the odd characters trivial on $\langle 3\rangle$, the subgroup of squares. The only such character is the Legendre symbol λ_{11}, which is odd; so $\pi = \pm\lambda_{11}$, and P is the set of nonzero squares or of nonsquares. A unit $t \equiv \sigma \pmod 3$, $t \equiv \pm 1 \pmod{11}$ gives $tA = O_{33}$.

So we may assume that every maximal order is 15 or 21.

$q = 7$. If $|c_0| = 2$, then $N_e = 6$, A consists of tops, and $m = 21$. Let $n(v)$ be the number of tops with residue v modulo 7. Since 3 generates $(\mathbb{Z}/7)^\times$, $\psi(3) \neq 1$ for odd ψ, and (1) gives $\sum_v n(v)\bar\psi(v) = 0$; so the odd function $n(v) - n(-v)$ vanishes, and $n(v) = 1$ for every v. Then A contains all six elements $\kappa(\sigma, v)$, which lie in $\Phi_7(\kappa(\sigma, 1))$ (Lemma 3.29), and Lemma 3.25 gives a move. So $|c_0| = 1$ and $N_{21} = 3$.

$q = 5$. If $|c_0| = 3$, then A consists of tops and $m = 15$; as $\psi(3) = \pm i \neq 1$ for the odd characters ψ modulo 5, (1) gives $n(v) = n(-v)$ as before, which is impossible since $n(v) + n(-v) = 3$. So $N_{15} \in \{2, 4\}$.

Both 15 *and* 21 *are maximal.* Then $m = 105$. As $N_{21} = 3$ and $N_{15} + N_{21} \leq 6$, $N_{15} = 2$, and there is one further element b. Let ρ_3 and ρ_5 be the reductions of $\mathbb{Z}/105$ modulo 3 and 5. The three tops at 21 have equal ρ_3, which therefore add up to 0, and $\rho_5 = 0$. The two tops at 15 have ρ_3 equal to the same nonzero value, and their ρ_5 are $2v_1$ and $2v_2$ with $v_1 \in \{\pm 1\}$ and $v_2 \in \{\pm 2\}$ (one top over each pair), whose sum $2(v_1 + v_2)$ is not 0 modulo 5. From $\rho_3(\sum_a a) = 0$ we get $\rho_3(b) \neq 0$, so $3 \mid f_b$; as f_b divides 15 or 21 and is neither, $f_b = 3$ and $\rho_5(b) = 0$. Then $\rho_5(\sum_a a) \neq 0$, a contradiction.

Only 15 *is maximal*, so $m = 15$, and the elements of A other than the $N = N_{15}$ tops have order 3 or 5. For a of order 3 write $a = 5z_a$; as A contains no pair, its r elements of order 3 are equal. The character χ_3 modulo 3 in (1) gives $(1 - \chi_3(5))N\sigma/8 + r\chi_3(z)/2 = 0$, that is $N\sigma = -2r\chi_3(z)$. If $N = 4$, then $r = 2$: the other two elements have order 3, and (1) for the odd characters modulo 5 involves only the tops, so $n(v) = n(-v)$ as before and $n(v) = 1$ for each $v \in (\mathbb{Z}/5)^\times$. The four tops then lie in $\Phi_5(\kappa(\sigma, 1))$, and Lemma 3.25 gives a move. If $N = 2$, then $r = 1$, and the other three elements $a = 3z_a$ have order 5, with no two of the z_a opposite. Let ψ be the odd character modulo 5 with $\psi(2) = i$, and $z_1 \in \{\pm 1\}$, $z_2 \in \{\pm 2\}$ the residues modulo 5 of the two tops. Then (1), multiplied by 8, reads

$$(1 - \psi(3))\big(\bar\psi(z_1) + \bar\psi(z_2)\big) + 2\sum_{f_a = 5} \bar\psi(z_a) = 0.$$

The first term has absolute value $|1 + i| \cdot |\pm 1 \pm i| = 2$. In the sum, the n_1 elements with $z_a \in \{\pm 1\}$ contribute $\pm n_1$ and the n_2 with $z_a \in \{\pm 2\}$ contribute $\pm i n_2$, with $n_1 + n_2 = 3$; so its absolute value is $(n_1^2 + n_2^2)^{1/2} \geq \sqrt{5} > 1$, a contradiction.

Only 21 *is maximal*, so $m = 21$, and the elements of A other than the three tops have order 3 or 7. Let $W \subseteq (\mathbb{Z}/7)^\times$ be the set of residues modulo 7 of the tops, one from each pair $\pm u$. For the character χ_3 the product in (1) vanishes for the tops, as $\chi_3(7) = 1$, so the elements of order 3, which are equal, have $\sum \chi_3(z_a) = 0$: there are none. So the other three elements are $a = 3z_a$ of order 7, and the multiset Z of the z_a contains no two opposite residues. Let $E = \mathbf{1}_W - \mathbf{1}_{-W}$ and $B(v) = \mathrm{mult}_Z(v) - \mathrm{mult}_Z(-v)$; both are odd, $|E| = 1$, and $\sum_v |B(v)| = 6$. For odd ψ modulo 7, (1), multiplied by 24, reads $(1 - \psi(3))\hat E(\psi) + 2\hat B(\psi) = 0$, where $\hat E(\psi) = \sum_v E(v)\bar\psi(v)$; as $\psi(3)\hat E(\psi)$ is the transform of $v \mapsto E(3v)$, and all three functions are odd, $2B(v) = E(3v) - E(v)$

for every v. So $|B| \le 1$, hence $|B| = 1$ everywhere, and $E(3v) = -E(v)$. As 3 generates $(\mathbb{Z}/7)^\times$, $E = E(1)\lambda_7$ and $B = -E$: W is the set of nonzero squares or of nonsquares, and $Z = -W$. A unit $t \equiv \sigma \pmod 3$, $t \equiv \pm 1 \pmod 7$, with tW the set of squares, gives $tA = O_{21}$.

This proves Theorem 3.27, and with it Theorem B. □

4. Fermat varieties of degrees 114, 44, 52 and 51

This section proves Theorem A. Aoki reduced the Hodge conjecture for all Fermat varieties of a given degree m to finitely many classes, one for each divisor of m of a certain kind [Aok83, Aok00]. For $m = 114$, 44, 52 and 102 a single class remains (Section 4.1), and the degree 51 reduces to 102. In degrees 114, 52 and 102 the class lives on the Fermat eightfold and is the join of two characters of the Fermat threefold (Sections 4.2, 4.7 and 4.8); in degree 44 it lives on the Fermat sixfold and is the join of a character of the Fermat threefold and one of the Fermat curve (Section 4.6). In all four cases the main step is a proof that certain pieces of H^3 of the threefold are supported on divisors. For this we use families of curves on the Fermat threefold and a residue formula for the derivative of their Abel–Jacobi map. The method is Peterson's [Pet26, §3], who used one such family on the Fermat threefold of degree 35; we set it up for families of a general shape (Section 4.3) and then write down two families of degree 114 (Section 4.5), one of degree 44 (Section 4.6), two of degree 52 (Section 4.7) and two of degree 102 (Section 4.8).

4.1. **Aoki's groups.** In this subsection $M \ge 3$ is an integer, and we use the language of Section 3.7: balanced multisets of nonzero elements of $\mathbb{Z}/M$, pairs, and algebraic multisets. Let R_M be the free abelian group on the symbols (y), $y \in \mathbb{Z}/M \setminus \{0\}$. A multiset A of nonzero elements has the class $[A] = \sum_{a\in A}(a)$, so that $[A+B] = [A] + [B]$, and an element of R_M with nonnegative coefficients is the class of exactly one multiset. A unit t acts by $t \cdot (y) = (ty)$. For $M = gM'$ the homomorphism $g_*\colon R_{M'} \to R_M$, $(y) \mapsto (gy)$, maps $[A']$ to $[gA']$, in the notation of Lemma 3.22(d).

Following Aoki [Aok00, §2], let $B_M \subseteq R_M$ be the subgroup generated by the classes of the balanced multisets with an even number of elements, let D_M be the subgroup generated by the classes $(y) + (-y)$ of the pairs, and let S_M be the subgroup generated by D_M and the classes of the *standard* multisets

$$\sigma_{p,i} = \Phi_p(i) + \{-pi\} \quad (p \text{ odd}), \qquad \sigma_{2,i} = \{i,\ i + M/2,\ -2i,\ M/2\}.$$

Here p is a prime dividing M with $p < M$, $d = M/p$, $0 < i < d$, and $\Phi_p(i) = \{i + jd :\ 0 \le j < p\}$ as in Section 3.7; for odd p, $\sigma_{p,i}$ is the multiset $\sigma_{p,x}$ of that subsection with $x = i$.

Lemma 4.1. *Every standard multiset is a balanced multiset of an even number of nonzero elements, and it is algebraic. Hence $D_M \subseteq S_M \subseteq B_M$. These three groups are stable under the units, and for $M = gM'$ the map g_* carries each of them at level M' into the same group at level M.*

Proof. For $p = 2$ the elements of $\sigma_{2,i}$ are nonzero because $0 < i < M/2$, and $\sigma_{2,i}$ is the multiset $T(i)$ of the proof of Lemma 3.26, which is balanced; having four elements, it is algebraic by Lemma 3.22(a). Let p be odd. Then $pi \ne 0$, as $0 < i < d$, and the computation in the proof of Lemma 3.22(e), which does not use that M is odd, shows that $\sigma_{p,i}$ is a balanced multiset of $p+1$ nonzero elements. For $p = 3$ it is algebraic by Lemma 3.22(a). For $p \ge 5$ let $g = \gcd(i, d)$, $i' = i/g$, $d' = d/g$ and $M' = M/g$. Then $\sigma_{p,i} = g\,\sigma_{p,i'}$, where $\sigma_{p,i'}$ is the standard multiset at level M', $\gcd(i', d') = 1$ and $d' \ge 2$. If $d' \ge 3$, Aoki constructs a subvariety of $X_{M'}^{p-1}$ of codimension $(p-1)/2$ that represents $\sigma_{p,i'}$ [Aok87, Theorem 2-1]. If $d' = 2$, then $i' = 1$, $M' = 2p$, and $\sigma_{p,1} = \{1, 3, \dots, 2p-1\} + \{p\}$ is the union of the pairs $\{y, 2p-y\}$ with $y < p$ odd

and of the pair $\{p,p\}$, so it is algebraic by Lemma 3.22(a),(b). In both cases $\sigma_{p,i}$ is algebraic by Lemma 3.22(d).

For a unit t, multiplication by t maps $\Phi_p(i) = i + d\mathbb{Z}/M$ onto $\Phi_p(i'')$, where $0 < i'' < d$ and $i'' \equiv ti \pmod d$, and it maps $-pi$ to $-pi''$, since $p(ti - i'') \in pd\mathbb{Z} = M\mathbb{Z}$; it fixes $M/2$ when M is even, as t is then odd. So $t\sigma_{p,i} = \sigma_{p,i''}$. Similarly $g\sigma_{p,i}$, formed at level M', is $\sigma_{p,gi}$ at level M. As balanced multisets and pairs are preserved by units and by g_* (Lemma 3.22(c),(d)), the last assertions follow. □

Lemma 4.2. *Let $\mathcal{C}$ be a set of algebraic balanced multisets, each with an even number of elements, and let A be a balanced multiset of an even number of nonzero elements of $\mathbb{Z}/M$ with $[A] \in D_M + \sum_{C\in\mathcal{C}} \mathbb{Z}[C]$. Then A is algebraic. If $B_M = S_M + \sum_{C\in\mathcal{C}} \mathbb{Z}[C]$, then every balanced multiset of an even number of nonzero elements of $\mathbb{Z}/M$ is algebraic, and the Hodge conjecture holds for X^n_M for every n.*

Proof. Write $[A] = \sum_l \epsilon_l[C_l] + \sum_y k_y\big((y) + (-y)\big)$ with $C_l \in \mathcal{C}$, repetitions allowed, $\epsilon_l = \pm 1$ and $k_y \in \mathbb{Z}$. When $\epsilon_l = -1$ we use $-[C_l] = [-C_l] - \sum_{c\in C_l}\big((c) + (-c)\big)$, where $-C_l$ is algebraic by Lemma 3.22(c). This gives $[A] = \sum_l [C'_l] + \sum_y k'_y\big((y) + (-y)\big)$ with algebraic C'_l and integers k'_y. Let P_- and P_+ be the unions of $|k'_y|$ copies of the pair $\{y,-y\}$ over the y with $k'_y < 0$ and with $k'_y > 0$. Then $[A + P_-] = [C'_1 + \cdots + C'_L + P_+]$, so $A + P_- = C'_1 + \cdots + C'_L + P_+$ as multisets. The right side is algebraic by Lemma 3.22(a),(b), and so A is algebraic by Lemma 3.22(b).

If $B_M = S_M + \sum_C \mathbb{Z}[C]$, we apply this with the standard multisets added to $\mathcal{C}$; they are algebraic by Lemma 4.1. For even n, the Hodge classes in $H^n(X^n_M, \mathbb{Q})$ are spanned by a power of the hyperplane class and the primitive Hodge classes, and by Proposition 3.6(ii) the latter span, over $\mathbb{C}$, the lines $V(a)$ whose entries form a balanced multiset; these lines are algebraic. In every other degree $2k$ the group $H^{2k}(X^n_M, \mathbb{Q})$ is spanned by a power of the hyperplane class, by the Lefschetz hyperplane theorem and Poincaré duality. □

Lemma 4.3. *Let ℓ be an odd prime with $\ell \parallel M$, and let $\nu_\ell\colon R_M \to \mathbb{Z}/2$ be the homomorphism with $\nu_\ell((y)) = 1$ if y has order ℓ and $\nu_\ell((y)) = 0$ otherwise. Then ν_ℓ vanishes on S_M.*

Proof. For a divisor k of M let $\Gamma_k \subseteq \mathbb{Z}/M$ be the subgroup of order k. A pair has 0 or 2 elements of order ℓ, since y and $-y$ have the same order. Let $\sigma = \sigma_{p,i}$ be standard and $d = M/p$. The elements of σ are those of the coset $i + \Gamma_p$, which does not contain 0, the element $-pi$, and, if $p = 2$, the element $M/2$, of order $2 \neq \ell$.

Let $p \neq \ell$. Then $\Gamma_p \cap \Gamma_\ell = 0$, so the coset $i + \Gamma_p$ contains at most one element of Γ_ℓ. It contains one exactly when $pi \in \Gamma_\ell$: if $z \in (i + \Gamma_p) \cap \Gamma_\ell$ then $pi = pz$, and conversely, multiplication by p is an automorphism of Γ_ℓ, so $pi = pz$ for some $z \in \Gamma_\ell$, and then $i - z \in \Gamma_p$. In that case $-pi = -pz$ has the same order as z. So σ has 0 or 2 elements of order ℓ.

Let $p = \ell$. Then $i \notin \Gamma_\ell = d\,\mathbb{Z}/M$, as $0 < i < d$, so $i + \Gamma_\ell$ does not meet Γ_ℓ; and $-\ell i \in \Gamma_\ell$ would mean that d divides ℓi, hence i, since $\gcd(\ell, d) = 1$. So σ has no element of order ℓ. □

Let P be the set consisting of 4 and the odd primes, and let $\mathcal{Q}(M)$ be the set of divisors $q > 1$ of M that are products of an even number of distinct elements of P.

Theorem 4.4 (Aoki)**.** *For every $q \in \mathcal{Q}(M)$ there is an element $\xi_q \in B_q$ such that*

$$B_M = S_M + \sum_{q\in\mathcal{Q}(M)} \sum_{t\in(\mathbb{Z}/M)^\times} \mathbb{Z}\, t\cdot(M/q)_*\xi_q.$$

Moreover $2x \in S_M$ and $x + t\cdot x \in S_M$ for every $x \in B_M$ and every unit t.

This is [Aok00, Theorem 2.1], with the second assertion taken from the proof of [Aok00, Corollary 2.3]; it rests on [Aok83, Theorem D]. In [Aok83] the groups are normalized differently,

and Peterson explains the translation [Pet26, §5.2]. Aoki also shows that $B_M/S_M \cong (\mathbb{Z}/2)^{|\mathcal{Q}(M)|}$ [Aok00, Remark 2.4]; we shall not need this.

Corollary 4.5. *Suppose that $\mathcal{Q}(M)$ has exactly one element, and let ℓ be an odd prime with $\ell \parallel M$. Let β be a balanced multiset of an even number of nonzero elements of $\mathbb{Z}/M$ with $\nu_\ell([\beta]) = 1$. Then $B_M = S_M + \mathbb{Z}[\beta]$. This applies to $M = 114$ with $\ell = 19$, to $M = 44$ with $\ell = 11$, to $M = 52$ with $\ell = 13$ and to $M = 102$ with $\ell = 17$, where $\mathcal{Q}(114) = \{57\}$, $\mathcal{Q}(44) = \{44\}$, $\mathcal{Q}(52) = \{52\}$ and $\mathcal{Q}(102) = \{51\}$.*

Proof. Let $\mathcal{Q}(M) = \{q\}$. By Theorem 4.4 the group B_M/S_M is generated by the classes of the elements $t \cdot (M/q)_*\xi_q$, and $t \cdot x \equiv -x \equiv x$ modulo S_M for every $x \in B_M$. So B_M/S_M is generated by one element of order at most 2. By Lemma 4.3, $[\beta] \notin S_M$; hence B_M/S_M has order 2 and is generated by the class of $[\beta]$. The elements of P dividing $114 = 2 \cdot 3 \cdot 19$ are 3 and 19, those dividing $44 = 4 \cdot 11$ are 4 and 11, those dividing $52 = 4 \cdot 13$ are 4 and 13, and those dividing $102 = 2 \cdot 3 \cdot 17$ are 3 and 17, since $2 \notin \mathsf{P}$; this gives the four values of $\mathcal{Q}$. □

For the multisets β of Lemmas 4.6, 4.22, 4.25 and 4.30 below, the equalities $B_M = S_M + \mathbb{Z}[\beta]$ for $M = 114$, 44, 52 and 102 have also been checked by an exact computation with integer matrices that does not use Theorem 4.4 (Appendix B).

4.2. **Two characters of the Fermat threefold.** Let $X = X_m^3$. Its cohomology $H^3(X, \mathbb{Q})$ is primitive, and by Proposition 3.6(i), $H^3(X, \mathbb{C})$ is the direct sum of the lines $V(a)$, $a \in \mathfrak{A}_m^3$, with $V(a) \subseteq H^{3-q,q}$ for $q = \langle a \rangle - 1$. As in the proof of Proposition 3.6(ii), for $a \in \mathfrak{A}_m^3$ the sum $\bigoplus_t V(ta)$ over the units t is defined over $\mathbb{Q}$; we write $M_a \subseteq H^3(X, \mathbb{Q})$ for the rational sub-Hodge structure with $M_a \otimes \mathbb{C} = \bigoplus_t V(ta)$. Since $\langle -a \rangle = 5 - \langle a \rangle$, the structure M_a has level one, that is, $M_a \otimes \mathbb{C} \subseteq H^{2,1} \oplus H^{1,2}$, exactly when $\langle ta \rangle \in \{2, 3\}$ for every unit t.

Lemma 4.6. *Let $m = 114$ and*

$$\alpha_1 = (1, 25, 43, 57, 102), \qquad \alpha_2 = (39, 63, 68, 80, 92).$$

(a) *$\alpha_1, \alpha_2 \in \mathfrak{A}_{114}^3$, and for every unit t, $\langle t\alpha_1 \rangle, \langle t\alpha_2 \rangle \in \{2, 3\}$ and $\langle t\alpha_1 \rangle + \langle t\alpha_2 \rangle = 5$. In particular $\langle \alpha_1 \rangle = \langle 13\alpha_2 \rangle = 2$, where $13\alpha_2 = (51, 21, 86, 14, 56)$.*

(b) *The ten entries of α_1 and α_2 form a balanced multiset β with $\nu_{19}([\beta]) = 1$, and β contains no pair.*

(c) *M_{α_1} and M_{α_2} have dimension 36.*

Proof. (a) The entries are nonzero, and they add up to 228 and 342. Since $\langle -ta \rangle = 5 - \langle ta \rangle$, it suffices to consider the eighteen units $t < 57$, and for them the values are

t	1	5	7	11	13	17	23	25	29	31	35	37	41	43	47	49	53	55
$\langle t\alpha_1 \rangle$	2	2	2	2	3	2	2	2	3	3	2	2	3	2	2	3	3	2
$\langle t\alpha_2 \rangle$	3	3	3	3	2	3	3	3	2	2	3	3	2	3	3	2	2	3

each entry being the sum of the five representatives in $\{1, \dots, 113\}$ of the entries of $t\alpha_i$, divided by 114. For instance, the entries of $13\alpha_2 = (51, 21, 86, 14, 56)$ add up to 228.

(b) For every unit t the representatives of the entries of $t\beta$ add up to $114(\langle t\alpha_1 \rangle + \langle t\alpha_2 \rangle) = 570$, so β is balanced. The elements of order 19 in $\mathbb{Z}/114$ are the nonzero multiples of 6, and $102 = 6 \cdot 17$ is the only entry of this kind. No two of the ten entries add up to 114.

(c) If $t\alpha_1 = \alpha_1$, then $t \equiv 1$, since 1 is an entry. If $t\alpha_2 = \alpha_2$, then $39t \equiv 39$ and $68t \equiv 68$, so $t \equiv 1$ modulo 38 and modulo 57, hence modulo 114. So the 36 characters $t\alpha_1$, and the 36 characters $t\alpha_2$, are distinct. □

By (a), $V(\alpha_1)$ and $V(13\alpha_2)$ lie in $H^{2,1}(X^3_{114})$, and the types of M_{α_1} and M_{α_2} are complementary: $V(t\alpha_1)\otimes V(t\alpha_2)$ has type $(3,3)$ for every t. By Lemma 4.3 and Corollary 4.5, β is not a combination of pairs and standard multisets, so it is not reached by linear subspaces and Aoki's cycles. The rest of the section shows that it is algebraic.

4.3. **Families of curves and coniveau.** In this subsection $m \geq 3$, $X = X^3_m \subseteq \mathbb{P}^4$ is the Fermat threefold with equation $F = \sum_k x_k^m$, $F_k = \partial F/\partial x_k = m\, x_k^{m-1}$, $G = \mu_m^5/\mu_m$ acts on X as in Section 3.5, and $U_k = X \cap \{x_k \neq 0\}$, an affine open set. We fix $\alpha \in \mathfrak{A}^3_m$ with $\langle\alpha\rangle = 2$, and write $\bar\alpha_k \in \{1,\dots,m-1\}$ for its entries, so that $\sum_k \bar\alpha_k = 2m$; then $V(\alpha) \subseteq H^{2,1}(X)$. Let

$$\Pi = \{z_0 + \cdots + z_4 = 0\} \subseteq \mathbb{P}^4, \qquad \Phi\colon X \to \Pi, \quad \Phi(x) = (x_0^m : \cdots : x_4^m).$$

The morphism Φ is finite and surjective, it is the quotient by G, and it is étale over $\Pi^\circ = \Pi \cap \{z_0 \cdots z_4 \neq 0\}$.

Definition 4.7. Let T be a smooth connected complex algebraic curve, not necessarily complete. A *block family of character α over T* consists of binary forms $B_1,\dots,B_r$ in (s,t) of degrees $\delta_1,\dots,\delta_r \geq 1$ and functions $c_0,\dots,c_4$, all with coefficients in the ring $\mathcal{O}(T)$ of regular functions, and of integers $e_{kl} \geq 0$ $(0 \leq k \leq 4,\ 1 \leq l \leq r)$, such that:

(B1) for every $\tau \in T$ the forms $B_l(\tau)$ are nonzero and no two of them have a common zero on $\mathbb{P}^1$, and $c_k(\tau) \neq 0$ for every k;

(B2) $D = \sum_l e_{kl}\delta_l$ does not depend on k; for every l some e_{kl} is positive and some e_{kl} vanishes; and the weight $w_l = \sum_k \bar\alpha_k e_{kl}$ of every block is divisible by m;

(B3) the forms $u_k = c_k \prod_l B_l^{e_{kl}}$ of degree D satisfy $u_0 + u_1 + u_2 + u_3 + u_4 = 0$.

We put $A = \prod_l B_l^{w_l/m}$, a form of degree $2D$, and $\Psi = \prod_k c_k^{\bar\alpha_k} \in \mathcal{O}(T)$; then

$$u_0^{\bar\alpha_0} u_1^{\bar\alpha_1} u_2^{\bar\alpha_2} u_3^{\bar\alpha_3} u_4^{\bar\alpha_4} = \Psi\, A^m. \tag{3}$$

A *chart* is an ordered pair (i,j) of distinct indices such that no block occurs in both u_i and u_j, that is, $e_{il}e_{jl} = 0$ for every l, and such that no block occurring in u_i vanishes at $(s:t) = (1:0)$.

Let $\tau \in T$. A common zero of the $u_k(\tau)$ would be a zero of some B_l, by (B1), and B_l is absent from some u_k by (B2); so $f_\tau = (u_0(\tau) : \cdots : u_4(\tau))$ is a morphism $\mathbb{P}^1 \to \Pi$, by (B3). Let

$$\mathcal{C}_\tau = \mathbb{P}^1 \times_\Pi X = \{((s:t), x) \in \mathbb{P}^1 \times X :\ f_\tau(s:t) = \Phi(x)\},$$

a curve finite over $\mathbb{P}^1$. For an irreducible component of $\mathcal{C}_\tau$ we write $\widehat{C}$ for its normalization, and $\pi_1\colon \widehat{C} \to \mathbb{P}^1$ and $\nu\colon \widehat{C} \to X$ for the two maps; π_1 is finite. Let $\widehat{C}^\circ$ be the preimage of the set of points $(s:1)$ with $u_0 \cdots u_4(s,1) \neq 0$. Since Φ is étale over Π°, near every point of $\widehat{C}^\circ$ the function s is a local coordinate, and ν is $s \mapsto x(s,\tau)$ with

$$x(s,\tau') = (u_0(s,1;\tau')^{1/m}, \dots, u_4(s,1;\tau')^{1/m}) \tag{4}$$

for suitable branches, defined also for τ' near τ.

Definition 4.8. Let (i,j) be a chart, let $a < b < c$ be the other three indices, let $\tau \in T$ and $v \in T_\tau T$. In the affine coordinate s (that is, $t = 1$), denote by a dot the derivative along v at fixed s and by a prime the derivative in s, and put

$$\Delta_v = \det \begin{pmatrix} 1 & 1 & 1 \\ \dot u_a/u_a & \dot u_b/u_b & \dot u_c/u_c \\ u_a'/u_a & u_b'/u_b & u_c'/u_c \end{pmatrix}, \qquad \eta_v = \frac{A\,\Delta_v}{u_i u_j}\, ds,$$

and

$$I_{ij}(\tau, v) = \sum_\rho \mathrm{Res}_{s=\rho}\, \eta_v,$$

the sum over the zeros $\rho \in \mathbb{C}$ of $u_i(s,1)$. All zeros of u_i lie in this affine line, because (i,j) is a chart.

Lemma 4.9. (a) *The poles of η_v in $\mathbb{C}$ lie at zeros of the blocks. At a zero ρ of B_l of multiplicity μ,*

$$\operatorname{ord}_\rho \eta_v \;\geq\; \mu\Big(\frac{w_l}{m} - 1 - e_{il} - e_{jl}\Big).$$

In particular η_v is regular at the zeros of the blocks that occur in neither u_i nor u_j.

(b) *The map $(\tau, v) \mapsto I_{ij}(\tau, v)$ is a holomorphic 1-form on T.*

Proof. (a) We have $\dot{u}_k/u_k = \dot{c}_k/c_k + \sum_l e_{kl}\dot{B}_l/B_l$ and $u_k'/u_k = \sum_l e_{kl}B_l'/B_l$. Write $\mathbf{1} = (1,1,1)$, $\gamma = (\dot{c}_k/c_k)_{k=a,b,c}$ and $\epsilon_l = (e_{kl})_{k=a,b,c}$. Expanding the determinant along its second and third rows,

$$\Delta_v = \sum_l \frac{B_l'}{B_l}\det(\mathbf{1}, \gamma, \epsilon_l) + \sum_{l \neq l'} \frac{\dot{B}_l}{B_l}\frac{B_{l'}'}{B_{l'}}\det(\mathbf{1}, \epsilon_l, \epsilon_{l'}),$$

the terms with $l = l'$ vanishing. The functions $\dot{c}_k/c_k$ are constants on $\mathbb{P}^1$, by (B1). At a zero ρ of B_l of multiplicity μ no other block vanishes, B_l'/B_l has a simple pole and $\dot{B}_l/B_l$ a pole of order at most μ; so $\operatorname{ord}_\rho \Delta_v \geq -\mu$. Moreover $\operatorname{ord}_\rho A = \mu w_l/m$ and $\operatorname{ord}_\rho(u_i u_j) = \mu(e_{il} + e_{jl})$. The last assertion holds because $w_l \geq m$, by (B2).

(b) Fix $\tau_0 \in T$. Let $Z_i(\tau) \subseteq \mathbb{C}$ be the set of zeros of the blocks that occur in u_i, and $Z'(\tau) \subseteq \mathbb{P}^1$ the set of zeros of the other blocks. These sets are disjoint by (B1), and they depend continuously on τ, since the blocks occurring in u_i do not vanish at $(1:0)$. Choose disjoint closed discs around the points of $Z_i(\tau_0)$ that do not meet $Z'(\tau_0)$, and let γ be the sum of their boundary circles. For τ near τ_0 the discs still contain $Z_i(\tau)$ and do not meet $Z'(\tau)$; so by (a) $I_{ij}(\tau, v) = (2\pi i)^{-1}\oint_\gamma \eta_v$, which is holomorphic in (τ, v) and linear in v. □

Peterson's argument now goes through for every block family; we give it in full, with his notation where possible. Let $\xi = \sum_k x_k\partial_k$ be the Euler vector field on $\mathbb{C}^5$, $\partial_k = \partial/\partial x_k$, $dV = dx_0 \wedge \cdots \wedge dx_4$ and $\Omega = \iota_\xi dV$. For $k \neq l$ put $\Omega_{kl} = \iota_{\partial_l}\iota_{\partial_k}\Omega$ and

$$\varphi_{kl} = \frac{x^{\bar{\alpha}-1}\,\Omega_{kl}}{F_k F_l}, \qquad x^{\bar{\alpha}-1} = \prod_{n=0}^{4} x_n^{\bar{\alpha}_n - 1}.$$

Lemma 4.10. (a) *φ_{kl} is the pullback of a rational 2-form on $\mathbb{P}^4$ that is regular where $x_k x_l \neq 0$; its restriction to X is a regular 2-form on $U_k \cap U_l$.*

(b) *$\varphi_{ln} - \varphi_{kn} + \varphi_{kl} = 0$ on $U_k \cap U_l \cap U_n$, and $\varphi_{lk} = -\varphi_{kl}$.*

(c) *$g^*\varphi_{kl} = \alpha(g)\varphi_{kl}$ for $g \in G$.*

So $\{\varphi_{kl}\}$ is a Čech 1-cocycle of Ω_X^2 for the cover $\{U_k\}$, and its class $[\varphi] \in H^1(X, \Omega_X^2)$ lies in the α-eigenspace.

Proof. (a) Under $x \mapsto \lambda x$ the factors $x^{\bar{\alpha}-1}$, Ω_{kl} and F_kF_l are multiplied by λ^{2m-5}, λ^3 and λ^{2m-2}, so φ_{kl} is invariant; and $\iota_\xi\Omega_{kl} = \iota_\xi\iota_{\partial_l}\iota_{\partial_k}\iota_\xi dV = 0$, since contractions anticommute. Hence φ_{kl} descends to $\mathbb{P}^4$.

(b) The second assertion is clear. Since $\iota_\xi dF = mF$, we have $dF \wedge \Omega = mF\,dV$. Contracting with ∂_k, ∂_l, ∂_n, and using that ι_{∂_p} is an antiderivation with $\iota_{\partial_p}dF = F_p$, gives

$$F_k\Omega_{ln} - F_l\Omega_{kn} + F_n\Omega_{kl} = dF \wedge \iota_{\partial_n}\Omega_{kl} + mF\,\iota_{\partial_n}\iota_{\partial_l}\iota_{\partial_k}dV$$

up to a sign common to both sides. On the cone over X both F and dF restrict to zero; multiplying by $x^{\bar{\alpha}-1}/(F_kF_lF_n)$ gives the cocycle relation.

(c) For $g = (\zeta_0 : \cdots : \zeta_4)$ we have $g^*x^{\bar\alpha-1} = \prod_n \zeta_n^{\bar\alpha_n-1}x^{\bar\alpha-1}$ and $g^*F_k = \zeta_k^{-1}F_k$, and $g^*\Omega_{kl} = \prod_{n\neq k,l}\zeta_n\,\Omega_{kl}$, because every term of Ω_{kl} is a multiple of $x_n\,dx_{n'}\wedge dx_{n''}$ with $\{n,n',n''\} = \{0,\dots,4\}\setminus\{k,l\}$. The product of the factors is $\prod_n \zeta_n^{\bar\alpha_n} = \alpha(g)$. □

This is the cocycle that Carlson and Griffiths attach to the residue of $x^{\bar\alpha-1}\Omega/F^2$ [CG80, §3(b)]; we do not need their normalization. By Proposition 3.6(i), the α-eigenspace of $H^{2,1}(X) \cong H^1(X,\Omega^2_X)$ is the line $V(\alpha)$.

We recall the Čech description of Dolbeault cohomology [Voi02, Chapter 4]. If θ is a $\bar\partial$-closed form of type $(2,1)$ on X, there are smooth $(2,0)$-forms χ_k on the Stein open sets U_k with $\bar\partial\chi_k = \theta|_{U_k}$, the forms $h_{kl} = \chi_l - \chi_k$ are holomorphic and form a Čech 1-cocycle of Ω^2_X, and $\theta \mapsto \{h_{kl}\}$ induces the isomorphism $H^{2,1}_{\bar\partial}(X) \cong H^1(X,\Omega^2_X)$, compatibly with the action of G.

Definition 4.11. Let $\tau \in T$, $v \in T_\tau T$, and let $\widehat C, \pi_1, \nu$ be as above. The vector $\partial x(s,\tau)/\partial v \in \mathbb{C}^5$, the derivative of (4) along v at fixed s, is tangent to the cone over X, since $\sum_k u_k = 0$ identically; so it defines a tangent vector to X at $\nu(s)$. A *first-order deformation of ν along v* is a family of holomorphic sections w_q of ν^*T_X over the members O_q of an open cover of $\widehat C$, such that $w_q - w_{q'}$ is tangent to $\widehat C$ on $O_q\cap O_{q'}$ and $w_q \equiv \partial x/\partial v$ modulo tangent vectors of $\widehat C$ on $O_q \cap \widehat C^\circ$.

For such w and a smooth form h of type $(2,0)$ or $(2,1)$ near $\nu(\widehat C)$, the form $\nu^*(\iota_w h)$, defined locally as $\nu^*(\iota_{w_q}h)$, does not depend on the choices: for ς tangent to $\widehat C$, $\nu^*(\iota_{d\nu(\varsigma)}h) = \iota_\varsigma\nu^*h = 0$, as ν^*h is a form of type $(2,0)$ or of degree three on a curve. Since the w_q are holomorphic, $\bar\partial\,\nu^*(\iota_w h) = -\nu^*(\iota_w\bar\partial h)$ for h of type $(2,0)$.

Lemma 4.12. *Let (i,j) be a chart, w a first-order deformation of ν along v, and $W_k = \nu^{-1}(U_k)$. For a Čech 1-cocycle h of Ω^2_X on $\{U_k\}$ put*

$$\mathcal{R}_w(h) = \sum_{o\in\widehat C\setminus W_i} \operatorname{Res}_o \nu^*(\iota_w h_{ij}).$$

Then $\mathcal{R}_w(h)$ depends only on the class of h, and for every $\bar\partial$-closed $(2,1)$-form θ with associated cocycle h,

$$\int_{\widehat C} \nu^*(\iota_w\theta) = -2\pi i\,\mathcal{R}_w(h).$$

Proof. As u_i and u_j have no common zero, x_i and x_j do not vanish together on $\nu(\widehat C)$; so $W_i\cup W_j = \widehat C$, and $\widehat C\setminus W_i$ is finite. Put $g_k = \nu^*(\iota_w\chi_k)$ on W_k for $k = i,j$. Then $\bar\partial g_k = -\nu^*(\iota_w\theta)$, and $g_j - g_i = \nu^*(\iota_w h_{ij})$ is holomorphic on $W_i\cap W_j$. Choose small disjoint discs $D_o \subseteq W_j$ around the points $o \in \widehat C\setminus W_i$. On a curve $\partial g_k = 0$, so $\nu^*(\iota_w\theta) = -dg_k$ on W_k, and Stokes' theorem gives

$$\int_{\widehat C}\nu^*(\iota_w\theta) = \sum_o \oint_{\partial D_o} g_i - \sum_o\oint_{\partial D_o} g_j = -2\pi i\sum_o \operatorname{Res}_o(g_j - g_i).$$

If h is a coboundary, $h_{kl} = \chi_l - \chi_k$ with holomorphic χ_k, and the same computation with $\theta = 0$ gives $\mathcal{R}_w(h) = 0$. □

Lemma 4.13. *Let (i,j) be a chart. There is a sign ϵ_{ij}, depending only on (i,j), such that on $\widehat C^\circ$, in the coordinate s,*

$$\nu^*(\iota_w\varphi_{ij}) = \epsilon_{ij}\,m^{-4}\,\nu^*\Big(\frac{x^{\bar\alpha}}{x_i^m x_j^m}\Big)\Delta_v\,ds, \qquad x^{\bar\alpha} = \prod_n x_n^{\bar\alpha_n}.$$

Moreover there is $r\in\mathbb{C}$ with $r^m = \Psi(\tau)$ such that $\nu^*(x^{\bar\alpha}/(x_i^m x_j^m)) = r\,\pi_1^*(A/(u_iu_j))$ on $\widehat C$.

Proof. Work on the cone, with the lift (4). The tangent vector and the deformation vector are

$$\frac{\partial x}{\partial s} = \Big(\frac{x_k u_k'}{m\,u_k}\Big)_k, \qquad \frac{\partial x}{\partial v} = \Big(\frac{x_k \dot{u}_k}{m\,u_k}\Big)_k.$$

Since φ_{ij} is the pullback of a form on $\mathbb{P}^4$, its value on the images of these vectors may be computed on the cone, and $\Omega_{ij}(\partial x/\partial v, \partial x/\partial s) = dV(\xi, \partial_i, \partial_j, \partial x/\partial v, \partial x/\partial s)$. Expanding this 5×5 determinant along the rows of ∂_i and ∂_j leaves, up to a sign ϵ_{ij}, the determinant of the rows ξ, $\partial x/\partial v$, $\partial x/\partial s$ in the columns a, b, c, which is $x_a x_b x_c m^{-2} \Delta_v$. Dividing by $F_i F_j = m^2 x_i^{m-1} x_j^{m-1}$ and multiplying by $x^{\bar\alpha - 1}$ gives the formula, since $w \equiv \partial x/\partial v$ modulo tangent vectors of $\widehat{C}$.

Both sides of the second identity are rational functions on $\widehat{C}$, of degree 0 in x and in (s,t). On $\widehat{C}^\circ$, by (4) and (3), the m-th power of their quotient is $\prod_k u_k^{\bar\alpha_k}/A^m = \Psi(\tau) \neq 0$. So the quotient is a constant on the connected curve $\widehat{C}$. □

Proposition 4.14 (Residue formula)**.** *Let ω be the harmonic representative of a nonzero class in $V(\alpha)$, and define $\lambda \in \mathbb{C}$ by $[\varphi] = \lambda[\omega]$ in $H^1(X, \Omega_X^2)$. Let (i,j) be a chart, $\tau \in T$, $v \in T_\tau T$, $\widehat{C}$ the normalization of a component of $\mathcal{C}_\tau$, and w a first-order deformation of ν along v. Then*

$$\lambda \int_{\widehat{C}} \nu^*(\iota_w \omega) = -2\pi i\, \epsilon_{ij}\, m^{-4}\, r\, \deg(\pi_1)\, I_{ij}(\tau, v),$$

with r as in Lemma 4.13. In particular, if $I_{ij}(\tau, v) \neq 0$, then $\lambda \neq 0$ and $\int_{\widehat{C}} \nu^*(\iota_w\omega) \neq 0$.

Proof. The class of ω spans the α-eigenspace of $H^1(X, \Omega_X^2)$, which contains $[\varphi]$ by Lemma 4.10; so λ is defined. Let h be the cocycle attached to ω. By Lemma 4.12, $\int_{\widehat{C}} \nu^*(\iota_w\omega) = -2\pi i\, \mathcal{R}_w(h)$ and $\mathcal{R}_w(\varphi) = \lambda \mathcal{R}_w(h)$. By Lemma 4.13, the holomorphic 1-form $\nu^*(\iota_w \varphi_{ij})$ on $W_i \cap W_j$ and the meromorphic 1-form $\epsilon_{ij} m^{-4} r\, \pi_1^* \eta_v$ agree on $\widehat{C}^\circ$, hence on the connected set $W_i \cap W_j$, which contains a punctured neighbourhood of every point of $\widehat{C} \setminus W_i$. These points are the preimages of the zeros of u_i. For a meromorphic 1-form η near $\rho \in \mathbb{P}^1$ one has $\sum_{o \in \pi_1^{-1}(\rho)} \operatorname{Res}_o \pi_1^* \eta = \deg(\pi_1) \operatorname{Res}_\rho \eta$. Therefore $\mathcal{R}_w(\varphi) = \epsilon_{ij} m^{-4} r \deg(\pi_1) I_{ij}(\tau, v)$, and the formula follows. Finally $r \neq 0$ and $\deg \pi_1 \geq 1$. □

Up to sign, $\int_{\widehat{C}} \nu^*(\iota_w\omega)$ is the derivative of the Abel–Jacobi map of the family of curves, paired with ω [CG80]; we only use it through the fibre integral below. Note that the formula concerns one component at a time: the components of $\mathcal{C}_\tau$ are permuted by G, and as α is a nontrivial character, their contributions add up to zero.

Lemma 4.15. *There are a smooth connected projective curve B with a nonconstant morphism $\pi\colon B \to \overline{T}$ to the smooth completion of T, a smooth projective surface S with morphisms $p\colon S \to B$ and $q\colon S \to X$, and a nonempty Zariski-open set $B^\circ \subseteq B$, such that for every $b \in B^\circ$:*

(i) *$\pi(b) \in T$, π is étale at b, and p is smooth over B°;*
(ii) *the fibre $S_b = p^{-1}(b)$, with the restriction of q and of the projection $S \to \mathbb{P}^1$, is the normalization of an irreducible component of $\mathcal{C}_{\pi(b)}$ with its maps ν and π_1;*
(iii) *for $v \in T_b B$, the images under dq of local holomorphic lifts of v to S form a first-order deformation of ν along $d\pi(v)$.*

Proof. Let $\mathcal{C} = \{(\tau, (s:t), x) \in T \times \mathbb{P}^1 \times X : \ f_\tau(s:t) = \Phi(x)\}$. The morphism Φ is finite and flat, being a finite morphism of smooth varieties of the same dimension, so $\mathcal{C} \to T \times \mathbb{P}^1$ is finite and flat, and every irreducible component of $\mathcal{C}$ is a surface mapping onto $T \times \mathbb{P}^1$. Fix one, $\mathcal{C}'$. Let B be the smooth projective curve whose function field is the algebraic closure of $\mathbb{C}(T)$ in $\mathbb{C}(\mathcal{C}')$, with the induced morphism π. The inclusion of function fields gives a rational

map $\mathcal{C}' \dashrightarrow B$, and with the projections to $\mathbb{P}^1$ and X it maps $\mathcal{C}'$ birationally onto a dense open subset of an irreducible surface $\mathcal{C}'' \subseteq B \times \mathbb{P}^1 \times X$, whose generic fibre over B is geometrically irreducible, since $\mathbb{C}(B)$ is algebraically closed in $\mathbb{C}(\mathcal{C}'')$. Let S be a resolution of singularities of $\mathcal{C}''$ that is an isomorphism over its smooth locus, with the induced p, q and $S \to \mathbb{P}^1$. By [Gro66, 9.7.7] and generic smoothness [Har77, III.10.7] there is a nonempty open B° satisfying (i) over which the fibres of p are smooth, connected and birational to the irreducible curves $\mathcal{C}''_b$. The image of $\mathcal{C}''_b$ in $\mathbb{P}^1 \times X$ lies in $\mathcal{C}_{\pi(b)}$ and is finite over $\mathbb{P}^1$, so it is a component; this gives (ii). Over the open set of $B \times \mathbb{P}^1$ where π is étale and $u_0 \cdots u_4 \neq 0$, the map $\mathcal{C}'' \to B \times \mathbb{P}^1$ is étale; so for a local coordinate z on B, the pair (z, s) is a system of local coordinates on S, and $q = x(s, \pi(z))$ with x as in (4). The vector field with s-component 0 and constant z-component equal to that of v is then a local lift $\tilde{v}$ of v with $dq(\tilde{v}) = \partial x/\partial(d\pi(v))$. Near the other points of S_b lifts exist because p is a submersion, and two lifts differ by a vertical field, whose image under dq is tangent to the fibre. This gives (iii). □

Theorem 4.16. *Let a block family of character α over T and a chart (i, j) be given, and suppose that $I_{ij}(\tau_0, v_0) \neq 0$ for some $\tau_0 \in T$ and $v_0 \in T_{\tau_0}T$. Then there are a smooth projective curve C, possibly disconnected, and an algebraic cycle Z of dimension 2 on $C \times X$, with the following property. Let $Z_*\colon H^1(C, \mathbb{Q}) \to H^3(X, \mathbb{Q})$ and $Z^*\colon H^3(X, \mathbb{Q}) \to H^1(C, \mathbb{Q})$ be the induced maps and $\Lambda = Z_*Z^*$. Then Λ preserves M_α, and there is a polynomial ℓ with rational coefficients such that $\ell(\Lambda)\Lambda$ is the identity on M_α. In particular $M_\alpha \subseteq Z_*H^1(C, \mathbb{Q})$, and M_α is supported on a divisor of X.*

Proof. Let B, S, p, q be as in Lemma 4.15 and $Z_1 = (p, q)_*[S]$, a cycle on $B \times X$, so that $Z_1^* = p_*q^*$. Let ω be the harmonic representative of a nonzero class $\omega_\alpha \in V(\alpha)$, a closed form of type $(2, 1)$. We first show that $Z_1^*\omega_\alpha \neq 0$. Over B° the map p is a proper submersion, and $\vartheta = p_*q^*\omega$ is a closed 1-form of type $(1, 0)$ there, hence holomorphic. For $b \in B^\circ$ and $v \in T_bB$, and any smooth lift $\tilde{v}$ of v along S_b, $\vartheta(v) = \pm \int_{S_b} \iota_{\tilde{v}} q^*\omega$, and the integrand does not change if $\tilde{v}$ changes by a vertical field; so $\vartheta(v) = \pm \int_{S_b} \nu^*(\iota_w \omega)$ with w as in Lemma 4.15(iii). The form ϑ extends holomorphically over each point $b_0 \in B \setminus B^\circ$: on a small punctured disc around b_0 its coefficient is integrable, being bounded by the fibre integral of an integrable form, so it has at most a simple pole; and the residue is $\pm(2\pi i)^{-1} \int_{p^{-1}\{|z| \leq \varepsilon\}} q^* d\omega = 0$. As $p^{-1}(B \setminus B^\circ)$ has measure zero, ϑ represents $Z_1^*\omega_\alpha$.

By Lemma 4.9(b) the 1-form I_{ij} on the connected curve T is holomorphic and not identically zero, so it vanishes only on a discrete set, while $\pi(B^\circ) \cap T$ is T minus a finite set. Choose $b \in B^\circ$ with $I_{ij}(\pi(b), \cdot) \neq 0$ and $v \in T_bB$ with $v \neq 0$. Since π is étale at b, $I_{ij}(\pi(b), d\pi(v)) \neq 0$, and by Proposition 4.14 and Lemma 4.15(ii),(iii), $\vartheta(v) \neq 0$. A nonzero holomorphic 1-form on a compact Riemann surface is not exact, so $Z_1^*\omega_\alpha = [\vartheta] \neq 0$.

Now let $C = G \times B$, a disjoint union of $|G|$ copies of B, and let Z restrict to $(\mathrm{id}_B \times g)_*Z_1$ on the copy indexed by g. Then $\Lambda = \sum_{g \in G} g_*Z_{1*}Z_1^*g^*$ commutes with the abelian group G, so it preserves each line $V(t\alpha)$ and acts on it by a scalar Λ_t. Since Λ is defined over $\mathbb{Q}$, the argument of Proposition 3.6(ii) shows that Λ_t is the image of $\Lambda_1 \in \mathbb{Q}(\mu_m)$ under $\zeta \mapsto \zeta^t$. Using $\int_X g_*x \cup x' = \int_X x \cup g^*x'$, $g^*\omega_\alpha = \alpha(g)\omega_\alpha$ and $Z_1^*\bar{\omega}_\alpha = \bar{\vartheta}$,

$$\Lambda_1 \int_X \omega_\alpha \cup \bar{\omega}_\alpha = \sum_{g \in G} |\alpha(g)|^2 \int_X Z_{1*}Z_1^*\omega_\alpha \cup \bar{\omega}_\alpha = |G| \int_B \vartheta \wedge \bar{\vartheta} \neq 0,$$

as $i \int_B \vartheta \wedge \bar{\vartheta} > 0$. So every Λ_t is nonzero, Λ is invertible on M_α, and by the Cayley–Hamilton theorem its inverse there is $\ell(\Lambda)$ for a polynomial ℓ with rational coefficients. Then $M_\alpha = \Lambda(M_\alpha) \subseteq Z_*H^1(C, \mathbb{Q})$, and $Z_*H^1(C, \mathbb{Q})$ vanishes on the complement of the image of the support of Z, a surface in X. □

4.4. **Joins with a threefold character.** We use the cohomological form of the inductive structure, due to Shioda [Shi79, Theorem II]; see also [SK79]. Let $r, s \geq 1$ and let Z be the blow-up of $X_m^r \times X_m^s$ along $Y \cong X_m^{r-1} \times X_m^{s-1}$, with the morphism $\psi\colon Z \to X_m^{r+s}$ of Proposition 3.4. Let $E \subseteq Z$ be the exceptional divisor, $j\colon E \to Z$ the inclusion and $\varpi\colon E \to Y$ the projection. Then $f = \psi_* j_* \varpi^*\colon H^{r+s-2}(Y, \mathbb{Q}) \to H^{r+s}(X_m^{r+s}, \mathbb{Q})$ maps $V(a') \otimes V(b')$ onto $V(a' * b')$ for $a' \in \mathfrak{A}_m^{r-1}$ and $b' \in \mathfrak{A}_m^{s-1}$, where $a' * b'$ is the juxtaposed character. Since f is a composite of pullbacks and pushforwards along morphisms, it maps algebraic classes to algebraic classes [Ful98, Chapter 19].

Proposition 4.17. *Let $\alpha, \alpha' \in \mathfrak{A}_m^3$ with $\langle t\alpha\rangle + \langle t\alpha'\rangle = 5$ for every unit t, and suppose that each of α and α' has the property of Theorem 4.16. Then the multiset of the ten entries of α and α' is balanced and algebraic.*

Proof. Let $X = X_m^3$. The multiset is balanced because the entries of $t\alpha$ and $t\alpha'$ add up to $5m$. For every t, one of $V(t\alpha)$, $V(t\alpha')$ lies in $H^{2,1}$ and the other in $H^{1,2}$; so $L = \bigoplus_t V(t\alpha) \otimes V(t\alpha') \subseteq H^6(X \times X, \mathbb{C})$ has type $(3,3)$. It is stable under $\mathrm{Gal}(\mathbb{Q}(\mu_m)/\mathbb{Q})$, so $L = L_{\mathbb{Q}} \otimes \mathbb{C}$ for a space $L_{\mathbb{Q}} \subseteq M_\alpha \otimes M_{\alpha'}$ of Hodge classes.

Let C, Z, Λ, ℓ and C', Z', Λ', ℓ' be as in Theorem 4.16 for α and α', and let $W = Z \times Z'$, a correspondence from $C \times C'$ to $X \times X$. On $H^3(X) \otimes H^3(X)$ the endomorphism $W_* W^*$ is $\pm\Lambda \otimes \Lambda'$, the sign coming from the Künneth isomorphism, so $\pm\ell(\Lambda) \otimes \ell'(\Lambda')$ inverts it on $M_\alpha \otimes M_{\alpha'}$. For $v \in L_{\mathbb{Q}}$ the class $v' = \pm W^*\big((\ell(\Lambda) \otimes \ell'(\Lambda'))v\big)$ lies in $H^2(C \times C', \mathbb{Q})$ and is a Hodge class, as W^* and $\ell(\Lambda) \otimes \ell'(\Lambda')$ are morphisms of Hodge structures up to a shift of type. By Theorem 2.3 on each component of the surface $C \times C'$, v' is algebraic, and so is $v = W_* v'$. Hence $V(\alpha) \otimes V(\alpha') \subseteq L \subseteq \mathrm{Alg}^3(X \times X) \otimes \mathbb{C}$. By the inductive structure with $r = s = 4$, the line $V(\alpha * \alpha') = f\big(V(\alpha) \otimes V(\alpha')\big)$ is algebraic. □

A character of the Fermat curve needs no family of curves: H^1 of a curve is already carried by a curve.

Proposition 4.18. *Let $\alpha \in \mathfrak{A}_m^3$ and $b \in \mathfrak{A}_m^1$ with $\langle t\alpha\rangle + \langle tb\rangle = 4$ for every unit t, and suppose that α has the property of Theorem 4.16. Then the multiset of the eight entries of α and b is balanced and algebraic.*

Proof. Let $X = X_m^3$ and $X' = X_m^1$. The multiset is balanced because the entries of $t\alpha$ and tb add up to $4m$. As $\langle tb\rangle \in \{1, 2\}$, we have $\langle t\alpha\rangle \in \{2, 3\}$, and $V(t\alpha) \subseteq H^{2,1}(X)$ exactly when $V(tb) \subseteq H^{0,1}(X')$; so $L = \bigoplus_t V(t\alpha) \otimes V(tb) \subseteq H^4(X \times X', \mathbb{C})$ has type $(2,2)$. As in the proof of Proposition 4.17, $L = L_{\mathbb{Q}} \otimes \mathbb{C}$ for a space $L_{\mathbb{Q}} \subseteq M_\alpha \otimes H^1(X', \mathbb{Q})$ of Hodge classes.

Let C, Z, Λ, ℓ be as in Theorem 4.16 for α, let $\Delta \subseteq X' \times X'$ be the diagonal, and let $W = Z \times \Delta$, a correspondence from $C \times X'$ to $X \times X'$. On $H^3(X) \otimes H^1(X')$ the endomorphism $W_* W^*$ is $\pm\Lambda \otimes 1$, so $\pm\ell(\Lambda) \otimes 1$ inverts it on $M_\alpha \otimes H^1(X')$. For $v \in L_{\mathbb{Q}}$ the class $v' = \pm W^*\big((\ell(\Lambda) \otimes 1)v\big)$ is a Hodge class in $H^2(C \times X', \mathbb{Q})$. By Theorem 2.3 on each component of the surface $C \times X'$ it is algebraic, and so is $v = W_* v'$. Hence $V(\alpha) \otimes V(b) \subseteq \mathrm{Alg}^2(X \times X') \otimes \mathbb{C}$, and by the inductive structure with $r = 4$ and $s = 2$ the line $V(\alpha * b) = f\big(V(\alpha) \otimes V(b)\big)$ is algebraic. □

4.5. **Two families.** We now take $m = 114$ and write down block families for the characters $13\alpha_2$ and α_1 of Lemma 4.6, both of which have $\langle\cdot\rangle = 2$. The first is explicit.

Proposition 4.19. *Let $\alpha = 13\alpha_2 = (51, 21, 86, 14, 56)$, $q(\lambda) = \lambda^3 - 3\lambda^2 + 1$, and let $T_1 \subseteq \mathbb{C}$ be the complement of $0, 1, 2/3, \pm 1/\sqrt{2}$ and the roots of q. Over T_1, with parameter λ, let the*

blocks be s, $s-t$, $s-\lambda t$ *and* t, *and*

$$u_0 = q(\lambda)\,(s-\lambda t)\,t^2, \qquad u_1 = -(\lambda-1)(s-\lambda t)^3, \qquad u_2 = (\lambda-1)\,s\,(s-t)^2,$$
$$u_3 = -(\lambda-1)(3\lambda-2)\,s^2t, \qquad u_4 = \lambda(2\lambda^2-1)(s-t)\,t^2.$$

This is a block family of character α *over* T_1, $(2,0)$ *is a chart, and*

$$I_{20}(\lambda,\partial_\lambda) = -\frac{2\lambda^3+12\lambda^2-21\lambda+8}{\lambda(\lambda-1)^2(3\lambda-2)(2\lambda^2-1)\,q(\lambda)}.$$

In particular $I_{20}(2,\partial_\lambda) = 5/28 \neq 0$.

Proof. On T_1 the blocks are pairwise coprime and the coefficients c_k do not vanish, which is (B1). The exponents of the blocks s, $s-t$, $s-\lambda t$, t in $(u_0,\dots,u_4)$ are $(0,0,1,2,0)$, $(0,0,2,0,1)$, $(1,3,0,0,0)$ and $(2,0,0,1,2)$, with weights $86+2\cdot14=114$, $2\cdot86+56=228$, $51+3\cdot21=114$ and $2\cdot51+14+2\cdot56=228$; every u_k is a cubic. This is (B2). For (B3), in the coordinate s ($t=1$) the coefficients of s^3, s^2, s and 1 in $\sum_k u_k$ are

$$-(\lambda-1)+(\lambda-1), \qquad 3\lambda(\lambda-1)-2(\lambda-1)-(\lambda-1)(3\lambda-2),$$
$$q(\lambda)-3\lambda^2(\lambda-1)+(\lambda-1)+\lambda(2\lambda^2-1), \qquad -\lambda q(\lambda)+\lambda^3(\lambda-1)-\lambda(2\lambda^2-1),$$

and all four vanish. The forms u_2 and u_0 share no block, and $u_2(1,0)=\lambda-1\neq0$; so $(2,0)$ is a chart.

Here $A = s(s-t)^2(s-\lambda t)t^2$, so in the coordinate s, $A/(u_2u_0) = 1/((\lambda-1)q(\lambda))$ is constant, and $\eta = \Delta\,ds/((\lambda-1)q)$, where Δ is formed with the rows $1,3,4$ and $v=\partial_\lambda$:

$$\frac{\dot u_1}{u_1} = \frac{1}{\lambda-1}-\frac{3}{s-\lambda}, \qquad \frac{\dot u_3}{u_3} = \frac{1}{\lambda-1}+\frac{3}{3\lambda-2}, \qquad \frac{\dot u_4}{u_4} = \frac{1}{\lambda}+\frac{4\lambda}{2\lambda^2-1} = \frac{6\lambda^2-1}{\lambda(2\lambda^2-1)},$$
$$\frac{u_1'}{u_1} = \frac{3}{s-\lambda}, \qquad \frac{u_3'}{u_3} = \frac{2}{s}, \qquad \frac{u_4'}{u_4} = \frac{1}{s-1}.$$

The zeros of u_2 are $s=0$ and $s=1$. At $s=0$ only u_3'/u_3 has a pole, and expanding the determinant along its third row shows that the residue of Δ is $2(\dot u_1/u_1(0)-\dot u_4/u_4) = 2(\frac{1}{\lambda-1}+\frac{3}{\lambda}-\frac{6\lambda^2-1}{\lambda(2\lambda^2-1)})$; over the common denominator $\lambda(\lambda-1)(2\lambda^2-1)$ the numerator is $\lambda(2\lambda^2-1)+3(\lambda-1)(2\lambda^2-1)-(\lambda-1)(6\lambda^2-1) = 2\lambda^3-3\lambda+2$. At $s=1$ only u_4'/u_4 has a pole, and the residue of Δ is $\dot u_3/u_3-\dot u_1/u_1(1) = \frac{1}{\lambda-1}+\frac{3}{3\lambda-2}-\frac{4}{\lambda-1} = -\frac{3(2\lambda-1)}{(\lambda-1)(3\lambda-2)}$. Hence

$$I_{20}(\lambda,\partial_\lambda) = \frac{2(2\lambda^3-3\lambda+2)(3\lambda-2)-3(2\lambda-1)\lambda(2\lambda^2-1)}{\lambda(\lambda-1)^2(3\lambda-2)(2\lambda^2-1)\,q(\lambda)},$$

and the numerator is $-(2\lambda^3+12\lambda^2-21\lambda+8)$. At $\lambda=2$ the numerator is -30 and the denominator is $2\cdot1\cdot4\cdot7\cdot(-3)=-168$. □

The second family has degree 24, and we know of no family of smaller degree for the orbit of α_1. Let (f,h,g,a,b,c,κ), with $f=(f_0,\dots,f_6)$, $h=(h_0,h_1)$ and $g=(g_0,\dots,g_{11})$, be the coordinates of $\mathbb{C}^{25}$, and consider the forms

$$F = s^7+\sum_{n<7} f_n s^n t^{7-n}, \quad H = s^2+h_1st+h_0t^2, \quad G = s^{12}+\sum_{n<12} g_n s^n t^{12-n}, \quad K = s(s-t)(s-\kappa t),$$

and

$$u_0 = F^3K, \quad u_1 = a\,HFK^5, \quad u_2 = b\,H^5F^2, \quad u_3 = -(1+a+b)\,G^2, \quad u_4 = c\,HK\,t^{19},$$

binary forms of degree 24. The coefficient of s^{24} in $E = \sum_k u_k$ is $1 + a + b - (1 + a + b) = 0$; let $E_0, \dots, E_{23}$ be the other coefficients, $\mathcal{P} \subseteq \mathbb{C}^{25}$ their common zero set, and J the Jacobian matrix of $(E_0, \dots, E_{23})$ with respect to the 24 coordinates other than κ.

Proposition 4.20. *There is a point $p^* \in \mathcal{P}$ with $\kappa(p^*) = (-17 + 3i)/10$ at which*

(i) *J is invertible;*
(ii) *$abc(1 + a + b) \neq 0$, and F, H, G, K are pairwise coprime;*
(iii) *the residue functional of the chart $(0, 3)$, defined below, does not vanish on the tangent vector v^* of $\mathcal{P}$ at p^* with $d\kappa(v^*) = 1$.*

The proof is a computation in interval arithmetic, described in Appendix B; with Propositions 4.23, 4.26 and 4.28 it is one of the four steps of the paper that depend on a computer. The point p^* is the unique zero of $E(\,\cdot\,, \kappa(p^*))$ in an explicit box of radius 10^{-95}, and the residue functional is enclosed in a disc of radius 10^{-10} around $75.2385455180 - 19.2657017116\,i$.

Corollary 4.21. *Let $\mathcal{P}_1$ be the irreducible component of $\mathcal{P}$ through p^*, and let $T_2 \subseteq \mathcal{P}_1$ be the set of points at which the conditions* (i) *and* (ii) *of Proposition 4.20 hold. Then T_2 is a smooth connected curve, the forms above are a block family of character α_1 over T_2, $(0, 3)$ is a chart, and $I_{03}(p^*, v^*) \neq 0$.*

Proof. At a point of $\mathcal{P}$ where J is invertible, $\mathcal{P}$ is smooth of dimension one, with κ as a local coordinate, by the Jacobian criterion; in particular only one component passes through it. So T_2 is a nonempty Zariski-open subset of the irreducible curve $\mathcal{P}_1$, and it is smooth and connected. The blocks are F, H, G, K and t, with exponents $(3, 1, 2, 0, 0)$, $(0, 1, 5, 0, 1)$, $(0, 0, 0, 2, 0)$, $(1, 5, 0, 0, 1)$ and $(0, 0, 0, 0, 19)$ in $(u_0, \dots, u_4)$ and weights

$$3 + 25 + 2 \cdot 43 = 114, \qquad 25 + 5 \cdot 43 + 102 = 342, \qquad 2 \cdot 57 = 114,$$
$$1 + 5 \cdot 25 + 102 = 228, \qquad 19 \cdot 102 = 1938 = 17 \cdot 114.$$

Every u_k has degree 24. The forms F, H, G, K are monic in s, so none of them vanishes at $(1 : 0)$, and with (ii) the blocks are pairwise coprime; the coefficients $1, a, b, -(1 + a + b), c$ do not vanish. This proves (B1)–(B3). The forms $u_0 = F^3K$ and $u_3 = -(1 + a + b)G^2$ share no block, so $(0, 3)$ is a chart, and (iii) is the last assertion. □

The residue functional is computed as follows. Here $A = FH^3GK^2t^{17}$, and in the coordinate s,

$$\eta_v = \frac{A\,\Delta_v}{u_0 u_3}\,ds = \frac{\mathcal{D}_v}{\mathcal{M}\,F^5}\,ds, \qquad \mathcal{D}_v = \det \begin{pmatrix} u_1 & u_2 & u_4 \\ \dot{u}_1 & \dot{u}_2 & \dot{u}_4 \\ u_1' & u_2' & u_4' \end{pmatrix}, \qquad \mathcal{M} = -abc(1 + a + b)\,K^5H^4G,$$

since $u_0u_1u_2u_3u_4 = \mathcal{M}F^5 \cdot FH^3GK^2$. The zeros of u_0 are the roots of F and of K. At a root of K the form η_v is regular, by Lemma 4.9(a): the block K has weight $2m$ and occurs once in u_0 and not in u_3. As $\mathcal{M}$ and F are coprime, there is a unique polynomial R_v of degree less than 35 with $R_v\mathcal{M} \equiv \mathcal{D}_v$ modulo F^5; then $\eta_v - R_vF^{-5}ds$ is regular at the roots of F, and R_vF^{-5} has no other poles in $\mathbb{C}$ and behaves like $r_{34}s^{-1}$ at infinity, where r_{34} is the coefficient of s^{34} in R_v. By the residue theorem,

$$I_{03}(p, v) = r_{34}.$$

This is the quantity enclosed by the certificate, for $p = p^*$ and $v = v^*$.

4.6. **Degree** 44**.** For $m = 44$ the class left by Corollary 4.5 can be taken to be the join of a threefold character and a curve character, so one family of curves suffices.

Lemma 4.22. *Let* $m = 44$ *and*

$$\alpha = (1, 5, 8, 33, 41), \qquad b = (9, 37, 42).$$

(a) $\alpha \in \mathfrak{A}^3_{44}$, $b \in \mathfrak{A}^1_{44}$, *and* $\langle t\alpha\rangle + \langle tb\rangle = 4$ *for every unit* t*. In particular* $\langle\alpha\rangle = 2$*.*
(b) *The eight entries of* α *and* b *form a balanced multiset* β *with* $\nu_{11}([\beta]) = 1$*, and* β *contains no pair.*

Proof. (a) The entries are nonzero, and both entry sums equal 88. Since $\langle -t\alpha\rangle = 5 - \langle t\alpha\rangle$ and $\langle -tb\rangle = 3 - \langle tb\rangle$, it suffices to consider the ten units $t < 22$, and for them the values are

t	1	3	5	7	9	13	15	17	19	21
$\langle t\alpha\rangle$	2	2	3	2	2	2	3	3	2	3
$\langle tb\rangle$	2	2	1	2	2	2	1	1	2	1

For instance $5\alpha = (5, 25, 40, 33, 29)$ and $5b = (1, 9, 34)$, with entry sums 132 and 44.

(b) For every unit t the representatives of the entries of $t\beta$ add up to $4 \cdot 44$, so β is balanced. The elements of order 11 in $\mathbb{Z}/44$ are the nonzero multiples of 4, and 8 is the only entry of this kind. No two of the eight entries add up to 44. □

By (a), $V(\alpha) \subseteq H^{2,1}(X^3_{44})$ and $V(b) \subseteq H^{0,1}(X^1_{44})$. By Lemma 4.3, β is not a combination of pairs and standard multisets. A direct enumeration shows that ν_{11} vanishes on every balanced multiset A of four or six nonzero elements of $\mathbb{Z}/44$. By Corollary 4.5, $[A] = x + k[\beta]$ with $x \in S_{44}$ and k even, so $[A] \in S_{44}$ by Theorem 4.4, and A is algebraic by Lemma 4.2. So for X^n_{44} with $n \le 4$ linear subspaces and Aoki's cycles suffice, and β, a class on the sixfold, is the first that needs more.

Let $(p, q, r_1, c_0, c_1, c_3, \kappa)$, with $p = (p_0, \dots, p_4)$ and $q = (q_0, q_1, q_2)$, be the coordinates of $\mathbb{C}^{13}$, and consider the forms

$$P = s^5 + \sum_{n<5} p_n s^n t^{5-n}, \qquad Q = s^3 + \sum_{n<3} q_n s^n t^{3-n}, \qquad S = s^2 + r_1 st + \kappa t^2,$$

and

$$u_0 = c_0\, s^3 PQ, \quad u_1 = c_1\, PQ^2, \quad u_2 = t^{11}, \quad u_3 = c_3\, QS^4, \quad u_4 = s\, P^2,$$

binary forms of degree 11. Let $E_0, \dots, E_{11}$ be the coefficients of $\sum_k u_k$, $\mathcal{P}' \subseteq \mathbb{C}^{13}$ their common zero set, and J' the Jacobian matrix of $(E_0, \dots, E_{11})$ with respect to the 12 coordinates other than κ.

Proposition 4.23. *There is a point* $p^\dagger \in \mathcal{P}'$ *with* $\kappa(p^\dagger) = (4 - 3i)/10$ *at which*

(i) J' *is invertible;*
(ii) $c_0 c_1 c_3 \neq 0$, $p_0 q_0 \kappa \neq 0$, *and* P, Q, S *are pairwise coprime;*
(iii) *the residue functional of the chart* $(0, 2)$*, described below, does not vanish on the tangent vector* $v^\dagger$ *of* $\mathcal{P}'$ *at* $p^\dagger$ *with* $d\kappa(v^\dagger) = 1$*.*

The proof is a computation in interval arithmetic of the same kind as for Proposition 4.20, with 12 unknowns in place of 24 (Appendix B). The point $p^\dagger$ is the unique zero of $E(\,\cdot\,, \kappa(p^\dagger))$ in an explicit box of radius 10^{-95}, and the residue functional is enclosed in a disc of radius 10^{-70} around $-0.1774444206 + 0.0654161493\, i$.

Corollary 4.24. *Let $\mathcal{P}'_1$ be the irreducible component of $\mathcal{P}'$ through $p^\dagger$, and let $T_3 \subseteq \mathcal{P}'_1$ be the set of points at which the conditions* (i) *and* (ii) *of Proposition 4.23 hold. Then T_3 is a smooth connected curve, the forms above are a block family of character α over T_3, $(0,2)$ is a chart, and $I_{02}(p^\dagger, v^\dagger) \neq 0$.*

Proof. As in the proof of Corollary 4.21, T_3 is a nonempty Zariski-open subset of the irreducible curve $\mathcal{P}'_1$, smooth and connected, with κ as a local coordinate at $p^\dagger$. The blocks are s, t, P, Q and S, with exponents $(3,0,0,0,1)$, $(0,0,11,0,0)$, $(1,1,0,0,2)$, $(1,2,0,1,0)$ and $(0,0,0,4,0)$ in $(u_0,\dots,u_4)$ and weights

$$3+41=44, \qquad 11\cdot 8=88, \qquad 1+5+2\cdot 41=88, \qquad 1+2\cdot 5+33=44, \qquad 4\cdot 33=132.$$

Every u_k has degree 11. The forms P, Q, S are monic in s, so none of them vanishes at $(1:0)$, the zero of t, and their values p_0, q_0, κ at $(0:1)$, the zero of s, do not vanish by (ii); with the rest of (ii) the five blocks are pairwise coprime, and the coefficients $c_0, c_1, 1, c_3, 1$ do not vanish. This proves (B1)–(B3). The forms u_0 and $u_2 = t^{11}$ share no block, and the blocks s, P, Q of u_0 do not vanish at $(1:0)$, so $(0,2)$ is a chart; the last assertion is (iii). □

The residue functional is computed as follows. Here $A = st^2P^2QS^3$, and since $u_1u_3u_4 = c_1c_3\, sP^3Q^3S^4$, in the coordinate s

$$\eta_v = \frac{A\,\Delta_v}{u_0u_2}\,ds = \frac{\mathcal{N}_v}{c_0c_1c_3\, s^3P^2Q^3S}\,ds, \qquad \mathcal{N}_v = \det\begin{pmatrix} u_1 & u_3 & u_4 \\ \dot u_1 & \dot u_3 & \dot u_4 \\ u'_1 & u'_3 & u'_4 \end{pmatrix}.$$

The zeros of u_0 are $s=0$ and the roots of P and of Q. By Lemma 4.9(a), η_v is regular at the roots of P, a block of weight $2m$ that occurs once in u_0 and not in u_2, and at the roots of S, which occurs in neither. So the poles of η_v in $\mathbb{C}$ lie at zeros of u_0, and $I_{02}(p,v)$ is the sum of all residues of η_v in $\mathbb{C}$, that is, minus its residue at infinity. As $\mathcal{N}_v$ has degree at most 32 and the denominator has degree 24, $I_{02}(p,v)$ is the coefficient of s^{-1} in the expansion of $\mathcal{N}_v/(c_0c_1c_3s^3P^2Q^3S)$ in powers of s^{-1}. This is the quantity enclosed by the certificate, for $p = p^\dagger$ and $v = v^\dagger$.

4.7. **Degree** 52**.** For $m = 52$ the class left by Corollary 4.5 can again be taken to be the join of two characters of the Fermat threefold, and both need a family of curves.

Lemma 4.25. *Let $m = 52$ and*

$$\alpha = (1,5,16,37,45), \qquad \alpha' = (29,18,42,50,17).$$

(a) *$\alpha, \alpha' \in \mathfrak{A}^3_{52}$, and for every unit t, $\langle t\alpha\rangle, \langle t\alpha'\rangle \in \{2,3\}$ and $\langle t\alpha\rangle + \langle t\alpha'\rangle = 5$. In particular $\langle\alpha\rangle = \langle 9\alpha'\rangle = 2$, where $9\alpha' = (1,6,14,34,49)$.*

(b) *The ten entries of α and α' form a balanced multiset β with $\nu_{13}([\beta]) = 1$, and β contains no pair.*

Proof. (a) The entries are nonzero, and they add up to 104 and 156. Since $\langle -ta\rangle = 5 - \langle ta\rangle$, it suffices to consider the twelve units $t < 26$, and for them the values are

t	1	3	5	7	9	11	15	17	19	21	23	25
$\langle t\alpha\rangle$	2	2	2	2	3	2	3	2	3	2	2	3
$\langle t\alpha'\rangle$	3	3	3	3	2	3	2	3	2	3	3	2

For instance, the entries of $9\alpha' = (1,6,14,34,49)$ add up to 104.

(b) For every unit t the representatives of the entries of $t\beta$ add up to $5 \cdot 52$, so β is balanced. The elements of order 13 in $\mathbb{Z}/52$ are the nonzero multiples of 4, and 16 is the only entry of this kind. No two of the ten entries add up to 52. □

By (a), $V(\alpha)$ and $V(9\alpha')$ lie in $H^{2,1}(X_{52}^3)$, and the types of M_α and $M_{\alpha'}$ are complementary. By Lemma 4.3, β is not a combination of pairs and standard multisets. A direct enumeration shows that ν_{13} vanishes on each of the 399 balanced multisets of four and the 4884 balanced multisets of six nonzero elements of $\mathbb{Z}/52$; as in Section 4.6, linear subspaces and Aoki's cycles therefore suffice for X_{52}^n with $n \le 4$. Balanced multisets of eight elements with $\nu_{13} = 1$ exist, so the first classes that need more lie on the sixfold; we reach them through β, a class on the eightfold (Remark 4.37(b)).

Let $(g, f, c_0, c_1, c_3, c_4, \kappa)$, with $g = (g_0, \dots, g_3)$ and $f = (f_0, \dots, f_6)$, be the coordinates of $\mathbb{C}^{16}$, and consider the forms

$$G = s^4 + \sum_{n<4} g_n s^n t^{4-n}, \qquad F = s^7 + \sum_{n<7} f_n s^n t^{7-n}, \qquad H = s^2 + \kappa st + t^2,$$

and

$$u_0 = c_0 F^2, \quad u_1 = c_1\, sFH^3, \quad u_2 = st^{13}, \quad u_3 = c_3\, G^3 H, \quad u_4 = c_4\, s^3 GF,$$

binary forms of degree 14. Let $E_0, \dots, E_{14}$ be the coefficients of $\sum_k u_k$, $\mathcal{P}_A \subseteq \mathbb{C}^{16}$ their common zero set, and J_A the Jacobian matrix of $(E_0, \dots, E_{14})$ with respect to the 15 coordinates other than κ.

Proposition 4.26. *There is a point $p^A \in \mathcal{P}_A$ with $\kappa(p^A) = -(3+2i)/10$ at which*

(i) *J_A is invertible;*
(ii) *$c_0c_1c_3c_4 \neq 0$, $g_0 f_0 \neq 0$, and G, F, H are pairwise coprime;*
(iii) *the residue functional of the chart $(0,2)$, described below, does not vanish on the tangent vector v^A of $\mathcal{P}_A$ at p^A with $d\kappa(v^A) = 1$.*

The proof is a computation in interval arithmetic of the same kind as for Proposition 4.20, with 15 unknowns (Appendix B). The point p^A is the unique zero of $E(\,\cdot\,, \kappa(p^A))$ in an explicit box of radius 10^{-95}, and the residue functional is enclosed in a disc of radius 10^{-67} around $-5.3886778524 + 19.2154179850\,i$.

Corollary 4.27. *Let $\mathcal{P}_{A,1}$ be the irreducible component of $\mathcal{P}_A$ through p^A, and let $T_4 \subseteq \mathcal{P}_{A,1}$ be the set of points at which the conditions* (i) *and* (ii) *of Proposition 4.26 hold. Then T_4 is a smooth connected curve, the forms above are a block family of character α over T_4, $(0,2)$ is a chart, and $I_{02}(p^A, v^A) \neq 0$.*

Proof. As in the proof of Corollary 4.21, T_4 is a nonempty Zariski-open subset of the irreducible curve $\mathcal{P}_{A,1}$, smooth and connected, with κ as a local coordinate at p^A. The blocks are G, s, F, t and H, with exponents $(0,0,0,3,1)$, $(0,1,1,0,3)$, $(2,1,0,0,1)$, $(0,0,13,0,0)$ and $(0,3,0,1,0)$ in $(u_0, \dots, u_4)$ and weights

$$3{\cdot}37{+}45 = 156, \qquad 5{+}16{+}3{\cdot}45 = 156, \qquad 2{+}5{+}45 = 52, \qquad 13{\cdot}16 = 208, \qquad 3{\cdot}5{+}37 = 52.$$

Every u_k has degree 14. The forms G, F, H are monic in s, so none of them vanishes at $(1:0)$, the zero of t, and their values g_0, f_0, 1 at $(0:1)$, the zero of s, do not vanish by (ii); with the rest of (ii) the five blocks are pairwise coprime, and the coefficients $c_0, c_1, 1, c_3, c_4$ do not vanish. This proves (B1)–(B3). The forms $u_0 = c_0F^2$ and $u_2 = st^{13}$ share no block, and F does not vanish at $(1:0)$, so $(0,2)$ is a chart; the last assertion is (iii). □

The residue functional is computed as follows. Here $A = G^3 s^3 F t^4 H$ and $u_0u_1u_2u_3u_4 = c_0c_1c_3c_4\, s^5F^4G^4H^4t^{13}$, so in the coordinate s

$$\eta_v = \frac{A\,\Delta_v}{u_0u_2}\,ds = \frac{\mathcal{N}_v}{c_0c_1c_3c_4\, s^2F^3GH^3}\,ds, \qquad \mathcal{N}_v = \det\begin{pmatrix} u_1 & u_3 & u_4 \\ \dot u_1 & \dot u_3 & \dot u_4 \\ u_1' & u_3' & u_4' \end{pmatrix}.$$

The zeros of u_0 are the roots of F. By Lemma 4.9(a), η_v is regular at $s=0$, since the block s has weight $3m$, occurs once in u_2 and not in u_0, and at the roots of G and H, which occur in neither. So $I_{02}(p,v)$ is the sum of all residues of η_v in $\mathbb{C}$. As $\mathcal{N}_v$ has degree at most 41 and the denominator has degree 33, $I_{02}(p,v)$ is the coefficient of s^{-1} in the expansion of η_v/ds in powers of s^{-1}. This is the quantity enclosed by the certificate, for $p = p^A$ and $v = v^A$.

The second family has degree 5. Let $(q_0, q_1, c_0, c_1, c_3, c_4, \kappa)$ be the coordinates of $\mathbb{C}^7$, and consider the forms $L = s + \kappa t$ and $Q = s^2 + q_1 st + q_0 t^2$, and

$$u_0 = c_0\, s^3(s-t)^2, \quad u_1 = c_1 L^3 Q, \quad u_2 = t^5, \quad u_3 = c_3\, t(s-t)^3 L, \quad u_4 = c_4\, sQ^2,$$

binary forms of degree 5. Let $E_0, \dots, E_5$ be the coefficients of $\sum_k u_k$, $\mathcal{P}_B \subseteq \mathbb{C}^7$ their common zero set, and J_B the Jacobian matrix of $(E_0, \dots, E_5)$ with respect to the six coordinates other than κ.

Proposition 4.28. *There is a point $p^B \in \mathcal{P}_B$ with $\kappa(p^B) = (3+6i)/10$ at which*

(i) *J_B is invertible;*
(ii) *$c_0c_1c_3c_4 \neq 0$, and s, $s-t$, L and Q are pairwise coprime;*
(iii) *the residue functional of the chart $(1,2)$, described below, does not vanish on the tangent vector v^B of $\mathcal{P}_B$ at p^B with $d\kappa(v^B) = 1$.*

Again the proof is a computation in interval arithmetic, now with six unknowns (Appendix B). The point p^B is the unique zero of $E(\,\cdot\,, \kappa(p^B))$ in a box of radius 10^{-95}, and the residue functional is enclosed in a disc of radius 10^{-86} around $-5.5394454947 - 5.3824696631\,i$.

Corollary 4.29. *Let $\mathcal{P}_{B,1}$ be the irreducible component of $\mathcal{P}_B$ through p^B, and let $T_5 \subseteq \mathcal{P}_{B,1}$ be the set of points at which the conditions* (i) *and* (ii) *of Proposition 4.28 hold. Then T_5 is a smooth connected curve, the forms above are a block family of character $9\alpha' = (1,6,14,34,49)$ over T_5, $(1,2)$ is a chart, and $I_{12}(p^B, v^B) \neq 0$.*

Proof. As before, T_5 is a nonempty Zariski-open subset of the irreducible curve $\mathcal{P}_{B,1}$, smooth and connected. The blocks are t, s, $s-t$, L and Q, with exponents $(0,0,5,1,0)$, $(3,0,0,0,1)$, $(2,0,0,3,0)$, $(0,3,0,1,0)$ and $(0,1,0,0,2)$ and, for the character $(1,6,14,34,49)$, weights

$$5\cdot 14 + 34 = 104, \qquad 3 + 49 = 52, \qquad 2 + 3\cdot 34 = 104, \qquad 3\cdot 6 + 34 = 52, \qquad 6 + 2\cdot 49 = 104.$$

Every u_k has degree 5. The forms s, $s-t$, L, Q do not vanish at $(1:0)$, the zero of t, so with (ii) the five blocks are pairwise coprime, and the coefficients $c_0, c_1, 1, c_3, c_4$ do not vanish. This proves (B1)–(B3). The forms $u_1 = c_1L^3Q$ and $u_2 = t^5$ share no block, and L, Q do not vanish at $(1:0)$, so $(1,2)$ is a chart; the last assertion is (iii). □

Here $A = t^2 s(s-t)^2 LQ^2$ and $u_0u_1u_2u_3u_4 = c_0c_1c_3c_4\, s^4(s-t)^5L^4Q^3t^6$, so in the coordinate s

$$\eta_v = \frac{A\,\Delta_v}{u_1u_2}\,ds = \frac{\mathcal{N}_v}{c_0c_1c_3c_4\, s^3(s-1)^3L^3Q}\,ds, \qquad \mathcal{N}_v = \det\begin{pmatrix} u_0 & u_3 & u_4 \\ \dot u_0 & \dot u_3 & \dot u_4 \\ u_0' & u_3' & u_4' \end{pmatrix}.$$

The zeros of u_1 are the roots of L and Q. By Lemma 4.9(a), η_v is regular at $s=0$ and $s=1$, the zeros of the blocks s and $s-t$, which occur in neither u_1 nor u_2. So $I_{12}(p,v)$ is the sum of all residues of η_v in $\mathbb{C}$, the coefficient of s^{-1} in the expansion of η_v/ds in powers of s^{-1}; here

$\mathcal{N}_v$ has degree at most 14 and the denominator degree 11. This is the quantity enclosed by the certificate, for $p = p^B$ and $v = v^B$.

4.8. **Degree** 51**.** For $m = 51$ we work at the level 102, where Corollary 4.5 leaves one class. It is again the join of two characters of the Fermat threefold, and both need a family of curves.

Lemma 4.30. *Let* $m = 102$ *and*

$$\alpha = (1, 7, 51, 60, 85), \qquad \alpha' = (91, 34, 26, 98, 57).$$

(a) $\alpha, \alpha' \in \mathfrak{A}^3_{102}$*, and for every unit* t*,* $\langle t\alpha\rangle, \langle t\alpha'\rangle \in \{2, 3\}$ *and* $\langle t\alpha\rangle + \langle t\alpha'\rangle = 5$*. In particular* $\langle 5\alpha\rangle = \langle 37\alpha'\rangle = 2$*, where* $5\alpha = (5, 35, 51, 96, 17)$ *and* $37\alpha' = (1, 34, 44, 56, 69)$*.*
(b) *The ten entries of* α *and* α' *form a balanced multiset* β *with* $\nu_{17}([\beta]) = 1$*, and* β *contains no pair.*

Proof. (a) The entries are nonzero, and they add up to 204 and 306. Since $\langle -ta\rangle = 5 - \langle ta\rangle$, it suffices to consider the sixteen units $t < 51$, and for them the values are

t	1	5	7	11	13	19	23	25	29	31	35	37	41	43	47	49
$\langle t\alpha\rangle$	2	2	2	2	3	2	2	3	2	2	2	3	2	3	2	3
$\langle t\alpha'\rangle$	3	3	3	3	2	3	3	2	3	3	3	2	3	2	3	2

The entries of 5α and of $37\alpha'$ add up to 204.

(b) For every unit t the representatives of the entries of $t\beta$ add up to $5 \cdot 102$, so β is balanced. The elements of order 17 in $\mathbb{Z}/102$ are the nonzero multiples of 6, and 60 is the only entry of this kind. No two of the ten entries add up to 102. □

By (a), $V(5\alpha)$ and $V(37\alpha')$ lie in $H^{2,1}(X^3_{102})$, and the types of M_α and $M_{\alpha'}$ are complementary. By Lemma 4.3, β is not a combination of pairs and standard multisets. A direct enumeration shows that ν_{17} vanishes on each of the 1456 balanced multisets of four and the 32490 balanced multisets of six nonzero elements of $\mathbb{Z}/102$; as in Section 4.6, linear subspaces and Aoki's cycles therefore suffice for X^n_{102} with $n \le 4$. Balanced multisets of eight elements with $\nu_{17} = 1$ exist, for instance $\{1, 13, 27, 31, 68, 80, 92, 96\}$, so the first classes that need more lie on the sixfold; we reach them through β, a class on the eightfold.

The first family has degree 6. Let $(q_0, q_1, f_1, c_1, c_2, c_3, c_4, \kappa)$ be the coordinates of $\mathbb{C}^8$, and consider the forms

$$Q = s^2 + q_1 st + q_0 t^2, \qquad F = s^3 + \kappa s^2 t + f_1 st^2 + t^3,$$

and

$$u_0 = F^2, \quad u_1 = c_1 Q^3, \quad u_2 = c_2\, t^3 F, \quad u_3 = c_3\, s^3 F, \quad u_4 = c_4\, s^2 t^4,$$

binary forms of degree 6. Let $E_0, \dots, E_6$ be the coefficients of $\sum_k u_k$, $\mathcal{P}_6 \subseteq \mathbb{C}^8$ their common zero set, and J_6 the Jacobian matrix of $(E_0, \dots, E_6)$ with respect to the seven coordinates other than κ.

Proposition 4.31. *There is a point* $p_6 \in \mathcal{P}_6$ *with* $\kappa(p_6) = -(637 + 53i)/500$ *at which*

(i) J_6 *is invertible;*
(ii) $c_1c_2c_3c_4 \neq 0$*,* $q_0 \neq 0$*, and* Q *and* F *are coprime;*
(iii) *the residue functional of the chart* $(0, 4)$*, described below, does not vanish on the tangent vector* v_6 *of* $\mathcal{P}_6$ *at* p_6 *with* $d\kappa(v_6) = 1$*.*

The proof is a computation in interval arithmetic of the same kind as for Proposition 4.20, with 7 unknowns (Appendix B). The point p_6 is the unique zero of $E(\,\cdot\,,\kappa(p_6))$ in an explicit box of radius 10^{-95}, and the residue functional is enclosed in a disc of radius 10^{-89} around $1.1329911782 + 1.0686851712\,i$.

Corollary 4.32. *Let $\mathcal{P}_{6,1}$ be the irreducible component of $\mathcal{P}_6$ through p_6, and let $T_6 \subseteq \mathcal{P}_{6,1}$ be the set of points at which the conditions* (i) *and* (ii) *of Proposition 4.31 hold. Then T_6 is a smooth connected curve, the forms above are a block family of character $37\alpha' = (1, 34, 44, 56, 69)$ over T_6, $(0, 4)$ is a chart, and $I_{04}(p_6, v_6) \neq 0$.*

Proof. As in the proof of Corollary 4.21, T_6 is a nonempty Zariski-open subset of the irreducible curve $\mathcal{P}_{6,1}$, smooth and connected, with κ as a local coordinate at p_6. The blocks are s, t, Q and F, with exponents $(0, 0, 0, 3, 2)$, $(0, 0, 3, 0, 4)$, $(0, 3, 0, 0, 0)$ and $(2, 0, 1, 1, 0)$ in $(u_0, \dots, u_4)$ and, for the character $(1, 34, 44, 56, 69)$, weights

$$3 \cdot 56 + 2 \cdot 69 = 306, \qquad 3 \cdot 44 + 4 \cdot 69 = 408, \qquad 3 \cdot 34 = 102, \qquad 2 + 44 + 56 = 102.$$

Every u_k has degree 6. The forms Q and F are monic in s, so neither vanishes at $(1 : 0)$, the zero of t, and their values q_0 and 1 at $(0 : 1)$, the zero of s, do not vanish by (ii); with the rest of (ii) the four blocks are pairwise coprime, and the coefficients $1, c_1, c_2, c_3, c_4$ do not vanish. This proves (B1)–(B3). The forms $u_0 = F^2$ and $u_4 = c_4 s^2 t^4$ share no block, and F does not vanish at $(1 : 0)$, so $(0, 4)$ is a chart; the last assertion is (iii). □

The residue functional is computed as for Proposition 4.26. Here $A = s^3 t^4 QF$ and $u_0 u_1 u_2 u_3 u_4 = c_1 c_2 c_3 c_4\, s^5 t^7 Q^3 F^4$, so in the coordinate s

$$\eta_v = \frac{A\,\Delta_v}{u_0 u_4}\,ds = \frac{\mathcal{N}_v}{c_1 c_2 c_3 c_4\, s^2 Q^2 F^3}\,ds, \qquad \mathcal{N}_v = \det \begin{pmatrix} u_1 & u_2 & u_3 \\ \dot{u}_1 & \dot{u}_2 & \dot{u}_3 \\ u_1' & u_2' & u_3' \end{pmatrix}.$$

The zeros of u_0 are the roots of F. By Lemma 4.9(a), η_v is regular at $s = 0$, since the block s has weight $3m$, occurs twice in u_4 and not in u_0, and at the roots of Q, which occurs in neither. So $I_{04}(p, v)$ is the sum of all residues of η_v in $\mathbb{C}$. As $\mathcal{N}_v$ has degree at most 17 and the denominator has degree 15, $I_{04}(p, v)$ is the coefficient of s^{-1} in the expansion of η_v/ds in powers of s^{-1}. This is the quantity enclosed by the certificate, for $p = p_6$ and $v = v_6$.

The second family has degree 18. Let $(f, g, q_1, c_1, c_2, c_3, c_4, \kappa)$, with $f = (f_0, \dots, f_8)$ and $g = (g_1, \dots, g_5)$, be the coordinates of $\mathbb{C}^{20}$, and consider the forms

$$Q = s^2 + q_1 st + \kappa t^2, \qquad F = s^9 + \sum_{n<9} f_n s^n t^{9-n}, \qquad G = s^6 + \sum_{0<n<6} g_n s^n t^{6-n} + t^6,$$

and

$$u_0 = G^3, \quad u_1 = c_1\, s^6 G^2, \quad u_2 = c_2 F^2, \quad u_3 = c_3\, st^{17}, \quad u_4 = c_4\, Q^6 G,$$

binary forms of degree 18. Let $E_0, \dots, E_{18}$ be the coefficients of $\sum_k u_k$, $\mathcal{P}_{18} \subseteq \mathbb{C}^{20}$ their common zero set, and J_{18} the Jacobian matrix of $(E_0, \dots, E_{18})$ with respect to the 19 coordinates other than κ.

Proposition 4.33. *There is a point $p_{18} \in \mathcal{P}_{18}$ with $\kappa(p_{18}) = -(147 + 117i)/500$ at which*

(i) *J_{18} is invertible;*
(ii) *$c_1 c_2 c_3 c_4 \neq 0$, $\kappa f_0 \neq 0$, and Q, F and G are pairwise coprime;*
(iii) *the residue functional of the chart $(2, 3)$, described below, does not vanish on the tangent vector v_{18} of $\mathcal{P}_{18}$ at p_{18} with $d\kappa(v_{18}) = 1$.*

Again the proof is a computation in interval arithmetic, now with 19 unknowns (Appendix B). The point p_{18} is the unique zero of $E(\,\cdot\,,\kappa(p_{18}))$ in a box of radius 10^{-95}, and the residue functional is enclosed in a disc of radius 10^{-76} around $-2.4644283791 - 3.8828103549\,i$.

Corollary 4.34. *Let $\mathcal{P}_{18,1}$ be the irreducible component of $\mathcal{P}_{18}$ through p_{18}, and let $T_7 \subseteq \mathcal{P}_{18,1}$ be the set of points at which the conditions* (i) *and* (ii) *of Proposition 4.33 hold. Then T_7 is a smooth connected curve, the forms above are a block family of character $5\alpha = (5, 35, 51, 96, 17)$ over T_7, $(2, 3)$ is a chart, and $I_{23}(p_{18}, v_{18}) \neq 0$.*

Proof. As before, T_7 is a nonempty Zariski-open subset of the irreducible curve $\mathcal{P}_{18,1}$, smooth and connected. The blocks are Q, t, F, s and G, with exponents $(0,0,0,0,6)$, $(0,0,0,17,0)$, $(0,0,2,0,0)$, $(0,6,0,1,0)$ and $(3,2,0,0,1)$ and, for the character $(5, 35, 51, 96, 17)$, weights

$$6{\cdot}17 = 102, \qquad 17{\cdot}96 = 1632, \qquad 2{\cdot}51 = 102, \qquad 6{\cdot}35{+}96 = 306, \qquad 3{\cdot}5{+}2{\cdot}35{+}17 = 102.$$

Every u_k has degree 18. The forms Q, F and G are monic in s, so none of them vanishes at $(1:0)$, the zero of t, and their values κ, f_0 and 1 at $(0:1)$, the zero of s, do not vanish by (ii); with the rest of (ii) the five blocks are pairwise coprime, and the coefficients $1, c_1, c_2, c_3, c_4$ do not vanish. This proves (B1)–(B3). The forms $u_2 = c_2F^2$ and $u_3 = c_3st^{17}$ share no block, and F does not vanish at $(1:0)$, so $(2,3)$ is a chart; the last assertion is (iii). □

Here $A = Qt^{16}Fs^3G$ and $u_0u_1u_2u_3u_4 = c_1c_2c_3c_4\, s^7t^{17}Q^6F^2G^6$, so in the coordinate s

$$\eta_v = \frac{A\,\Delta_v}{u_2u_3}\,ds = \frac{\mathcal{N}_v}{c_1c_2c_3c_4\, s^4Q^5FG^5}\,ds, \qquad \mathcal{N}_v = \det\begin{pmatrix} u_0 & u_1 & u_4 \\ \dot{u}_0 & \dot{u}_1 & \dot{u}_4 \\ u_0' & u_1' & u_4' \end{pmatrix}.$$

The zeros of u_2 are the roots of F. By Lemma 4.9(a), η_v is regular at $s = 0$, since the block s has weight $3m$, occurs once in u_3 and not in u_2, and at the roots of Q and G, which occur in neither u_2 nor u_3. So $I_{23}(p, v)$ is the sum of all residues of η_v in $\mathbb{C}$, the coefficient of s^{-1} in the expansion of η_v/ds in powers of s^{-1}; here $\mathcal{N}_v$ has degree at most 53, as has the denominator. This is the quantity enclosed by the certificate, for $p = p_{18}$ and $v = v_{18}$.

4.9. **Proof of Theorem A.**

Proof of Theorem A. Degree 114. By Proposition 4.19 at $\lambda = 2$ and Theorem 4.16, the character $13\alpha_2$ has the property of Theorem 4.16; so has α_2, as $M_{13\alpha_2} = M_{\alpha_2}$. By Corollary 4.21 and Theorem 4.16, so has α_1. By Lemma 4.6(a) and Proposition 4.17, the multiset β of Lemma 4.6(b) is algebraic. By Corollary 4.5, $B_{114} = S_{114} + \mathbb{Z}[\beta]$, and Lemma 4.2 gives the Hodge conjecture for X^n_{114} for every n, and the algebraicity of every balanced multiset of an even number of nonzero elements of $\mathbb{Z}/114$.

Degree 57. Let A' be a balanced multiset of an even number of nonzero elements of $\mathbb{Z}/57$. Then $2A' \subseteq \mathbb{Z}/114$ is balanced, hence algebraic, and so A' is algebraic by Lemma 3.22(d). As at the end of the proof of Lemma 4.2, this gives the Hodge conjecture for X^n_{57} for every n.

Degree 44. By Corollary 4.24 and Theorem 4.16, the character α of Lemma 4.22 has the property of Theorem 4.16. By Lemma 4.22(a) and Proposition 4.18, the multiset β of Lemma 4.22(b) is algebraic. By Corollary 4.5, $B_{44} = S_{44} + \mathbb{Z}[\beta]$, and Lemma 4.2 gives the Hodge conjecture for X^n_{44} for every n, and the algebraicity of every balanced multiset of an even number of nonzero elements of $\mathbb{Z}/44$.

Degree 52. By Corollary 4.27 and Theorem 4.16, the character α of Lemma 4.25 has the property of Theorem 4.16. By Corollary 4.29 and Theorem 4.16, so has $9\alpha'$, and hence α', as $M_{9\alpha'} = M_{\alpha'}$. By Lemma 4.25(a) and Proposition 4.17, the multiset β of Lemma 4.25(b) is algebraic. By Corollary 4.5, $B_{52} = S_{52} + \mathbb{Z}[\beta]$, and Lemma 4.2 gives the Hodge conjecture for

X^n_{52} for every n, and the algebraicity of every balanced multiset of an even number of nonzero elements of $\mathbb{Z}/52$.

Degree 102. By Corollary 4.32 and Theorem 4.16, the character $37\alpha'$ of Lemma 4.30 has the property of Theorem 4.16, and hence so has α', as $M_{37\alpha'} = M_{\alpha'}$. By Corollary 4.34 and Theorem 4.16, so has 5α, and hence α. By Lemma 4.30(a) and Proposition 4.17, the multiset β of Lemma 4.30(b) is algebraic. By Corollary 4.5, $B_{102} = S_{102} + \mathbb{Z}[\beta]$, and Lemma 4.2 gives the Hodge conjecture for X^n_{102} for every n, and the algebraicity of every balanced multiset of an even number of nonzero elements of $\mathbb{Z}/102$.

Degree 51. As for the degree 57: if A' is a balanced multiset of an even number of nonzero elements of $\mathbb{Z}/51$, then $2A' \subseteq \mathbb{Z}/102$ is balanced, hence algebraic, and so A' is algebraic by Lemma 3.22(d). This gives the Hodge conjecture for X^n_{51} for every n.

The general case. Let m be $2^a3^b19^c$ with $a \le 1$, or 2^a11^b, or 2^a13^b, or $2^a3^b17^c$ with $a \le 1$. For $m \le 2$ the variety X^n_m is a linear space or a quadric, whose cohomology is spanned by classes of linear subspaces; let $m \ge 3$. The elements of P that divide m are among 3 and 19 in the first case, among 4 and 11 in the second, among 4 and 13 in the third, and among 3 and 17 in the fourth. So $\mathcal{Q}(m)$ is empty, or it is $\{q\}$ with $q = 57$, 44, 52 or 51, according as q divides m. By Theorem 4.4, $B_m = S_m$ when $\mathcal{Q}(m)$ is empty, and $B_m = S_m + \sum_t \mathbb{Z}\, t \cdot (m/q)_*\xi_q$ when $\mathcal{Q}(m) = \{q\}$. The element $\xi_q \in B_q$ is an integral combination of classes of balanced multisets with an even number of elements, which are algebraic by the case of degree 57, 44, 52 or 51; their images under $(m/q)_*$ and then under a unit are classes of algebraic balanced multisets, by Lemma 3.22(c),(d). Lemma 4.2 gives the Hodge conjecture for X^n_m for every n.

Products and abelian varieties. By [Shi79, Theorem IV], the Hodge conjecture for $X^{r_1}_m \times \cdots \times X^{r_k}_m$ follows from the Hodge conjecture for X^r_m for the even $r \le r_1 + \cdots + r_k + 2(k-1)$. By [Aok00, Corollary 3.2], the Hodge conjecture for X^n_m for every n implies it for every abelian variety of Fermat type of degree m. □

Remark 4.35. (a) For the degrees $2^a3^b19^c$ with $a \le 1$ and $57 \nmid m$, Theorem A is due to Aoki [Aok00, Theorem 0.1], since m is then of the form 2^a3^b, 19^c or $2 \cdot 19^c$. For $m = 57$, the Hodge conjecture for X^n_{57} is stated in [MMRV26, Theorem 1.1]. Its proof writes ξ_{57}, modulo S_{114}, as a balanced sextuple of residues modulo 114 [MMRV26, Table 4], a Hodge class on X^4_{114}, and declares it algebraic by Kang's statement [Kan16, Corollary 3.2], which rests on the step discussed in Remark 3.21; the proof above does not use it. By Aoki's reduction, the case $m = 57$ implies the others, so the same remark applies to every m divisible by 57; for $m \ne 57$, the first of which are 114, 171, 342 and 513, the theorem has not been stated before. Jumagulov's census of the Fermat fourfolds of even degree at most 250 lists the degree 114 as open [Jum26a].

(b) The census [Jum26a], a computer-assisted preprint whose lattice computation we have repeated, finds that for even $m \le 250$ every Hodge class on X^4_m is algebraic by linear subspaces, Aoki's cycles and the inductive structure, except for classes attached to the levels 70, 110 and 114 and their multiples 140, 210, 220 and 228. Peterson's theorem covers 70, 140 and 210 [Pet26], and Theorem A, with Lemma 3.22(d), covers 114 and 228. So the even degrees up to 250 for which the Hodge conjecture for X^4_m remains open are 110 and 220, where the classes left over are those the census attaches to the level 110. They lie in the class of $2_*\xi_{55}$, and we have not found block families for them; see Remark 4.38(b) for the reason.

(c) The characters α_1, α_2 and the two families were found by computer searches over the level-one characters of X^3_{114} and over the possible block structures; the proofs do not depend on these searches. The searches found no family of degree less than 24 for the orbit of α_1, and no pair of characters as in Lemma 4.6 for which both families have degree 3, like the first one. With such a pair, no step of the proof would need a computer.

Remark 4.36. (a) When $44 \nmid m$, a degree $2^a 11^b$ is a power of 2, or 11^b, or $2 \cdot 11^b$, and Theorem A is due to Aoki [Aok87, Aok00]. The preprint [MMRV26] treats all dimensions in every degree below 65 except 44, 51 and 52. So for the degrees divisible by 44, the first of which are 44, 88, 176, 352 and 484, Theorem A has not been proved before, apart from the claim of [Ope26b], which covers every Fermat variety (Theorem E). For the fourfold X^4_{44} nothing new happens: by the paragraph after Lemma 4.22, linear subspaces and Aoki's cycles suffice for it, in agreement with the census [Jum26a], and the first class of degree 44 that needs the family of Proposition 4.23 lies on the sixfold.

(b) The characters α and b and the family were found by computer searches over the level-one characters of X^3_{44}, the characters of the Fermat curve of degree 44 and the possible block structures; the proofs do not depend on these searches.

Remark 4.37. (a) When $52 \nmid m$, a degree $2^a 13^b$ is a power of 2, or 13^b, or $2 \cdot 13^b$, and Theorem A is due to Aoki [Aok87, Aok00]. Since [MMRV26] leaves out the degrees 44, 51 and 52, Theorem A has not been proved before for the degrees divisible by 52, the first of which are 52, 104, 208, 416 and 676, apart from the claim of [Ope26b]. For the fourfold X^4_{52}, linear subspaces and Aoki's cycles suffice, by the paragraph after Lemma 4.25, in agreement with the census [Jum26a].

(b) The characters and the two families were found by computer searches over the level-one characters of X^3_{52} and the possible block structures; the proofs do not depend on these searches. In every pair of characters as in Lemma 4.25 that the searches found, one character lies in the orbit of α or of one of $(1, 4, 17, 37, 45)$, $(1, 5, 16, 33, 49)$, $(1, 5, 17, 32, 49)$ and $(1, 5, 24, 33, 41)$, and for none of these five orbits did they find a family of degree less than 14. The orbit of $(1, 12, 17, 28, 46)$ has a family of cubics, like that of Proposition 4.19, whose residue is an explicit rational function of the parameter; but its only partner is $(1, 5, 24, 33, 41)$, for which the smallest family we found has degree 19. For the threefold characters that occur in joins with a character of the Fermat curve, as in degree 44, the searches found no block family at all.

Remark 4.38. (a) When $51 \nmid m$, a degree $2^a 3^b 17^c$ with $a \leq 1$ is of the form $2^a 3^b$, 17^c or $2 \cdot 17^c$, and Theorem A is due to Aoki [Aok87, Aok00]. Since [MMRV26] leaves out the degrees 44, 51 and 52, Theorem A has not been proved before for the degrees $2^a 3^b 17^c$ with $a \leq 1$ that are divisible by 51, the first of which are 51, 102, 153, 306 and 459, apart from the claim of [Ope26b]. For the fourfold X^4_{102}, linear subspaces and Aoki's cycles suffice, by the paragraph after Lemma 4.30, in agreement with the census [Jum26a]; the fourfold X^4_{51} is covered by Theorem B.

(b) The characters and the two families were found by computer searches over the pairs of level-one characters of X^3_{102} as in Lemma 4.30 and over the possible block structures; the proofs do not depend on these searches. Many characters fail for the following reason. Let $a \in \mathfrak{A}^3_m$ with $\langle a \rangle = 2$, and suppose that some $t \in \mathbb{Z}/m$, not a unit, has $ta_k \neq 0$ for every k and $\sum_k \overline{ta_k} = m$, where $\overline{x} \in \{1, \dots, m-1\}$ represents x. Put $y_k = \overline{ta_k}/m$. For a block family of character a, the weight condition (B2) gives $\sum_k y_k e_{kl} \in \mathbb{Z}$, and this number is positive; hence

$$\sum_l \delta_l \leq \sum_l \delta_l \sum_k y_k e_{kl} = \sum_k y_k D = D.$$

Up to scaling the blocks and the c_k and up to the action of $\mathrm{PGL}_2(\mathbb{C})$, the forms have $\sum_l \delta_l + 1$ free coefficients, subject to the $D + 1$ equations (B3); so the solutions form a set of expected dimension $\sum_l \delta_l - D \leq 0$, and a family with a nonzero residue functional, which cannot be constant up to $\mathrm{PGL}_2(\mathbb{C})$, needs a component of larger dimension than expected. In the families of Propositions 4.31 and 4.33 the block degrees add up to $7 > 6$ and $19 > 18$. In each pair of characters of the Fermat threefold that our searches found at a level $M \in \{110, 165, 220\}$ whose

join represents the class of $(M/55)_*\xi_{55}$, one of the two characters has such a t; this is why we found no block families there (Remark 4.35(b)).

5. Abelian varieties: CM points and propagation

5.1. **The statement (CM).** Let A be an abelian variety of CM type. Its Hodge group is a torus, and after extending scalars to a Galois CM field E that splits it, $H^1(A)$ is a sum of lines indexed by embeddings, each of type $(1,0)$ or $(0,1)$. Deligne and André showed that every Hodge class on A is a sum of pull-backs, along homomorphisms $A \to A_\Delta$, of Weil classes on abelian varieties A_Δ of split Weil type [Mil20, Theorem 1]. So (CM) is equivalent to the algebraicity of these split Weil classes, and it is open in the refereed literature.

The preprint [Ope26b] claims (CM) [Ope26b, Theorem 1.1]. Its proof has four steps. (1) The theorem is reduced [Ope26b, Proposition 2.6] to the algebraicity of a line of type $(2,2)$ on a product of four CM abelian varieties $B(v_1),\dots,B(v_4)$, attached to odd sign functions $v_i\colon \mathrm{Gal}(E/\mathbb{Q}) \to \{\pm 1\}$ with $v_1+v_2 = v_3+v_4$ [Ope26b, Theorem 2.5]. (2) That line is shown to be algebraic as soon as four degree-one classes on a surface, coming from the four CM Hodge structures, have a nonzero mixed period [Ope26b, Proposition 3.1]. (3) Such a period is produced by theta forms on compact quotients of the complex two-ball [Ope26b, §4]. (4) The theta classes are shown to come from the prescribed CM Hodge structures, through Hecke operators and a Frobenius relation on the reductions of the ball quotients [Ope26b, Theorem 5.1 and Corollary 6.7].

We have checked the combinatorial part of step (1): its core is Lemma 5.1 below, and its contraction formula [Ope26b, (2.6)] carries the sign $(-1)^{p(p-1)/2}$, which we recompute in Appendix A. We have not checked steps (2) to (4), and the claim rests on step (4). Accordingly, every result of this paper that uses (CM) states it as a hypothesis.

Lemma 5.1 (Two-row switches)**.** *Let G be a finite set with a fixed-point-free involution c, let $s = |G|/2$, and call $v\colon G \to \{\pm 1\}$* odd *if $v(cg) = -v(g)$. Let $x = (x_1,\dots,x_p)$ and $y = (y_1,\dots,y_p)$ be lists of odd functions with $\sum_i x_i = \sum_i y_i$. Then there is a chain of lists of odd functions $x = x^{(0)}, x^{(1)}, \dots, x^{(N)} = y$ with $N \le s\lfloor p/2 \rfloor$, in which $x^{(k+1)}$ differs from $x^{(k)}$ in exactly two entries $i \ne j$, and $x_i^{(k+1)} + x_j^{(k+1)} = x_i^{(k)} + x_j^{(k)}$.*

Proof. Choose a set T of representatives of the orbits of c; an odd function is determined by its values on T. Treat the elements $\tau \in T$ one at a time. Let P_τ be the set of indices i with $x_i(\tau) = 1$ and $y_i(\tau) = -1$, and N_τ the set with $x_i(\tau) = -1$ and $y_i(\tau) = 1$. Then $\sum_i x_i(\tau) - \sum_i y_i(\tau) = 2|P_\tau| - 2|N_\tau|$, so $|P_\tau| = |N_\tau| \le \lfloor p/2 \rfloor$. Pair the two sets, and for each pair (i,j) exchange the values of the i-th and j-th functions at τ and at $c\tau$. Each exchange keeps the functions odd, changes exactly the entries i and j, and keeps their sum, since it only swaps two values at each of two points. After the pairs for τ are exchanged, the list agrees with y at τ and $c\tau$, and the later steps do not change these values. □

In [Ope26b] the lists come from the types of the factors of a Hodge monomial, a two-row switch is exactly a relation $v_1+v_2 = v_3+v_4$, and the algebraic tensors of consecutive switches are composed by contracting the middle factors against an ample class [Ope26b, (2.6)].[9] Nothing in this combinatorial part depends on the geometry of steps (2) to (4).

[9] The contraction multiplies by $(-1)^{p(p-1)/2}\prod_i q_i$, where the sign comes from regrouping $\beta_1\cdots\beta_p\gamma_1\cdots\gamma_p$ as $\beta_1\gamma_1\cdots\beta_p\gamma_p$ for classes of degree one, and $q_i = \int_{B(b_i)} \beta_i \wedge \gamma_i \wedge \ell_i^{s-1}$ is nonzero by the Hodge–Riemann relations, because γ_i is a nonzero multiple of $\bar\beta_i$.

5.2. **Propagation from CM points.**

Lemma 5.2 (CM points are dense)**.** *Let G be a connected reductive group over $\mathbb{Q}$, let $u_0\colon \mathrm{U}(1) \to G_{\mathbb{R}}$ be a homomorphism, and let $\mathcal{D}$ be its orbit under conjugation by $G(\mathbb{R})^+$. The points $u \in \mathcal{D}$ with $u(\mathrm{U}(1)) \subseteq T_{\mathbb{R}}$ for a maximal torus T of G defined over $\mathbb{Q}$ are dense in $\mathcal{D}$.*

Proof. Let $u_1 \in \mathcal{D}$ and let W be a neighbourhood of u_1. A maximal torus T_1 of $G_{\mathbb{R}}$ contains the torus $u_1(\mathrm{U}(1))$; let $\mathfrak{t}_1$ be its Lie algebra and $Y_1 \in \mathfrak{t}_1$ a regular element. The map $\phi\colon G(\mathbb{R})^+ \times \mathfrak{t}_1 \to \mathfrak{g}_{\mathbb{R}}$, $(g, Y) \mapsto \mathrm{Ad}(g)Y$, has differential $(Z, Y') \mapsto [Z, Y_1] + Y'$ at $(1, Y_1)$, which is onto because $\mathfrak{g}_{\mathbb{R}} = \mathfrak{t}_1 \oplus [\mathfrak{g}_{\mathbb{R}}, Y_1]$ for a regular semisimple Y_1. So for small neighbourhoods N_1 of 1, with $N_1 u_1 N_1^{-1} \subseteq W$, and N_2 of Y_1 consisting of regular elements, $\phi(N_1 \times N_2)$ contains a nonempty open set, and hence a rational element $Y = \mathrm{Ad}(g)Y'$ with $g \in N_1$, $Y' \in N_2$. The identity component T of the centralizer of Y is a maximal torus defined over $\mathbb{Q}$, and $T_{\mathbb{R}} = gT_1g^{-1}$. Then $u = gu_1g^{-1} \in W$ and $u(\mathrm{U}(1)) \subseteq T_{\mathbb{R}}$. □

Proposition 5.3 (A CM member through every Hodge class)**.** *Let A_0 be an abelian variety and γ a Hodge class of degree $2p$ on A_0. There are a polarized abelian scheme $\pi\colon \mathcal{A} \to S$ over a smooth connected quasi-projective variety, a global section ξ of $R^{2p}\pi_*\mathbb{Q}$, and points $s_0, c \in S$, such that $\mathcal{A}_{s_0} \cong A_0$ with $\xi_{s_0} = \gamma$, ξ_s is a Hodge class for every s, and $\mathcal{A}_c$ is of CM type.*

Proof. Replacing γ by a multiple, we may assume it integral. Fix a polarization of A_0 with form ψ on $V = H_1(A_0, \mathbb{Q})$, and a level $N \geq 3$. The complex structures J on $V_{\mathbb{R}}$ that are compatible with ψ and positive for it form the Siegel space $\mathcal{H}$, and its quotient $\mathcal{S}$ by the principal congruence subgroup of level N in $\mathrm{Sp}(H_1(A_0, \mathbb{Z}), \psi)$ is a smooth quasi-projective variety with a universal polarized abelian scheme $f\colon \mathcal{A}_g \to \mathcal{S}$ [MFK94, Chapter 7]. Let $\mathbb{V} = R^{2p}f_*\mathbb{Z}$. By the theorem of Cattani, Deligne and Kaplan [CDK95, Theorem 1.1], the set of pairs (a, v), with $a \in \mathcal{S}$ and $v \in \mathbb{V}_a$ an integral Hodge class with the same self-intersection as γ, is a closed algebraic subvariety of the total space of the vector bundle $\mathbb{V} \otimes \mathcal{O}_{\mathcal{S}}$, finite over $\mathcal{S}$; let $\mathcal{C}$ be its connected component through the point (a_0, γ) given by A_0 and a lift $J_0 \in \mathcal{H}$.

Let $G \subseteq \mathrm{Sp}(V, \psi)$ be the Hodge group of A_0, the smallest algebraic subgroup over $\mathbb{Q}$ whose real points contain the image of the homomorphism $u_0\colon \mathrm{U}(1) \to \mathrm{GL}(V_{\mathbb{R}})$, $e^{i\theta} \mapsto \cos\theta + J_0 \sin\theta$. It is connected and reductive, and it fixes every Hodge class of A_0, in particular γ [Del82, §3]. For $g \in G(\mathbb{R})^+$ the structure $J = gJ_0g^{-1}$ lies in $\mathcal{H}$, and the Hodge group of A_J is contained in G, so γ is a Hodge class on A_J. The orbit $\mathcal{D} = \{gJ_0g^{-1}\}$ is connected, so its image $J \mapsto (a_J, \gamma)$ lies in $\mathcal{C}$. By Lemma 5.2 there are $J_k \in \mathcal{D}$ converging to J_0 such that $u_{J_k}(\mathrm{U}(1))$ lies in a torus defined over $\mathbb{Q}$; the Hodge group of A_{J_k} is then commutative, and A_{J_k} is of CM type.[10] One of the finitely many irreducible components of $\mathcal{C}$, say $\mathcal{C}_0$, contains infinitely many of the points (a_{J_k}, γ), and, being closed, also their limit (a_0, γ).

Let $S \to \mathcal{C}_0$ be a resolution of singularities [Hir64], so that S is smooth, connected and quasi-projective, and let $\mathcal{A} \to S$ be the pull-back of $\mathcal{A}_g$. The total space of a local system is a covering space of the base, so $(a, v) \mapsto v$ is a flat section on $\mathcal{C}_0$, and its pull-back is a global section ξ of $R^{2p}\pi_*\mathbb{Z}$ that is a Hodge class at every point. Take s_0 over (a_0, γ) and c over some (a_{J_k}, γ). □

Proposition 5.4. (F2) *holds if and only if both* (CM) *and* (IP) *hold.*

Proof. (F2) contains (CM), and it implies (IP): if ξ_c is algebraic, then every ξ_s is a Hodge class by Lemma 2.10, hence algebraic. Conversely, let γ be a Hodge class on an abelian variety A_0,

[10] The torus T lies in a maximal torus T' of $\mathrm{GL}(V)$ over $\mathbb{Q}$, the units of a commutative semisimple subalgebra $L \subseteq \mathrm{End}(V)$ of dimension $2g$. Since L commutes with $T' \supseteq \mathrm{Hg}(A_{J_k})$, it lies in the commutant of the Hodge group, which is $\mathrm{End}(A_{J_k}) \otimes \mathbb{Q}$.

and take π, ξ, s_0 and c from Proposition 5.3. Then ξ_c is a Hodge class on an abelian variety of CM type, so it is algebraic by (CM), and by (IP) the class $\gamma = \xi_{s_0}$ is algebraic. □

The proposition separates the two inputs of the argument of Deligne and André as Milne records it [Mil20, §5]: the classes at CM points, and their transport along families. The variational conjecture (V) supplies both at once [Mil20, Theorem 3 and Remark 2], because the split Weil classes are algebraic at a suitable member of each family [Del82, 4.8]. (IP) supplies only the transport, and it does so only from CM members.

5.3. **Proof of Theorem C.** By Lemma 2.11, (i) implies each of (ii)–(vi). If (F2) holds, every γ in the definition of $\mathrm{Ab}^p(X)$ is algebraic, so $\mathrm{Ab}^p(X) = \mathrm{Alg}^p(X)$, and (F3′) becomes (HC); so (ii) implies (i). (L) implies (F2) by the theorem of Abdulali and André [Mil20, Theorem 4][11], (V) implies (F2) [Mil20, Theorem 3 and Remark 2], and (CM) with (IP) implies (F2) by Proposition 5.4; so (iii), (iv) and (v) imply (ii). Finally (vi) implies (i) by Proposition 2.8. □

5.4. **Weil families and degrees at CM points.** Let K be an imaginary quadratic field and $n \geq 1$. An abelian variety A of dimension $2n$ with an action of K is *of Weil type* if both embeddings of K occur with multiplicity n on $H^{1,0}(A)$. Then the *Weil plane* $W(A) = \bigwedge_K^{2n} H^1(A, \mathbb{Q}) \subseteq H^{2n}(A, \mathbb{Q})$ is a $\mathbb{Q}$-plane of Hodge classes [Wei79], [Del82, §4]. For a polarization whose Rosati involution induces complex conjugation on K, the Riemann form is $\mathrm{Tr}_{K/\mathbb{Q}}(fH)$ for a K-hermitian form H of signature (n, n), and A is *of split Weil type* if H has a totally isotropic K-subspace of dimension n [Mil20, §2.1].

Fix such data and a level $N \geq 3$. The abelian varieties of split Weil type for (K, n) with this hermitian form and level structure are parametrized by a smooth connected quasi-projective variety S, a quotient $\Gamma\backslash\mathcal{D}$ of the hermitian symmetric domain $\mathcal{D}$ of $\mathrm{SU}(n, n)$, of dimension n^2, by a neat arithmetic subgroup Γ of the unitary group; it is a connected Shimura variety of PEL type and carries a polarized abelian scheme $\pi\colon \mathcal{A} \to S$ with an action of $\mathcal{O}_K$ [Wei79, vG94].[12] The special points of S are the members of CM type. Fix a nonzero global section ξ of the Weil planes. An element $\alpha \in \mathcal{O}_K$ acts on $\mathcal{A}_s$, and its pull-back multiplies the K-line $W(\mathcal{A}_s)$ by α^{2n}. Choose α with $\alpha^{2n} \notin \mathbb{Q}$.[13] Then α^* moves any nonzero class of $W(\mathcal{A}_s)$ to a class that spans $W(\mathcal{A}_s)$ with it, and pull-back by an endomorphism preserves algebraic classes; so $W(\mathcal{A}_s)$ consists of algebraic classes exactly when ξ_s is algebraic.

Definition 5.5. Let $\Sigma \subseteq S$ be the set of s at which ξ_s is algebraic. For $d \geq 1$ let Σ_d be the set of s for which there are effective algebraic cycles Z^+, Z^- of codimension n on $\mathcal{A}_s$, each of degree at most d for the polarization, and an integer $1 \leq e \leq d$, with $e\,\xi_s = \mathrm{cl}(Z^+) - \mathrm{cl}(Z^-)$.

Lemma 5.6. *Each Σ_d is Zariski closed in S, $\Sigma_d \subseteq \Sigma_{d+1}$, and $\Sigma = \bigcup_d \Sigma_d$.*

Proof. The last two assertions are clear. Effective cycles of codimension n and degree at most d in the fibres of π are parametrized by finitely many relative Chow varieties $\mathcal{C}_i \to S$, which are projective over S [Kol96, Chapter I, §§3–4]. On a connected component of $\mathcal{C}_i \times_S \mathcal{C}_j$ the class $\mathrm{cl}(Z^+) - \mathrm{cl}(Z^-)$ of the fibre is a flat section of the pulled-back local system, and so is $e\,\xi$. Two flat sections over a connected base that agree at one point agree everywhere, so the set where

[11] Milne's hypothesis is the Lefschetz standard conjecture in the form that L^{d-2i} maps the algebraic classes of degree $2i$ onto those of degree $2d-2i$. It follows from (L), since the inverse of L^{d-2i} is then induced by an algebraic cycle and maps algebraic classes to algebraic classes.

[12] The unitary group acts on the Weil plane through the determinant. For $\gamma \in \Gamma$ the determinant is a unit of $\mathcal{O}_K$ of norm one, hence a root of unity, and it is 1 because Γ is neat. So the Weil planes of the members form a trivial local system on S, and every Weil class on one member extends to a global section.

[13] The $\alpha \in K \otimes \mathbb{R} = \mathbb{C}$ with α^{2n} real lie on $2n$ lines through 0, and the lattice $\mathcal{O}_K$ is not contained in finitely many lines.

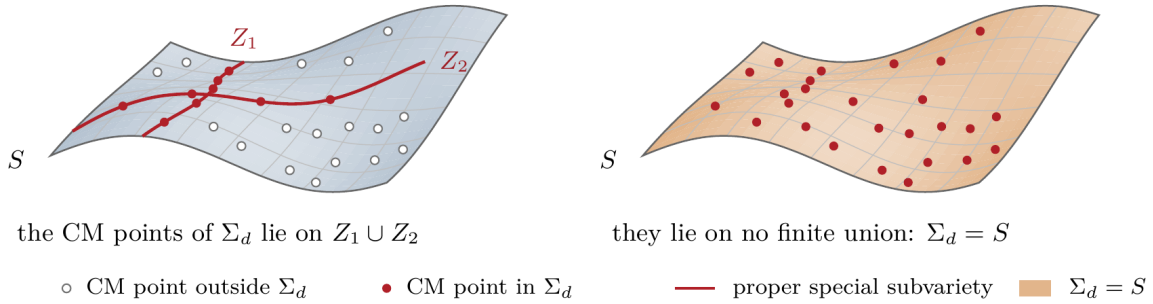


FIGURE 3. The criterion of Theorem F, drawn schematically (S has dimension n^2). CM points are dense in S. On the left the CM points of Σ_d lie on two proper special subvarieties Z_1 and Z_2, and condition (iii) fails for this d. On the right they lie on no finite union of proper special subvarieties; by the André–Oort theorem their Zariski closure is then S, and $\Sigma_d = S$.

they agree is a union of connected components. Its image in S is closed, because the projection is proper. Σ_d is the union of finitely many such images. □

Proof of Theorem F. The first sentence is Lemma 5.6. (ii) implies (i) trivially. If (i) holds, S^{an} is the countable union of the closed sets Σ_d, so by the Baire category theorem one of them has an interior point; a Zariski closed subset of the irreducible variety S with an interior point is S, which gives (ii). If (ii) holds, the CM points in $\Sigma_d = S$ are dense in S^{an} by Lemma 5.2, so they lie in no finite union of proper closed subvarieties, which gives (iii). Assume (iii), and let Z be the Zariski closure of the set of CM points in Σ_d; it lies in Σ_d. Every irreducible component of Z contains a Zariski dense set of special points, so it is a special subvariety, by the André–Oort conjecture, which is a theorem [PST21]. By (iii) one of these components is not proper, so $S \subseteq Z \subseteq \Sigma_d$, which is (ii). Finally, at a CM point ξ_s is a Hodge class on an abelian variety of CM type, so under (CM) it is algebraic. □

Remark 5.7. Without any hypothesis Σ is dense. It contains the members K-isogenous to $E^n \times \bar{E}^n$, which is of split Weil type and whose Weil classes are polynomials in divisor classes (Lemma 6.10); and with every point it contains its Hecke orbit, which is dense. Algebraic representatives move along isogenies by push-forward and pull-back, and their degree grows with the degree of the isogeny. (CM) adds every CM point, but it gives no bound on the degrees, and by Theorem F a bound on a set of CM points that is not trapped in finitely many proper Shimura subvarieties is exactly what remains. Neither [Ope26b] nor any other source we know gives such a bound.

6. Weil classes and Schoen's subvariety

In this section K is an imaginary quadratic field with a fixed embedding $\chi\colon K \to \mathbb{C}$ and its conjugate $\bar{\chi}$. Let A be an abelian variety of Weil type for K of dimension $2n$ (Section 5.4) on which $\mathcal{O}_K$ acts, with a polarization class η whose Rosati involution induces complex conjugation on K. Endomorphisms act on cohomology by pull-back, which makes $V = H^1(A, \mathbb{Q})$ a K-vector space of dimension $2n$, and

$$V_{\mathbb{C}} = V \otimes_{\mathbb{Q}} \mathbb{C} = V_{\chi} \oplus V_{\bar{\chi}},$$

where $\alpha \in K$ acts on V_{χ} by $\chi(\alpha)$ and on $V_{\bar{\chi}}$ by $\bar{\chi}(\alpha)$. Each summand has dimension $2n$ and, because A is of Weil type, meets $H^{1,0}$ and $H^{0,1}$ in subspaces of dimension n. The Weil plane

satisfies

$$W(A)\otimes\mathbb{C} = \textstyle\bigwedge^{2n} V_\chi \oplus \bigwedge^{2n} V_{\bar\chi} \subseteq H^{2n}(A,\mathbb{C})$$

[Del82, §4], and both lines are of type (n,n). We call A *Weil-algebraic* if every class in $W(A)$ is algebraic. Two facts are used repeatedly. First, an endomorphism $\alpha\in\mathcal{O}_K$ multiplies η by the norm $N(\alpha)=\chi(\alpha)\bar\chi(\alpha)$, since its Rosati adjoint is $\bar\alpha$; comparing with the action of α on $\bigwedge^2 V_\mathbb{C} = \bigwedge^2 V_\chi \oplus (V_\chi\otimes V_{\bar\chi}) \oplus \bigwedge^2 V_{\bar\chi}$ for any $\alpha\notin\mathbb{Q}$ shows that $\eta\in V_\chi\otimes V_{\bar\chi}$, and hence

$$\eta^k \in \textstyle\bigwedge^k V_\chi \otimes \bigwedge^k V_{\bar\chi}. \tag{5}$$

Second, if $W(A)$ contains one nonzero algebraic class, then A is Weil-algebraic, by the argument given for ξ_s in Section 5.4.

The Hodge group $\mathrm{Hg}(A)$ commutes with K and preserves the Riemann form, so it lies in the unitary group $\mathrm{U}(V,H)$ of the hermitian form H of the polarization. It fixes the Hodge classes in $W(A)$, on which $\mathrm{U}(V,H)$ acts through the determinant, so $\mathrm{Hg}(A)\subseteq\mathrm{SU}(V,H)$. We say that A has *maximal Hodge group* if equality holds.

6.1. **The Hodge ring of a general member.**

Proposition 6.1. *If* $\mathrm{Hg}(A)=\mathrm{SU}(V,H)$*, then* $\mathrm{Hdg}^k(A)=\mathbb{Q}\,\eta^k$ *for* $k\neq n$*, and*

$$\mathrm{Hdg}^n(A) = \mathbb{Q}\,\eta^n \oplus W(A).$$

Moreover $\eta^n\cup w=0$ *for every* $w\in W(A)$*.*

Proof. The Hodge classes in $H^{2k}(A,\mathbb{Q})=\bigwedge^{2k}V$ are the invariants of $\mathrm{Hg}(A)$ [Del82, §3]. Over $\mathbb{C}$ the group $\mathrm{SU}(V,H)$ becomes $\mathrm{SL}(V_\chi)$, acting on V_χ by the standard representation and on $V_{\bar\chi}$ by its dual, because the Riemann form pairs $V_{\bar\chi}$ perfectly with V_χ. The invariants in

$$\textstyle\bigwedge^{2k} V_\mathbb{C} = \bigoplus_{a+b=2k} \bigwedge^a V_\chi \otimes \bigwedge^b V_\chi^\vee = \bigoplus_{a+b=2k} \mathrm{Hom}(\bigwedge^b V_\chi, \bigwedge^a V_\chi)$$

form a line in each summand with $a=b$ or $\{a,b\}=\{0,2n\}$, and vanish in the others, because the representations $\bigwedge^a V_\chi$, $0\le a\le 2n$, are irreducible and pairwise non-isomorphic except for $\bigwedge^0\cong\bigwedge^{2n}$.[14] By (5) the line for $a=b=k$ contains η^k, which is nonzero for $k\le 2n$ because η^{2n} is a positive multiple of the orientation class. In degree $2n$ the two remaining lines span $W(A)\otimes\mathbb{C}$. This proves the first two assertions. By (5), $\eta^n\cup\bigwedge^{2n}V_\chi$ lies in $\bigwedge^{3n}V_\chi\otimes\bigwedge^n V_{\bar\chi}=0$, and likewise for $V_{\bar\chi}$. □

6.2. **What is known.** For $n=1$ the Weil plane lies in H^2 and Theorem 2.3 applies. For $n=2$ the hermitian form is split exactly when its discriminant is trivial, and Floccari and Fu prove that the Weil classes of every Weil fourfold of discriminant one are algebraic [FF26]. Schoen proves algebraicity for the Weil fourfolds of trivial discriminant for $\mathbb{Q}(\sqrt{-3})$ and for the Weil sixfolds of a family for $\mathbb{Q}(\sqrt{-3})$ in which his Prym varieties are dense [Sch88, Sch98], and for further families of fourfolds [Sch98]; Koike treats families for $\mathbb{Q}(\sqrt{-1})$ built in the same way [Koi04]. Markman's preprint proves algebraicity for every Weil fourfold and every split Weil sixfold [Mar25a]. Since the Hodge ring of an abelian variety of dimension at most five is generated by divisor classes and the Weil classes of its quotients [MZ99], the fourfold case gives (F2) in dimension at most five. For $2n\ge 8$ we know of no family of Weil type whose general member has maximal Hodge group and on which the Weil classes are known to be algebraic at every member.

[14] The representation $\bigwedge^a$ of SL_{2n} is irreducible with highest weight the a-th fundamental weight for $1\le a\le 2n-1$, and these weights are distinct; $\bigwedge^0$ and $\bigwedge^{2n}$ are trivial.

6.3. **Semiregularity.** Let Z be a smooth projective variety of dimension N and $\mathcal{E}$ a coherent sheaf on Z, with Atiyah class $\mathrm{at}(\mathcal{E}) \in \mathrm{Ext}^1(\mathcal{E}, \mathcal{E} \otimes \Omega^1_Z)$. The *semiregularity map* of Buchweitz and Flenner is

$$\sigma_{\mathcal{E}} \colon \mathrm{Ext}^2(\mathcal{E}, \mathcal{E}) \longrightarrow \bigoplus_{q=0}^{N-2} H^{q+2}(Z, \Omega^q_Z), \qquad e \longmapsto \Big(\mathrm{tr}\big(\mathrm{at}(\mathcal{E})^q \circ e\big)/q!\Big)_q,$$

and $\mathcal{E}$ is *semiregular* if $\sigma_{\mathcal{E}}$ is injective [BF03]. A subvariety $Y \subseteq Z$ is semiregular if $\mathcal{O}_Y$ is.[15] The point of the definition is the following theorem. If an infinitesimal deformation of Z keeps $\mathrm{ch}(\mathcal{E})$ of Hodge type, the obstruction to deforming $\mathcal{E}$ along it is sent by $\sigma_{\mathcal{E}}$ to the obstruction for $\mathrm{ch}(\mathcal{E})$ to stay of Hodge type, which vanishes; injectivity of $\sigma_{\mathcal{E}}$ then kills the first obstruction.

Theorem 6.2 (Buchweitz–Flenner [BF03, Theorem 1.5], after Bloch [Blo72])**.** *Let $f \colon \mathcal{Z} \to T$ be a smooth projective morphism to a connected complex manifold, let $0 \in T$, and let $\mathcal{E}$ be a semiregular coherent sheaf on $\mathcal{Z}_0$. Suppose that over a contractible neighbourhood of 0 the flat continuation of every component of $\mathrm{ch}(\mathcal{E})$ is a Hodge class on every fibre. Then $\mathcal{E}$ extends to a coherent sheaf on $f^{-1}(U)$, flat over U, for some open neighbourhood U of 0.*

We return to the split Weil family $\pi \colon \mathcal{A} \to S$ of Section 5.4, with its global section ξ of Weil planes and the sets $\Sigma_d \subseteq \Sigma \subseteq S$ of Definition 5.5.

Proposition 6.3. *Let $s_0 \in S$ and let $\mathcal{E}$ be a semiregular coherent sheaf on $A = \mathcal{A}_{s_0}$ such that $\mathrm{ch}_k(\mathcal{E}) \in \mathbb{Q}\,\eta^k$ for $k \neq n$ and $\mathrm{ch}_n(\mathcal{E}) = a\,\eta^n + w$ with $a \in \mathbb{Q}$ and $0 \neq w \in W(A)$. Then $\Sigma = S$: the Weil classes are algebraic on every member of the family.*

Proof. The powers of η extend to flat sections that are Hodge classes on every member, and so does w, since the Weil planes form a trivial local system of Hodge classes on S (Section 5.4). So the hypothesis of Theorem 6.2 holds, and $\mathcal{E}$ extends to a sheaf on $\pi^{-1}(U)$, flat over U, for a contractible open neighbourhood U of s_0 in S^{an}. For $t \in U$ its restriction $\mathcal{E}_t$ is algebraic by GAGA and has a finite locally free resolution, so $\mathrm{ch}(\mathcal{E}_t)$ is algebraic, and it is the flat continuation of $\mathrm{ch}(\mathcal{E})$. Hence $w_t = \mathrm{ch}_n(\mathcal{E}_t) - a\,\eta_t^n$ is a nonzero algebraic class in $W(\mathcal{A}_t)$, and $t \in \Sigma$. Thus $U \subseteq \Sigma = \bigcup_d \Sigma_d$. By the Baire category theorem some Σ_d contains a nonempty open subset of S^{an}; it is Zariski closed by Lemma 5.6 and S is irreducible, so $\Sigma_d = S$. □

6.4. **Schoen's subvariety.** From now on $K = \mathbb{Q}(\sqrt{-3})$, and $\zeta \in \mathcal{O}_K$ is a primitive cube root of unity. Let X be a curve of genus $g = n + 1 \geq 3$, let $L \in \mathrm{Pic}^0(X)$ have exact order three, and let $\pi \colon C \to X$ be the étale cyclic triple cover defined by L, with covering automorphism σ; thus $\pi_*\mathcal{O}_C = \mathcal{O}_X \oplus L \oplus L^2$, and C has genus $3g - 2$. In $J = J(C)$ put $N = 1 + \sigma + \sigma^2$ and $P = 3 - N$, and let $B = P(J)$, the *Prym variety* of the cover, with the polarization η induced by the principal polarization of J. Since $N^2 = 3N$, we have $NP = 0$ and $P^2 = 3P$; so P is multiplication by 3 on B, and σ acts on B with $\sigma^2 + \sigma + 1 = 0$, which lets ζ act as σ.

Lemma 6.4. (a) *B is an abelian variety of Weil type for K of dimension $2n$.*
(b) *The hermitian form of η is split.*
(c) *For very general (X, L), B has maximal Hodge group.*

Proof. (a) By the decomposition of $\pi_*\mathcal{O}_C$, the space $H^0(C, K_C)$ is the sum of $H^0(X, K_X \otimes L^j)$, $j = 0, 1, 2$, on which σ^* acts by the three cube roots of unity. By Riemann–Roch these spaces have dimensions g, $g - 1$ and $g - 1$, since L^j is a nontrivial bundle of degree zero for $j = 1, 2$.

[15] Bloch introduced semiregularity for local complete intersection subvarieties, through a map $H^1(Y, N_{Y/Z}) \to H^{p+1}(Z, \Omega_Z^{p-1})$ in codimension p, and proved the deformation theorem below in that setting [Blo72]. Buchweitz and Flenner extend both the map and the theorem to coherent sheaves.

Let $P'\colon J \to B$ be P with its target restricted and $\iota\colon B \to J$ the inclusion. Since $P'\iota = 3$ and $P = \iota P'$, pull-back by P' identifies $H^{1,0}(B)$ with the image of $P^* = 3 - N^*$ on $H^{1,0}(J)$, which is the sum of the two nontrivial eigenspaces. So $\dim B = 2(g-1) = 2n$, and both embeddings of K occur with multiplicity n.

(b) The Riemann form of η is the intersection form E on $H_1(B,\mathbb{Q}) = P\,H_1(C,\mathbb{Q})$, and a K-subspace is isotropic for the hermitian form exactly when it is isotropic for E. The cover corresponds to a surjection $\phi\colon H_1(X,\mathbb{Z}) \to \mathbb{Z}/3$. Since $\mathrm{Sp}_{2g}(\mathbb{Z})$ maps onto $\mathrm{Sp}_{2g}(\mathbb{Z}/3)$, which is transitive on nonzero vectors, there is a symplectic basis $a_1, b_1, \dots, a_g, b_g$ with $\phi(x) = \langle x, b_1\rangle \bmod 3$; represent it by simple closed curves in standard position. For $i \geq 2$ the curves a_i and b_i have trivial monodromy, so each lifts to three disjoint closed curves, permuted by σ. Choose lifts $\tilde{a}_i$ and $\tilde{b}_i$ that meet. Then $E(\sigma^k\tilde{a}_i, \sigma^l\tilde{a}_j) = 0$ and $E(\sigma^k\tilde{a}_i, \sigma^l\tilde{b}_j) = \delta_{ij}\delta_{kl}$ for $i,j \geq 2$. The first relation shows that the $\mathbb{Z}[\sigma]$-span of the classes $P\tilde{a}_i$, $2 \leq i \leq g$, is isotropic. These classes are K-linearly independent: if $\sum_i \alpha_i P\tilde{a}_i = 0$ with $\alpha_i = \sum_k c_{ik}\sigma^k \in \mathbb{Z}[\sigma]$, pairing with $\sigma^l\tilde{b}_j$ gives $3c_{jl} - \sum_k c_{jk} = 0$ for every l, so $c_{j0} = c_{j1} = c_{j2}$ and α_j is a multiple of N, which is zero on B. So they span an isotropic subspace of dimension $n = g-1$.

(c) The monodromy of the Prym varieties over the moduli of pairs (X,L) is an arithmetic subgroup of the unitary group [Loo97], so by Borel density its Zariski closure contains $\mathrm{SU}(V,H)$. By André's theorem [And92] the connected monodromy group is contained in the Hodge group of a very general member. Together with $\mathrm{Hg}(B) \subseteq \mathrm{SU}(V,H)$ this gives equality. □

The pairs (X,L) of genus g have $3g-3 = 3n$ moduli, while the split Weil family of K in dimension $2n$ has dimension n^2. For $n \leq 3$ Schoen uses these Prym varieties to reach every member [Sch88, Sch98]. For $n \geq 4$ they form a proper subvariety (Figure 4).

The construction. Let $h = 2g-2 = 2n$. The group $G = (\mathbb{Z}/3)^h \rtimes S_h$ acts on C^h by $(v,\tau)\cdot(c_i)_i = (\sigma^{v_i}c_{\tau^{-1}(i)})_i$, and the quotient is the symmetric product $X^{(h)}$; write $\mu\colon C^h \to X^{(h)}$. The map $\varepsilon(v,\tau) = \sum_i v_i$ is a homomorphism $G \to \mathbb{Z}/3$, with kernel G'. Following Schoen [Sch88, §2], put

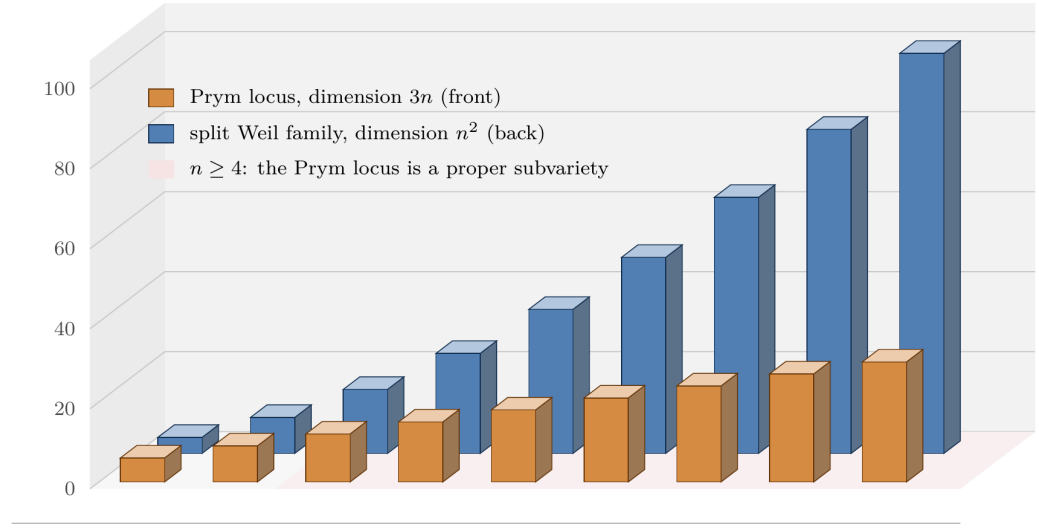
$$W_0 = C^h/G', \qquad \rho\colon C^h \to W_0, \qquad \nu\colon W_0 \to X^{(h)},$$

and let $\delta = \sigma \times 1 \times \cdots \times 1$, so that $\varepsilon(\delta) = 1$ and δ induces an automorphism $\bar{\delta}$ of W_0 over $X^{(h)}$.

Lemma 6.5. (a) *No element of $G \setminus G'$ has a fixed point on C^h. Hence ν is a connected étale Galois cover with group $\mathbb{Z}/3$, generated by $\bar{\delta}$.*
(b) *G acts freely on $\mu^{-1}(D)$ for every reduced divisor $D \in X^{(h)}$.*
(c) *μ is finite and flat, so μ^{-1} of a subvariety of pure dimension r has pure dimension r.*

Proof. Suppose (v,τ) fixes (c_i), so that $c_{\tau(i)} = \sigma^{v_{\tau(i)}}c_i$ for all i. Going once around a cycle $(i_1,\dots,i_k)$ of τ gives $c_{i_1} = \sigma^{v_{i_1}+\cdots+v_{i_k}}c_{i_1}$; since σ has no fixed points, the sum of the v_i over every cycle is 0 in $\mathbb{Z}/3$, and so is $\varepsilon(v,\tau)$. This is (a): W_0 is connected as a quotient of C^h, and $G/G' \cong \mathbb{Z}/3$ acts freely on it with quotient $X^{(h)}$. If moreover the points $\pi(c_i)$ are distinct, then $\pi(c_{\tau(i)}) = \pi(c_i)$ forces $\tau = 1$, and then $v = 0$; this is (b). The map $C^h \to X^h$ is étale and $X^h \to X^{(h)}$ is a finite map between smooth varieties of the same dimension, hence flat. By going down for flat maps, every component of the preimage of a subvariety Z dominates a component of Z, and the fibres are finite; this gives (c). □

Let $u\colon X^{(h)} \to \mathrm{Pic}^h(X)$ send D to $\mathcal{O}_X(D)$. Its fibre over $[K_X]$ is the canonical system $F = |K_X| \cong \mathbb{P}^n$. Since F is simply connected, the restriction of ν to $\nu^{-1}(F)$ is a trivial cover: $\nu^{-1}(F) = Q_0 \sqcup Q_1 \sqcup Q_2$ with each Q_t mapped isomorphically onto F, and $\bar{\delta}$, which has no fixed points, permutes the Q_t cyclically; number them so that $\bar{\delta}(Q_t) = Q_{t+1}$. These are the three



n	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
$3n$	6	9	12	15	18	21	24	27	30
n^2	4	9	16	25	36	49	64	81	100
n^2-3n	−2	0	4	10	18	28	40	54	70

FIGURE 4. The dimension $3n$ of the locus of Prym varieties of étale cyclic triple covers of curves of genus $n+1$, against the dimension n^2 of the split Weil family of $\mathbb{Q}(\sqrt{-3})$ in dimension $2n$. For $n \geq 4$ the Prym locus is a proper subvariety, and a semiregular Y would have to deform in the n^2-3n remaining directions.

components of Schoen's construction.[16] Put

$$\Lambda_0 = \rho^{-1}(Q_0) \subseteq C^h,$$

with its reduced structure. It has pure dimension n by Lemma 6.5(c), and since the general member of $|K_X|$ is reduced, ρ is étale over a dense open subset of Q_0 by Lemma 6.5(b), so that $\rho_*[\Lambda_0] = |G'|\,[Q_0]$. The construction is drawn in Figure 5.

Fix $c_0 \in C$ and let

$$\Psi\colon C^h \to B, \qquad (c_1,\dots,c_h) \mapsto P\Big(\sum_i (c_i - c_0)\Big).$$

The key input is a computation of cohomology with twisted coefficients.

Lemma 6.6. *Let $\mathcal{L}$ be a nontrivial local system of rank one on $X^{(h)}$. Then restriction $H^h(X^{(h)},\mathcal{L}) \to H^h(F,\mathcal{L})$ is an isomorphism, and $H^h(F,\mathcal{L}) \cong H^{2n}(\mathbb{P}^n,\mathbb{C}) \cong \mathbb{C}$.*

Proof. The fibres of u are projective spaces, so $\mathcal{L}$ is trivial on each of them, and $\mathcal{M} = u_*\mathcal{L}$ is a local system of rank one on $T = \mathrm{Pic}^h(X)$ with $\mathcal{L} = u^*\mathcal{M}$.[17] It is nontrivial, because $\mathcal{L}$ is.

[16] Patel and Zhang obtain the same three components by base change of the Abel–Jacobi map along an isogeny onto $J(X)$ [PZ27].

[17] By proper base change the stalk of $u_*\mathcal{L}$ at y is $H^0(u^{-1}(y),\mathcal{L}) \cong \mathbb{C}$, so a nonzero section over the fibre extends over $u^{-1}(D)$ for a small ball $D \ni y$. This set is connected, since u is proper with connected fibres, and a section of a local system of rank one over a connected set is determined by its value at one point and vanishes nowhere unless it is zero. So restriction from $u^{-1}(D)$ to each fibre over D is an isomorphism, $u_*\mathcal{L}$ is locally constant, and $u^*u_*\mathcal{L} \to \mathcal{L}$ is an isomorphism on stalks.

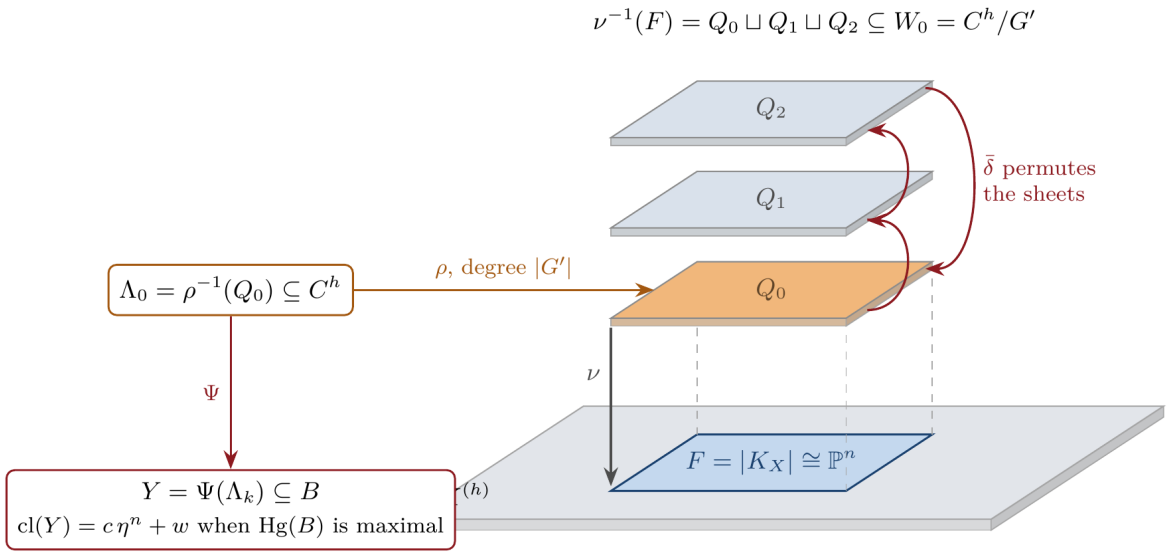


FIGURE 5. Schoen's construction. Over the canonical system $F = |K_X|$ of X, the étale triple cover $\nu\colon W_0 \to X^{(h)}$ splits into three sheets, which $\bar\delta$ permutes cyclically. $\Lambda_0 \subseteq C^h$ is the preimage of one sheet, and Y is the image under Ψ of one of its components (Theorem G).

As a real manifold T is a torus $(S^1)^{2g}$, and $\mathcal{M}$ is an exterior product of local systems on the circles, at least one of them nontrivial; a nontrivial local system of rank one on a circle has no cohomology, so $H^*(T, \mathcal{M}) = 0$ by the Künneth formula. Let $U = T \setminus \{[K_X]\}$. The exact sequence

$$\cdots \to H^k_c(U, \mathcal{M}) \to H^k(T, \mathcal{M}) \to H^k(\{[K_X]\}, \mathcal{M}) \to H^{k+1}_c(U, \mathcal{M}) \to \dots$$

gives $H^1_c(U, \mathcal{M}) \cong \mathbb{C}$ and $H^k_c(U, \mathcal{M}) = 0$ for $k \neq 1$. For $M \in U$ Riemann–Roch gives $h^0(M) = g - 1 = n$, so over U the map u is the projective bundle of a vector bundle of rank n [ACGH85, Chapter IV], and by the Leray–Hirsch theorem $H^*_c(u^{-1}(U), \mathcal{L}) \cong H^*_c(U, \mathcal{M}) \otimes H^*(\mathbb{P}^{n-1})$, which vanishes outside the degrees $1, 3, \dots, 2n-1$. The exact sequence of the pair $(X^{(h)}, F)$,

$$H^h_c(u^{-1}(U), \mathcal{L}) \to H^h(X^{(h)}, \mathcal{L}) \to H^h(F, \mathcal{L}) \to H^{h+1}_c(u^{-1}(U), \mathcal{L}),$$

has zero outer terms, since $h = 2n$. Finally $\mathcal{L}$ is trivial on the simply connected $F \cong \mathbb{P}^n$. □

Proof of Theorem G. Write $V_\chi, V_{\bar\chi} \subseteq H^1(B, \mathbb{C})$ for the eigenspaces of σ^* with eigenvalues $\lambda = \chi(\zeta)$ and $\bar\lambda$, and choose a basis $\bar y_1, \dots, \bar y_h$ of $V_{\bar\chi}$; put $\bar\omega = \bar y_1 \wedge \dots \wedge \bar y_h$, which spans $\bigwedge^{2n} V_{\bar\chi} \subseteq W(B) \otimes \mathbb{C}$.

Step 1: the pull-back of $\bar\omega$. Write $\Psi = P' \circ \Xi$, where $\Xi(c_1, \dots, c_h) = \sum_i \mathrm{aj}(c_i)$ and $\mathrm{aj}(c) = [c - c_0]$ is the Abel–Jacobi map. On H^1 the pull-back by the addition map $J^h \to J$ is the sum of the pull-backs by the projections, so $\Psi^* \bar y_j = \sum_i p_i^* \tilde y_j$, where p_i is the i-th projection of C^h and $\tilde y_j = \mathrm{aj}^*\, P'^* \bar y_j \in H^1(C, \mathbb{C})$. Since P'^* is injective and aj^* is an isomorphism, both commuting with σ^*, the $\tilde y_j$ are independent and lie in the $\bar\lambda$-eigenspace of σ^* on $H^1(C, \mathbb{C})$. That eigenspace has dimension h, because the invariant part of $H^1(C, \mathbb{C})$ is $\pi^* H^1(X, \mathbb{C})$, of dimension $2g$, and

the two other eigenspaces are conjugate; so the $\tilde{y}_j$ form a basis of it. Now

$$u := \Psi^*\bar{\omega} = \prod_{j=1}^{h}\Big(\sum_{i=1}^{h} p_i^*\tilde{y}_j\Big).$$

A product $\tilde{y}_j \cup \tilde{y}_k$ lies in the $\bar{\lambda}^2$-eigenspace of $H^2(C,\mathbb{C})$, which is zero because σ acts trivially on $H^2(C)$ and $\bar{\lambda}^2 \neq 1$. So in the expansion only the terms with one factor on each p_i survive: $u = \sum_{s\in S_h} \pm \prod_i p_i^*\tilde{y}_{s(i)}$. By the Künneth formula these terms are linearly independent, and $u \neq 0$. Each term is multiplied by $\bar{\lambda}^{\sum v_i}$ under $(\sigma^{v_1}\times\cdots\times\sigma^{v_h})^*$, and u is symmetric because Ψ is; so u is invariant under G', and $\delta^* u = \bar{\lambda}u$.

Step 2: restriction to Q_0. By transfer, ρ^* identifies $H^*(W_0,\mathbb{C})$ with $H^*(C^h,\mathbb{C})^{G'}$, so $u = \rho^*\bar{u}$ for a class $\bar{u} \neq 0$ with $\bar{\delta}^*\bar{u} = \bar{\lambda}\bar{u}$. The decomposition $\nu_*\mathbb{C} = \bigoplus_\psi \mathcal{L}_\psi$ into eigensheaves of $\bar{\delta}$, indexed by the characters ψ of $\mathbb{Z}/3$, identifies the $\psi(1)$-eigenspace of $\bar{\delta}^*$ on $H^*(W_0,\mathbb{C})$ with $H^*(X^{(h)},\mathcal{L}_\psi)$, compatibly with restriction to $\nu^{-1}(F)$ and F. For $\psi \neq 1$ the local system $\mathcal{L}_\psi$ is nontrivial, since $H^0(W_0,\mathbb{C})$ is the invariant line. By Lemma 6.6, $\bar{u}$ restricts to a nonzero class on $\nu^{-1}(F) = Q_0 \sqcup Q_1 \sqcup Q_2$. As $\bar{\delta}$ maps Q_0 isomorphically onto Q_t and $\bar{u}$ is an eigenvector, $\bar{u}|_{Q_0} = 0$ would force $\bar{u}|_{Q_t} = 0$ for every t. Hence $\int_{Q_0}\bar{u} \neq 0$, and

$$\int_B \Psi_*[\Lambda_0]\cup\bar{\omega} = \int_{[\Lambda_0]}\rho^*\bar{u} = |G'|\int_{Q_0}\bar{u} \neq 0.$$

Step 3: one component. The class $\Psi_*[\Lambda_0]$ is $\sum_k e_k\,\mathrm{cl}(\Psi(\Lambda_k))$, the sum over the irreducible components Λ_k of Λ_0 on which Ψ is generically finite, with e_k the degree of $\Lambda_k \to \Psi(\Lambda_k)$. By Step 2 one of them, say $Y = \Psi(\Lambda_k)$, satisfies $\int_B \mathrm{cl}(Y)\cup\bar{\omega} \neq 0$. It is irreducible of dimension n. Since $W(B)\otimes\mathbb{C}$ is spanned by $\bar{\omega}$ and its conjugate, $\mathrm{cl}(Y)$ pairs nontrivially with $W(B)$.

Step 4: the shape of the class. If $\mathrm{Hg}(B) = \mathrm{SU}(V,H)$, then Proposition 6.1 gives $\mathrm{cl}(Y) = c\,\eta^n + w$ with $c \in \mathbb{Q}$ and $w \in W(B)$. Since $\eta^n\cup\bar{\omega} = 0$, Step 3 gives $w \neq 0$. Since $\eta^n \cup w = 0$,

$$c\int_B \eta^{2n} = \int_B \mathrm{cl}(Y)\cup\eta^n = \deg_\eta Y > 0,$$

so $c > 0$. The last assertion of Theorem G is Corollary 6.8 below. □

Corollary 6.7 (Schoen [Sch88])**.** *For every étale cyclic triple cover of a curve of genus at least three, the Prym variety B is Weil-algebraic.*

Proof. If B has maximal Hodge group, $w = \mathrm{cl}(Y) - c\,\eta^n$ is a nonzero algebraic class in $W(B)$, and B is Weil-algebraic. By Lemma 6.4(c) this covers every (X,L) outside a countable union of proper closed subsets Z_i of a smooth connected base T over which, after a finite étale base change, the Prym varieties form a polarized abelian scheme with a trivial local system of Weil planes.[18] The proof of Lemma 5.6 shows that the locus where the Weil classes have algebraic representatives of degree at most d is Zariski closed in T. These countably many closed sets, together with the Z_i, cover T, so by the Baire category theorem one of them has interior. It is not a Z_i, so it is one of the degree-d loci, and it equals T. □

Corollary 6.8. *Suppose that for one (X,L) of genus $n+1 \geq 3$ with maximal Hodge group the subvariety Y of Theorem G is semiregular. Then every abelian variety of split Weil type for $\mathbb{Q}(\sqrt{-3})$ of dimension $2n$ is Weil-algebraic.*

[18]Take for T a connected component of a finite étale cover of the moduli space of curves of genus g with level-3 structure, on which L is the section given by a fixed nonzero vector of $(\mathbb{Z}/3)^{2g}$. The monodromy acts on the Weil planes through the determinant, which takes values in the roots of unity of $\mathcal{O}_K$, so it becomes trivial on a finite cover.

Proof. The sheaf $\mathcal{O}_Y$ has $\mathrm{ch}_k(\mathcal{O}_Y) = 0$ for $k < n$ and $\mathrm{ch}_n(\mathcal{O}_Y) = \mathrm{cl}(Y) = c\,\eta^n + w$ with $w \neq 0$ [Ful98, Chapter 15]; for $k > n$ the class $\mathrm{ch}_k(\mathcal{O}_Y)$ is algebraic, hence a multiple of η^k by Proposition 6.1. By Lemma 6.4(b), B is a member of a split Weil family S as in Section 5.4, and Proposition 6.3 gives $\Sigma = S$. Every abelian variety of split Weil type for K of dimension $2n$ is K-isogenous to a member of S,[19] and a K-isogeny f carries Weil planes to Weil planes and algebraic classes to algebraic classes in both directions, through f^* and $(\deg f)^{-1} f_*$. □

6.5. **Transfer by correspondences.** Schoen's cycles reach a family of dimension $3n$ inside one of dimension n^2. One way to spread them is to carry the Weil classes to other varieties by correspondences. This gives a theorem, but a count of dimensions shows that it cannot reach the general member by itself.

Lemma 6.9. *Let B, A_1 and A_2 be of Weil type for K, of dimensions $2(m_1 + m_2)$, $2m_1$ and $2m_2$, and let $f\colon B \to A_1 \times A_2$ be an isogeny that commutes with $\mathcal{O}_K$.*

(a) *f^* carries a plane in $W(A_1) \otimes W(A_2)$ onto $W(B)$; over $\mathbb{C}$ the plane is spanned by $\omega_1 \otimes \omega_2$ and $\bar\omega_1 \otimes \bar\omega_2$, where ω_i spans $\bigwedge^{2m_i} V_\chi(A_i)$.*

(b) *If two of B, A_1, A_2 are Weil-algebraic, so is the third.*

Proof. As in the proof of Corollary 6.8, we may take $B = A_1 \times A_2$. Then $V_\chi(B) = V_\chi(A_1) \oplus V_\chi(A_2)$, and the top exterior power of a sum is the tensor product of the top exterior powers; this is (a). If A_1 and A_2 are Weil-algebraic, every class in $W(A_1) \otimes W(A_2)$ is a sum of products $\mathrm{pr}_1^*\alpha_1 \cup \mathrm{pr}_2^*\alpha_2$ of algebraic classes, so B is Weil-algebraic by (a). Suppose that B and A_2 are Weil-algebraic. On A_2 we have $\omega_2 \cup \omega_2 = 0$, since $\bigwedge^{4m_2} V_\chi(A_2) = 0$, while $p = \int \omega_2 \cup \bar\omega_2 \neq 0$, since $\bigwedge^{2m_2} V_\chi \otimes \bigwedge^{2m_2} V_{\bar\chi}$ is the top exterior power of $H^1(A_2, \mathbb{C})$; and p is real, because classes of even degree commute. Nonzero rational classes $x \in W(B)$ and $\gamma \in W(A_2)$ have the form $x = s\,\omega_1 \otimes \omega_2 + \bar s\,\bar\omega_1 \otimes \bar\omega_2$ and $\gamma = t\,\omega_2 + \bar t\,\bar\omega_2$ with $s, t \neq 0$. The class

$$\mathrm{pr}_{1*}(x \cup \mathrm{pr}_2^*\gamma) = p\,(s\bar t\,\omega_1 + \bar s t\,\bar\omega_1)$$

is rational, algebraic, nonzero, and lies in $W(A_1)$. □

Lemma 6.10. *Let K be any imaginary quadratic field, E an elliptic curve with complex multiplication by $\mathcal{O}_K$, and $\bar E$ the same curve with $\mathcal{O}_K$ acting through complex conjugation. For $k \geq 1$ the variety $E^k \times \bar E^k$, with the product polarization, is of split Weil type, and its Weil classes are polynomials in divisor classes.*

Proof. On $H_1(E, \mathbb{Q}) = K$ the polarization is $\mathrm{Tr}_{K/\mathbb{Q}}(f\,x\bar y)$ for some $f \in K$ with $\bar f = -f$, so its hermitian form is $x\bar y$. For $\bar E$ the K-structure is conjugated, and the same Riemann form is $\mathrm{Tr}_{K/\mathbb{Q}}(f \cdot (-\bar x y))$, with hermitian form $-\bar x y$. So the form of $E^k \times \bar E^k$ is $\mathrm{diag}(1, \dots, 1, -1, \dots, -1)$, for which the vectors $e_i + e_{k+i}$ span an isotropic subspace of dimension k, and both embeddings of K occur k times on $H^{1,0}$. With coordinates $z_1, \dots, z_{2k}$ on the factors, V_χ is spanned by $dz_1, \dots, dz_k$ and $d\bar z_{k+1}, \dots, d\bar z_{2k}$, so up to sign $\omega_\chi = \bigwedge_{i=1}^k (dz_i \wedge d\bar z_{k+i})$. Each factor is a class of type $(1,1)$ on $E \times \bar E \cong E \times E$, whose Néron–Severi group has rank $4 = h^{1,1}$ because E has complex multiplication; so it is a complex combination of divisor classes, and so are ω_χ and its conjugate. □

Theorem 6.11. *Let A be an abelian variety of Weil type for $K = \mathbb{Q}(\sqrt{-3})$, of dimension $2m \geq 2$, on which $\mathcal{O}_K$ acts. There are a smooth curve $C \subseteq A$, stable under ζ and with ζ acting freely on it, so that $C \to X = C/\zeta$ is an étale cyclic triple cover of a curve of some genus*

[19] Two split hermitian forms of the same dimension are isometric, being orthogonal sums of hyperbolic planes. An isometry carries the complex structure of the first variety to a point of the symmetric domain of the second form, and the two lattices are commensurable in the same K-vector space.

$g \geq m+1$; an abelian variety A' of Weil type of dimension $2(g-1-m)$; and an isogeny commuting with $\mathcal{O}_K$ between the Prym variety of $C \to X$ and $A \times A'$. For every such curve, A is Weil-algebraic if and only if A' is.

Proof. The curve. The automorphism ζ has finitely many fixed points, the kernel of the isogeny $1-\zeta$. Let $\mathcal{N}$ be ample on A. The bundle $\mathcal{N}' = \mathcal{N} \otimes \zeta^*\mathcal{N} \otimes \zeta^{2*}\mathcal{N}$ is isomorphic to its pull-back by ζ, so it carries a linearization for the cyclic group generated by ζ;[20] its cube descends to an ample line bundle $\bar{\mathcal{M}}$ on the normal quotient $q\colon A \to A/\zeta$, since the stabilizers, of order dividing three, act trivially on its fibres. For k large, $2m-1$ general members D_i of $|\bar{\mathcal{M}}^k|$ meet in a smooth irreducible curve X that avoids the images of the fixed points, by Bertini's theorem. Then $C = q^{-1}(X)$ is étale over X, hence smooth, and it is the complete intersection of the ample divisors q^*D_i, hence connected. Schoen observes the existence of such curves in [Sch88, §3].

The surjection. By the Lefschetz hyperplane theorem, restriction $H^1(A,\mathbb{Q}) \to H^1(C,\mathbb{Q})$ is injective, so the homomorphism $\alpha\colon J(C) \to A$ induced by the inclusion is surjective; it commutes with ζ. Since $1+\zeta+\zeta^2 = 0$ in $\mathcal{O}_K$, $\alpha N = 0$ and $\alpha P = 3\alpha$. So α maps the Prym variety $B = P(J(C))$ onto A, and $g-1 \geq m$.

The complement. Let A' be the connected component of the kernel of $\alpha|_B$ and A'' its complement in B for the induced polarization. Both are stable under σ, which preserves the polarization, so $\mathcal{O}_K$ acts on them, and $A'' \to A$ and $A' \times A'' \to B$ are isogenies commuting with $\mathcal{O}_K$. By Lemma 6.4(a) both embeddings of K occur $g-1$ times on $H^{1,0}(B)$, and they occur m times on $H^{1,0}(A)$; so they occur $g-1-m$ times on $H^{1,0}(A')$, and A' is of Weil type.

The equivalence. B is Weil-algebraic by Corollary 6.7, and Lemma 6.9(b) gives the equivalence. □

Corollary 6.12. *A is Weil-algebraic as soon as, for one curve as in Theorem 6.11, the complement A' is K-isogenous to a product of Weil-algebraic varieties, such as Prym varieties of étale cyclic triple covers (Corollary 6.7), abelian surfaces of Weil type (Theorem 2.3), and the varieties $E^k \times \bar{E}^k$ (Lemma 6.10).*

Proof. Products of Weil-algebraic varieties are Weil-algebraic by Lemma 6.9(b), and Theorem 6.11 applies. □

When $g-1 = m$ the complement is zero and A is K-isogenous to a Prym variety. For larger g the theorem trades the Weil classes of A for those of A', and since curves of higher degree give complements of every size, one might hope to choose the curve so that A' is of a known kind. The count below says that for $m \geq 4$ this hope fails at the general member unless an unexpected intersection occurs.

Remark 6.13. Pryms of étale triple covers of curves of genus $m+k+1$ form a family of dimension $3(m+k)$ inside the Weil family of dimension $(m+k)^2$. The varieties K-isogenous to $A \times A'$, with A in the family of dimension m^2 and A' in a family of dimension f, form a subvariety of dimension m^2+f. For A to be general the two must meet in dimension at least m^2, which needs $3(m+k)+f \geq (m+k)^2$ unless they meet in excess dimension. Products of the varieties of Corollary 6.12 of total dimension $2k$ have at most $3k$ moduli, and $3(m+k)+3k < (m+k)^2$ for every $k \geq 0$ once $m \geq 4$. For $m = 3$ only $k = 0$ passes, the Prym locus itself, and for $m = 2$ exactly $k \leq 2$ pass. So Corollary 6.12 reaches only a meagre part of the family once $m \geq 4$, unless the Prym locus meets these product loci in excess dimension, as Schoen anticipated [Sch88, §3]. This is a count, not a proof that no such intersection exists.

[20] Choose an isomorphism $\phi\colon \zeta^*\mathcal{N}' \to \mathcal{N}'$. The composite of ϕ, $\zeta^*\phi$ and $\zeta^{2*}\phi$ is an automorphism of $\mathcal{N}'$, hence a scalar c, and $c^{-1/3}\phi$ is a linearization.

6.6. **Where it stops.** Everything on the Weil route now rests on one question.

Question 6.14. Let $n \geq 4$ and let (X, L) be very general of genus $n+1$. Is the subvariety $Y \subseteq B$ of Theorem G semiregular?

A positive answer forces Y to deform with B to first order along all n^2 directions of the Weil family, and not only along the $3n$ directions of the Prym locus; so it asks for $n^2 - 3n$ unexpected first order deformations. We have not been able to decide the question, and the analogues point both ways: the Abel–Jacobi curve, discussed next, and the Abel–Prym curve of Proposition 6.16 are rigid, while the theta divisors discussed after Proposition 6.17 deform.

The Abel–Jacobi curve $C \subseteq J(C)$ of a curve of genus $g \geq 4$ is built from the same map as Y. Its class $\theta^{g-1}/(g-1)!$ is a Hodge class on every principally polarized abelian variety. Yet by the criterion of Matsusaka and Ran [Mat59, Ran81] a curve with this class exists only on Jacobians and products of Jacobians, which form a proper subvariety of the moduli space $\mathcal{A}_g$. So the curve does not deform in the directions normal to the Jacobian locus, and by Theorem 6.2 it is not semiregular. The number of those directions is classical. The tangent space to $\mathcal{A}_g$ at $J(C)$ is dual to $\operatorname{Sym}^2 H^0(K_C)$, the tangent space to the Jacobian locus is dual to $H^0(2K_C)$, and by Max Noether's theorem the multiplication map between them is surjective for non-hyperelliptic C, with kernel the space I_2 of quadrics containing the canonical curve. So the Jacobian locus has codimension

$$\dim I_2 = \binom{g+1}{2} - (3g-3) = \frac{(g-2)(g-3)}{2}$$

at $J(C)$. Schoen's subvariety is the analogue one step up, built from the fibre of the Abel–Jacobi map over the canonical class, and the corresponding codimension, that of the Prym locus in the Weil family, is $n^2 - 3n$. Nothing in the construction of Y moves it off the Prym locus. This is an analogy and not a proof.

Nor can a branched cover take the place of the étale one, because branching does not enlarge the Prym locus enough.

Proposition 6.15. *Let $\pi\colon C \to X$ be a cyclic triple cover of a curve of genus q, branched at r points, with covering automorphism σ, and let its Prym variety, the image of $3 - N$ on $J(C)$ with $N = 1 + \sigma + \sigma^2$, have dimension $2n$. Then $2q + r = 2n + 2$, and the covers with these invariants form finitely many families, each of dimension $q + 2n - 1 \leq 3n$, with equality only for étale covers of curves of genus $n+1$. In particular, for $n \geq 4$ a very general member of the split Weil family of K in dimension $2n$ is not K-isogenous to the Prym variety of any cyclic triple cover, étale or branched.*

Proof. Since 3 is prime, every branch point is totally ramified, and the Riemann–Hurwitz formula $2g(C) - 2 = 3(2q-2) + 2r$ gives $g(C) = 3q - 2 + r$. The invariant part of $H^0(C, K_C)$ under σ^* is $\pi^* H^0(X, K_X)$, of dimension q, and as in the proof of Lemma 6.4(a) the space $H^{1,0}$ of the Prym variety is the sum of the two other eigenspaces. So the Prym variety has dimension $g(C) - q = 2q - 2 + r$, and $2q + r = 2n + 2$. Given X and the r branch points, the cover corresponds to a surjection from the fundamental group of the punctured curve onto $\mathbb{Z}/3$, and there are finitely many of these. So the covers form finitely many families, of dimension $3q - 3 + r$ when $q \geq 2$, r when $q = 1$, and $r - 3$ when $q = 0$; by $r = 2n + 2 - 2q$ each of these equals $q + 2n - 1$. As $r \geq 0$, $q \leq n + 1$, so the dimension is at most $3n$, and it equals $3n$ exactly when $q = n + 1$ and $r = 0$.

The Prym varieties of each family form the image of a variety of dimension at most $3n$ in the moduli space, and the K-isogenies of a given degree carry it to finitely many subvarieties of at most the same dimension. The members of the split Weil family that are K-isogenous to

such a Prym variety therefore lie in a countable union of subvarieties of dimension at most $3n$, which for $n \geq 4$ is less than n^2. □

The analogy with the Abel–Jacobi curve can be tested on B itself, in dimension one. Following Lange and Ortega [LO11], call

$$\psi\colon C \to B, \qquad \psi(c) = (1-\sigma)(c - c_0),$$

the *Abel–Prym map*. As $P = (1-\sigma)(1-\sigma^2)$, the restriction of Ψ to one factor is $(1-\sigma^2)\circ\psi$, and $1-\sigma^2$ is an isogeny of B. Put $V_j = H^0(X, K_X \otimes L^j)$, so that $H^0(C, K_C) = V_0 \oplus V_1 \oplus V_2$ as in the proof of Lemma 6.4(a), and let

$$\mu_L\colon V_1 \otimes V_2 \longrightarrow H^0(X, K_X^{\otimes 2})$$

be multiplication of sections, through an isomorphism $L^3 \cong \mathcal{O}_X$. A first-order deformation of X with Kodaira–Spencer class $u \in H^1(X, T_X)$ carries the étale cover along with it, uniquely, and with it $J(C)$, σ and B. Write $d\mathcal{P}(u) \in H^1(B, T_B)$ for the Kodaira–Spencer class of the resulting deformation of B. Lange and Ortega identify the dual of $d\mathcal{P}$ with a multiplication map [LO11], and we derive the form of this that we need in the proof below.

Proposition 6.16. *Let $v \in H^1(B, T_B)$ be the Kodaira–Spencer class of a first-order deformation $\mathcal{B}$ of B to which σ extends. Then ψ extends to a morphism into $\mathcal{B}$ from some first-order deformation of C if and only if v lies in the image of $d\mathcal{P}$, and this image has dimension $\operatorname{rk}\mu_L \leq 3n$. Suppose moreover that $n \geq 4$, that B has maximal Hodge group, and that S is a split Weil family containing B (Lemma 6.4(b)). Then the tangent vectors to S at B along which ψ extends form a subspace of codimension at least $n^2 - 3n$, although $\psi_*[C]$ is a rational multiple of η^{2n-1}, whose flat continuation is algebraic on every member of S.*

Proof. All deformations are over $\operatorname{Spec}\mathbb{C}[t]/(t^2)$. Suppose that ψ extends to $\psi_t\colon \mathcal{C} \to \mathcal{B}$, where $\mathcal{C}$ is a deformation of C with class $w \in H^1(C, T_C)$. Cover C and B by affine open sets U_i and W_i with $\psi(U_i) \subseteq W_i$ over which $\mathcal{C}$ and $\mathcal{B}$ are trivial, with transition automorphisms $1 + t\theta_{ij}$ and $1 + t\xi_{ij}$, where (θ_{ij}) and (ξ_{ij}) are cocycles of vector fields representing w and v. On U_i the map ψ_t is $\psi + ts_i$ for a section s_i of ψ^*T_B, and compatibility on $U_i \cap U_j$ reads $s_j - s_i = \psi^*\xi_{ij} - d\psi(\theta_{ij})$ [Hor74]. Hence

$$\psi^* v = H^1(d\psi)(w) \quad \text{in } H^1(C, \psi^*T_B), \tag{6}$$

where $d\psi\colon T_C \to \psi^*T_B$ is the differential.

We compute both sides by Serre duality. The bundle T_B is trivial with fibre T_0B, and ψ^* identifies $H^0(B, \Omega^1_B)$ with $V_- = V_1 \oplus V_2$. Indeed, since $(1-\sigma)N = 0$, the pull-back by $1-\sigma\colon J \to B$ maps $H^{1,0}(B)$ injectively into the sum of the nontrivial eigenspaces of σ^* on $H^{1,0}(J)$, which has dimension $2n = \dim B$, and the Abel–Jacobi map identifies this sum with V_-. So $H^1(C, \psi^*T_B) = H^1(C, \mathcal{O}_C) \otimes T_0B$ is dual to $H^0(C, K_C) \otimes V_-$, and $H^1(C, T_C)$ is dual to $H^0(C, K_C^{\otimes 2})$. The dual of $d\psi$ is evaluation $\mathcal{O}_C \otimes V_- \to K_C$, so the dual of $H^1(d\psi)$ is multiplication

$$m\colon H^0(C, K_C) \otimes V_- \longrightarrow H^0(C, K_C^{\otimes 2})$$

[LO11]. Thus (6) says that ψ^*v, as a linear form on $H^0(C, K_C) \otimes V_-$, is $w \circ m$.

Now use σ. We have $\psi\sigma = t_a\sigma\psi$, where t_a is the translation by $a = (1-\sigma)(\sigma c_0 - c_0)$. Translations act trivially on $H^1(B, \mathcal{O}_B)$ and on T_0B, so ψ^* commutes with the actions of σ, and m is equivariant. As σ extends to $\mathcal{B}$, the class v is invariant, and so is the form ψ^*v; an invariant form vanishes on every eigenspace of σ with an eigenvalue other than 1. The automorphism σ^* acts on V_j by κ^j for a primitive cube root of unity κ, so the invariant part of $H^0(C, K_C) \otimes V_-$ is $E = (V_1 \otimes V_2) \oplus (V_2 \otimes V_1)$. Since $\pi_* K_C^{\otimes 2} = K_X^{\otimes 2} \otimes \pi_*\mathcal{O}_C$, the map m

sends E to $H^0(C, K_C^{\otimes 2})^\sigma = \pi^* H^0(X, K_X^{\otimes 2})$, by $(x, y) \mapsto \pi^* M(x, y)$ with $M(x, y) = \mu_L(x + \tau y)$, where $\tau\colon V_2 \otimes V_1 \to V_1 \otimes V_2$ is the transposition. So $\psi^* v$ is determined by its restriction to E, which by (6) is $f \circ M$, where f is the linear form $q \mapsto \langle w, \pi^* q\rangle$ on $H^0(X, K_X^{\otimes 2})$, that is, an element of $H^1(X, T_X)$.

Apply this to the deformation of B induced by $u \in H^1(X, T_X)$. The deformation of C has class $\pi^* u$, and ψ extends after c_0 is extended to a section. Since π has degree 3, $\langle \pi^* u, \pi^* q\rangle = 3\langle u, q\rangle$, so the restriction of $\psi^* d\mathcal{P}(u)$ to E is $3\, u \circ M$. Returning to v, the invariant forms $\psi^* v$ and $\psi^* d\mathcal{P}(f/3)$ agree on E, hence everywhere. The map ψ^* on $H^1(B, T_B) = H^1(B, \mathcal{O}_B) \otimes T_0 B$ is injective, since on $H^1(B, \mathcal{O}_B)$ it is the complex conjugate of the injective map on $H^0(B, \Omega^1_B)$. So $v = d\mathcal{P}(f/3)$. Conversely, if $v = d\mathcal{P}(u)$, then $\mathcal{B}$ is isomorphic to the deformation induced by u, since first-order deformations are determined by their Kodaira–Spencer classes, and ψ extends. Finally $d\mathcal{P}(u) = 0$ exactly when $u \circ M = 0$, that is, when u vanishes on the image of μ_L; so $d\mathcal{P}$ has rank $\operatorname{rk} \mu_L \le h^0(X, K_X^{\otimes 2}) = 3g - 3 = 3n$.

For the last assertion, the element $\zeta \in \mathcal{O}_K$ acts on the abelian scheme over S as σ on B, so the Kodaira–Spencer map $T_{S,B} \to H^1(B, T_B)$ has invariant image. It is injective, because S is locally isomorphic to the symmetric domain $\mathcal{D}$, the period map of the family lifts to the inclusion of $\mathcal{D}$ in the period domain, and the differential of the period map is the Kodaira–Spencer map followed by the isomorphism $H^1(B, T_B) \cong \operatorname{Hom}(H^{1,0}(B), H^{0,1}(B))$ [Voi02, Chapter 10]. So the tangent vectors along which ψ extends form a subspace of dimension at most $3n$ in the n^2-dimensional space $T_{S,B}$. By Proposition 6.1, $\psi_*[C] \in \operatorname{Hdg}^{2n-1}(B) = \mathbb{Q}\,\eta^{2n-1}$. □

So the curve $\psi(C)$, whose class stays algebraic on the whole family, moves with B only along the Prym locus. When X has Clifford index at least 2, as a very general curve of genus at least 5 does [ACGH85, Chapter V], the map ψ is a closed embedding [LO11], and then, for $n \ge 4$ and B with maximal Hodge group, the curve $\psi(C)$ is not semiregular. Otherwise Theorem 6.2 would extend $\mathcal{O}_{\psi(C)}$ along every tangent vector of S; a first-order deformation of $\mathcal{O}_{\psi(C)}$ is locally generated by one section, so it is a line bundle on the subscheme defined by its annihilator, which is a first-order deformation of $\psi(C)$ and so carries an extension of ψ, against Proposition 6.16.

This makes the analogy precise on B, but it does not decide Question 6.14. For Y itself we have the following description, which does not mention Λ_k: up to an étale map, Y is a Brill–Noether locus of the Prym variety. Let $\mathrm{Nm}\colon \mathrm{Pic}^{2n}(C) \to \mathrm{Pic}^{2n}(X)$ be the norm map, let $N_K = \mathrm{Nm}^{-1}([K_X])$, and let $V \subseteq N_K$ be the closed subset of the line bundles M with $h^0(C, M) > 0$. On J the endomorphism N is $\pi^* \circ \mathrm{Nm}$. Put

$$\phi\colon N_K \to B, \qquad \phi(M) = P\big(M \otimes \mathcal{O}_C(-2nc_0)\big).$$

Proposition 6.17. (a) *On each connected component of N_K the map ϕ is a translate of multiplication by 3, an étale map onto B of degree 3^{4n}, and $Y = \phi(V_k)$ for an irreducible component V_k of V of dimension n.*

(b) *Let M be a general point of V_k. Then $h^0(C, M) = 1$, and the divisor E of a nonzero section s of M contains no two points of one fibre of π. The varieties V_k and Y are smooth of dimension n at M and at $\phi(M)$, and $T_M V_k \subseteq H^1(C, \mathcal{O}_C)$ is the annihilator of the subspace*

$$\pi^* H^0(X, K_X) + s \cdot H^0(C, K_C(-E))$$

of $H^0(C, K_C)$, which has dimension $2n + 1$: the two summands have dimension $n + 1$ and meet in the line spanned by $\pi^*\omega$, where $\operatorname{div}\omega = \pi_* E$.

Proof. (a) Since $\pi^* \circ \mathrm{Nm} = N$ on J and $NP = 0$, the norm maps B into the kernel of π^* on $J(X)$, which is finite; as B is connected, $B \subseteq \ker \mathrm{Nm}$. This kernel has dimension

$(3n+1)-(n+1) = 2n = \dim B$, so B is its identity component, and every connected component of N_K is a coset $M_1 + B$. For $b \in B$ we have $\phi(M_1 + b) = \phi(M_1) + Pb = \phi(M_1) + 3b$, and $\phi(M_1) \in P(J) = B$. For $(c_1, \dots, c_h) \in \Lambda_0$ put $E = \sum_i c_i$; then $\pi_* E = \mu(c_1, \dots, c_h)$ lies in $F = |K_X|$, so $\mathcal{O}_C(E) \in V$, and $\Psi(c_1, \dots, c_h) = \phi(\mathcal{O}_C(E))$. Let V_k be the image of Λ_k under $(c_i) \mapsto \mathcal{O}_C(\sum_i c_i)$. It is irreducible and closed, and $Y = \phi(V_k)$. Its dimension is at most $\dim \Lambda_k = n$ and at least $\dim Y = n$. On the other hand, if $M \in V$ and E is the divisor of a nonzero section, then $\mathcal{O}_X(\pi_* E) \cong \mathrm{Nm}(M) \cong K_X$; so V is the image of $Z = \{E \in C^{(2n)} : \pi_* E \in F\}$, the preimage of F under the finite map $C^{(2n)} \to X^{(2n)}$, of dimension at most n. Hence V_k is an irreducible component of V.

(b) Let $Z_k \subseteq Z$ be the image of Λ_k; it has dimension n, so it is an irreducible component of Z, and it maps onto V_k. The map $\Lambda_k \to Q_0 \cong F$ is finite between irreducible varieties of the same dimension, hence surjective, so for E general in Z_k the divisor $\pi_* E$ is a general member of $|K_X|$. It is reduced, because $|K_X|$ has no base points, so the $2n$ points of E lie over distinct points of X. If $h^0(\mathcal{O}_C(E)) \geq 2$ for general $E \in Z_k$, the linear system $|E|$ lies in Z, since $\mathcal{O}_X(\pi_* E') \cong \mathrm{Nm}\, \mathcal{O}_C(E)$ for $E' \in |E|$; being irreducible, it lies in Z_k, the only component through a general point. Then the fibres of $Z_k \to V_k$ have positive dimension and $\dim V_k < n$, which is false. So $h^0(M) = 1$ for general $M = \mathcal{O}_C(E)$ in V_k.

Let $W \subseteq \mathrm{Pic}^{2n}(C)$ be the image of $C^{(2n)}$. Since $h^0(M) = 1$, W is smooth at M and $T_M W$ is the annihilator of $s \cdot H^0(C, K_C(-E))$ [ACGH85, Chapter IV]. The differential of the norm is the trace $H^1(C, \mathcal{O}_C) \to H^1(X, \mathcal{O}_X)$, which is Serre dual to $\pi^* \colon H^0(X, K_X) \to H^0(C, K_C)$; so N_K, a coset of $\ker \mathrm{Nm}$, is smooth with $T_M N_K$ the annihilator of $\pi^* H^0(X, K_X)$. As $V_k \subseteq W \cap N_K$, the tangent space of V_k at M lies in the annihilator of the sum. By Riemann–Roch $h^0(C, K_C(-E)) = h^0(C, \mathcal{O}_C(E)) - \deg E + g(C) - 1 = n + 1$, and $h^0(X, K_X) = n + 1$. A form $\pi^* \omega$ in both summands vanishes on E; since the points of E lie over distinct points of X, ω vanishes on $\pi_* E$, which has degree $2n = \deg K_X$, so $\operatorname{div} \omega = \pi_* E$. Conversely this ω lies in both, since $\operatorname{div} \pi^* \omega = E + \sigma E + \sigma^2 E$. So the sum has dimension $2n + 1$ and its annihilator in the space $H^1(C, \mathcal{O}_C)$ of dimension $3n + 1$ has dimension $n = \dim V_k$. Hence V_k is smooth at M with this tangent space. Finally $\phi^{-1}(Y)$ meets the component of N_K containing V_k in the union of the translates $V_k + t$, $t \in B[3]$. Those different from V_k meet it in a proper closed subset, which the general point M avoids, so the étale map ϕ carries a neighbourhood of M in V_k isomorphically onto a neighbourhood of $\phi(M)$ in Y. □

So Y is the analogue, for cyclic triple covers, of a Prym–Brill–Noether locus, and the comparison with étale double covers points the other way from the Abel–Prym curve. For an étale double cover of a curve of genus g, Mumford shows that $\mathrm{Nm}^{-1}([K_X])$ has two components, on which h^0 is even and odd, and that the locus $h^0 \geq 2$ in the even one is a theta divisor Ξ of the principally polarized Prym variety [Mum74]. So there V is the union of the odd component and Ξ, and Ξ deforms with every principally polarized abelian variety of dimension $g - 1$, although for $g \geq 7$ the Prym varieties of double covers, with $3g - 3$ moduli, form a proper subvariety of that moduli space. The analogy is loose in one respect: Ξ is a divisor whose class is a principal polarization, while Y has codimension n. Likewise, the theta divisor of a Jacobian is the image of the $(g-1)$-st symmetric product of the curve under the Abel–Jacobi map, and it deforms with every principally polarized abelian variety, while the Abel–Jacobi curve does not.

Proposition 6.17 also shows what a direct computation would need. On the open subset of V_k where (b) holds, the normal space of V_k at M is dual to $H^0(C, K_C(-E))/\mathbb{C}t$, where t is the section with divisor $\sigma E + \sigma^2 E$; so there the normal bundle comes from the bundle with fibres $H^0(C, K_C \otimes M^{-1})$. The obstruction to extending $\mathcal{O}_Y$ along a tangent vector of S is a class in a

global cohomology group, and computing it needs the normal sheaf on all of V_k, including the locus where $h^0 \geq 2$ and the divisors E that meet a fibre of π twice. We have not determined it.

A weaker hypothesis would do if the semiregularity theorem were sharper. Markman asks whether Theorem 6.2 holds when $\sigma_{\mathcal{E}}$ is injective only on the image of the evaluation map from the second Hochschild cohomology $HH^2(Z)$ to $\mathrm{Ext}^2(\mathcal{E},\mathcal{E})$ [Mar25b, Question 11.4]. A positive answer would replace the semiregularity of $\mathcal{O}_Y$ in Corollary 6.8 by injectivity of $\sigma_{\mathcal{O}_Y}$ on that image, a condition that involves only the deformations of B and of its category of sheaves. For a general imaginary quadratic field, cycles representing Weil classes at members with maximal Hodge group are known in dimension four, and for split forms in dimension six [Mar25a]. For nonsplit forms in dimension six, and in dimension $2n \geq 8$ for fields other than $\mathbb{Q}(\sqrt{-3})$ and $\mathbb{Q}(\sqrt{-1})$, we know of none, so the route has nothing to deform.

7. The routes

We now treat the statements as propositional letters and the results of the paper as rules between them. Besides the seven statements of Definition 2.7 and (HC) there are three more: (SR), that the subvariety Y of Theorem G is semiregular for some (X, L) of each genus $n+1 \geq 5$ with maximal Hodge group; (W3), that the Weil classes are algebraic on every abelian variety of split Weil type for $\mathbb{Q}(\sqrt{-3})$; and (Fer), the Hodge conjecture for every Fermat variety. The rules are listed in Table 1.

Implication	Source
$(\mathrm{F2}) \wedge (\mathrm{F3}') \Rightarrow (\mathrm{HC})$	Section 5.3: if (F2) holds then $\mathrm{Ab}^p(X) = \mathrm{Alg}^p(X)$
$(\mathrm{HC}) \Rightarrow (\mathrm{F2}), (\mathrm{F3}'), (\mathrm{L}), (\mathrm{M}), (\mathrm{V})$	Lemma 2.11
$(\mathrm{L}) \wedge (\mathrm{M}) \Rightarrow (\mathrm{HC})$	Proposition 2.8, after André [And96]
$(\mathrm{L}) \Rightarrow (\mathrm{F2})$	Abdulali, André [Mil20, Theorem 4]
$(\mathrm{V}) \Rightarrow (\mathrm{F2})$	[Mil20, Theorem 3 and Remark 2]
$(\mathrm{V}) \Rightarrow (\mathrm{IP})$	(IP) is a case of (V)
$(\mathrm{F3}') \Rightarrow (\mathrm{M})$	Proposition 2.9
$(\mathrm{CM}) \wedge (\mathrm{IP}) \Rightarrow (\mathrm{F2})$	Proposition 5.4
$(\mathrm{F2}) \Rightarrow (\mathrm{CM}), (\mathrm{IP})$	Proposition 5.4
$(\mathrm{CM}) \Rightarrow (\mathrm{Fer})$, $(\mathrm{HC}) \Rightarrow (\mathrm{Fer})$	Theorem E
$(\mathrm{SR}) \Rightarrow (\mathrm{W3})$	Corollary 6.8 for $2n \geq 8$; Theorem 2.3 and [Sch88, Sch98] for $2n \leq 6$
$(\mathrm{F2}) \Rightarrow (\mathrm{W3})$	(W3) is a case of (F2)

Table 1. The implications between the statements. A set of statements *implies* a statement if the statement is reached from the set by repeated use of these rules.

For $2n \leq 6$ the conclusion of (SR) is already known, which is why (SR) only concerns the genera $n+1 \geq 5$. A set of statements is *closed* if it contains the conclusion of every rule whose premises it contains. The statements implied by a set S form the smallest closed set containing S, so a closed set that contains S and not (HC) shows that (HC) does not follow from S.

Lemma 7.1. *Let $\mathcal{U}$ be the set of all eleven statements. The four sets*

$$T_1 = \mathcal{U} \setminus \{(\mathrm{HC}), (\mathrm{M}), (\mathrm{F3}')\}, \qquad T_2 = \mathcal{U} \setminus \{(\mathrm{HC}), (\mathrm{L}), (\mathrm{F3}')\},$$
$$T_3 = \mathcal{U} \setminus \{(\mathrm{HC}), (\mathrm{F2}), (\mathrm{L}), (\mathrm{V}), (\mathrm{IP})\}, \quad T_4 = \mathcal{U} \setminus \{(\mathrm{HC}), (\mathrm{F2}), (\mathrm{L}), (\mathrm{V}), (\mathrm{CM})\}$$

are closed, and so is $\{(\mathrm{CM}), (\mathrm{Fer})\}$.

Proof. A rule can fail to be respected by a set only if its conclusion is missing from the set while all its premises are present. We check the rules whose conclusions are missing. In T_1 the missing conclusions are (HC), (M) and (F3′); every rule concluding one of them has a premise among (HC), (F3′) and (M), which are missing. In T_2 the rules concluding (HC), (L) or (F3′) need one of (F3′), (HC) or (L). In T_3 every rule concluding (HC), (F2), (L), (V) or (IP) has a premise among (F2), (HC), (L), (V) and (IP); the rule (CM) $\wedge$ (IP) $\Rightarrow$ (F2) is blocked by (IP). T_4 is the same with (CM) in place of (IP). In $\{(\mathrm{CM}), (\mathrm{Fer})\}$ the only rule with all premises present is (CM) $\Rightarrow$ (Fer). □

Proof of Theorem D. Let S be a set of statements, not containing (HC), that implies (HC). Then S is contained in none of $T_1, \ldots, T_4$, so it meets each of

$$\{(\mathrm{M}), (\mathrm{F3}')\}, \quad \{(\mathrm{L}), (\mathrm{F3}')\}, \quad \{(\mathrm{F2}), (\mathrm{L}), (\mathrm{V}), (\mathrm{IP})\}, \quad \{(\mathrm{F2}), (\mathrm{L}), (\mathrm{V}), (\mathrm{CM})\}.$$

If (F3′) $\notin S$, the first two give (M), (L) $\in S$, which is (b). If (F3′) $\in S$ and S contains none of (F2), (L), (V), the last two give (IP), (CM) $\in S$. This is (a).

Conversely, $\{(\mathrm{F2}), (\mathrm{F3}')\}$ and $\{(\mathrm{L}), (\mathrm{M})\}$ imply (HC) directly, and $\{(\mathrm{L}), (\mathrm{F3}')\}$, $\{(\mathrm{V}), (\mathrm{F3}')\}$ and $\{(\mathrm{CM}), (\mathrm{IP}), (\mathrm{F3}')\}$ imply (F2) and then (HC). From (HC) the rules give (F2), (F3′), (L), (M), (V), then (CM), (IP), (W3) and (Fer): every statement except (SR). No minimal set contains (SR), because (SR) is a premise only of the rule that concludes (W3), and (W3) is a premise of no rule. □

Corollary 7.2. (i) *A set of statements that implies* (HC) *and contains neither* (F2) *nor* (F3′) *contains* (L) *and* (M)*; by Theorem C this pair is equivalent to* (HC)*.*
(ii) *Every set that implies* (HC) *also implies* (F2) *and* (F3′)*.*
(iii) *A set that implies* (HC) *still implies it when* (SR) *is removed.*
(iv) (CM) *implies* (Fer) *and nothing else; neither* $\{(\mathrm{CM}), (\mathrm{F3}')\}$ *nor* $\{(\mathrm{CM}), (\mathrm{SR})\}$ *implies* (HC)*.*

Proof. (i) is Theorem D(b), since (a) requires (F3′). (ii) holds because (HC) implies (F2) and (F3′). (iii) is the last sentence of the proof of Theorem D. For (iv), $\{(\mathrm{CM}), (\mathrm{Fer})\}$ is closed by Lemma 7.1, and $\{(\mathrm{CM}), (\mathrm{F3}')\}$ and $\{(\mathrm{CM}), (\mathrm{SR})\}$ lie in the closed set T_3, which does not contain (HC). □

So the conjecture can be reached without assuming (F2) or (F3′), but only through (L) and (M), which carry the whole conjecture between them; and it cannot be reached without proving (F2) and (F3′) along the way. Schoen's subvariety, if it is semiregular, settles a family of Weil classes in every dimension, but it lies on no route to (HC). The closed set $\{(\mathrm{CM}), (\mathrm{Fer})\}$ is a valuation of the eleven statements in which every rule of Table 1 holds, (CM) holds, and (HC) fails: the rules alone cannot derive the conjecture from the preprint [Ope26b].

8. What is proved and what is not

Proved. The following hold without any unproved hypothesis.

(1) The Hodge conjecture for the Fermat variety X_m^n of every dimension and every degree $m = 2^a 3^b 19^c$ or $m = 2^a 3^b 17^c$ with $a \leq 1$, $m = 2^a 11^b$ or $m = 2^a 13^b$, for every product of Fermat varieties of such a degree, and for every abelian variety of Fermat type of

such a degree (Theorem A). Besides the results of this paper, the proof uses Aoki's reduction [Aok83, Aok00], Shioda's inductive structure [Shi79], Aoki's cycles [Aok87] and the Lefschetz theorem on $(1,1)$-classes, and it rests on six certified computations in interval arithmetic (Propositions 4.20, 4.23, 4.26, 4.28, 4.31 and 4.33 and Appendix B). When $57 \mid m$ it is the first proof that does not depend on the statement discussed in Remark 3.21, on which the earlier argument of [MMRV26] for $m = 57$ rests, and when $44 \mid m$, $52 \mid m$ or $51 \mid m$ it is the first proof.

(2) The Hodge conjecture for the Fermat fourfold X_m^4 of every odd degree m (Theorem B); for m prime to 6, the classification of the Hodge characters behind it: each contains a pair or is 5-standard (Theorem 3.9); and for $3 \mid m$, the reduction of every Hodge character without a pair, by moves, to three exceptional orbits (Theorem 3.27 and Lemma 3.26). Besides the results of this paper, the proof uses the inductive structure of Shioda and Ran, Aoki's cycles [Aok87], and the classical nonvanishing of $B_{1,\chi}$ for odd primitive Dirichlet characters [Was97].

(3) The equivalences of Theorem C, and the description in Theorem D and Corollary 7.2 of every set of the statements of Table 1 from which the conjecture follows by those implications.

(4) The statement (F3′) for every Fermat variety, every product of Fermat varieties and every variety onto which such a product maps surjectively (Theorem E), together with the description of the Hodge classes of Fermat varieties by characters (Proposition 3.6).

(5) The closedness of the bounded-degree loci Σ_d in a split Weil family, and the criterion of Theorem F for $\Sigma = S$.

(6) For every étale cyclic triple cover of a curve of genus $n+1 \geq 3$: its Prym variety is of split Weil type (Lemma 6.4); Schoen's construction gives an irreducible n-dimensional subvariety Y whose class pairs nontrivially with the Weil plane; at a member with maximal Hodge group $\mathrm{cl}(Y) = c\,\eta^n + w$ with $c > 0$ and $w \neq 0$ (Theorem G); and the Prym variety is Weil-algebraic (Corollary 6.7, first proved by Schoen).

(7) The transfer of Theorem 6.11: an abelian variety of Weil type for $\mathbb{Q}(\sqrt{-3})$ is, up to isogeny, a factor $A \times A'$ of a Prym variety, and A is Weil-algebraic if and only if A' is.

(8) The count of Proposition 6.15: Prym varieties of cyclic triple covers, étale or branched, form families of dimension at most $3n$, so for $n \geq 4$ they miss the very general member of the split Weil family.

(9) The rigidity of Proposition 6.16: the Abel–Prym curve of an étale cyclic triple cover deforms with its Prym variety only along the Prym locus, although its class stays algebraic on the whole split Weil family; so for $n \geq 4$ and maximal Hodge group, when it is embedded, it is not semiregular.

(10) The description of Y in Proposition 6.17: up to an étale map, it is a component of the locus of line bundles with a nonzero section in the fibre of the norm map over $[K_X]$. It is smooth of dimension n at its general point, with an explicit tangent space.

Proved under a stated hypothesis.

(11) Under (CM): the Hodge conjecture for every Fermat variety, every product of Fermat varieties and every variety such a product dominates (Theorem E), and the algebraicity of the Weil classes at every CM point of a split Weil family (Theorem F).

(12) Under the semiregularity of one subvariety Y of genus $n+1$ with maximal Hodge group: the Weil classes of every abelian variety of split Weil type for $\mathbb{Q}(\sqrt{-3})$ of dimension $2n$ (Corollary 6.8).

Not proved. The Hodge conjecture; each of (F2), (F3′), (L), (M), (V), (IP); the semiregularity of Y for any $n \geq 4$ (Question 6.14); (CM), which is claimed in the unrefereed preprint [Ope26b] and used here only as a hypothesis; and the Hodge conjecture for the Fermat fourfolds of degrees 110 and 220, the only degrees up to 250 left open by Theorems A and B, [Pet26] and the census [Jum26a].

The first open cases. By Theorem D every route passes through (F2) and (F3′), and each has a first case that no known method reaches.

Weil classes in dimensions six and eight. (F2) holds in dimension at most five by [MZ99] and the Weil fourfolds of the preprint [Mar25a] (Section 6.2). In dimension six the Weil classes of split forms are algebraic, for $\mathbb{Q}(\sqrt{-3})$ by [Sch88, Sch98] and in general by [Mar25a], and those of nonsplit forms are open. For $2n = 8$ we know of no family of Weil type whose general member has maximal Hodge group and which carries algebraic Weil classes at every member. For the split family of $\mathbb{Q}(\sqrt{-3})$ this is Question 6.14 with $n = 4$, where Y would have to deform in $n^2 - 3n = 4$ directions beyond the Prym locus. Proposition 6.15 shows that Prym varieties of branched cyclic triple covers do not reach the general member either, and Remark 6.13 shows the same for Corollary 6.12. The Abel–Prym curve, built from the same map as Y, does not deform in those directions (Proposition 6.16), while the Prym theta divisor of an étale double cover, the analogue of Y by Proposition 6.17, deforms with every principally polarized abelian variety; neither decides the question for Y. Preprints announcing the split eightfolds are cited in [BMB26]; we could not locate them in the public release of [Ope26b], and we do not use them.

Beyond Weil classes. On the square $A \times A$ of a Mumford fourfold A [Mum69] there are Hodge classes that are not generated by divisor classes and Weil classes. They are the first instance of (F2) that the Weil route does not cover. In [BMB26, Theorem 21.4] they are reduced to the Kuga–Satake class of one K3 surface, which is not known to be algebraic.

The first case of (F3′). For a K3 surface S whose transcendental lattice has a totally real field of degree at least two as its algebra of Hodge endomorphisms [vG08], the classes on $S \times S$ that induce these endomorphisms are not known to be algebraic in general [BFvGS24]; $S \times S$ lies in the reach of abelian varieties by algebraic correspondences once the Kuga–Satake correspondence of S is algebraic [BMB26, Remark 20.36]. They are algebraic on several explicit families, where the real multiplication comes from double planes or from isogenies [Sch10, vGS25], and on the general member of the first four-dimensional families with real multiplication by a real quadratic field, through symplectic automorphisms [Var23]. For real multiplication by $\mathbb{Q}(\sqrt{d})$ it would suffice to find an algebraic correspondence from S to some K3 surface S' that induces a Hodge similarity $T(S) \to T(S')$ of multiplier d: composing its inverse with $\sqrt{d}$ gives a rational Hodge isometry, which is algebraic by the theorem of Buskin and Huybrechts [Bus19, Huy19]. For a real quadratic field the maximal families have Picard number 2 and dimension 8 [vG08], and their very general member is the first open case.

So the conjecture is equivalent to two statements, and each has an explicit first open case. What this paper adds to the Weil case is that, for $\mathbb{Q}(\sqrt{-3})$ in every dimension, it now depends on a single, explicitly constructed subvariety of a Prym variety and the question whether it is semiregular.

A note on preparation. This paper was prepared with substantial help from an AI assistant, working under the author's direction; the author checked the arguments and is responsible for the content. The assistant took part in every section, and we list the main ideas it proposed. In Section 4: the extension of Peterson's method to block families (Section 4.3), the characters α_1, α_2 and the join argument (Sections 4.2 and 4.4), the characters of degree 44 and the join

with a curve character (Lemma 4.22 and Proposition 4.18), the characters of degrees 52 and 102 (Lemmas 4.25 and 4.30), the obstruction in Remark 4.38(b), and the seven families with the certificates of Appendix B, found by computer searches that the assistant designed. In Section 3: the classification of Hodge sextuples (Theorem 3.9) and the moves of Theorem 3.27. In Section 6: the twisted-cohomology computation behind Theorem G and the description of Y in Proposition 6.17. In Section 7: the enumeration of routes in Theorem D. The assistant also drafted much of the text and wrote the programs that check the finite steps.

Appendix A. Machine checks

Apart from Propositions 4.20, 4.23, 4.26 and 4.28, whose proofs are the computations of Appendix B, no proof in this paper depends on a computer. The finite steps have been repeated by machine, in several independent implementations. The programs are in the author's repository `https://github.com/creelie/e84hc`, each release of which is archived on Zenodo under the concept DOI 10.5281/zenodo.23234362, and one script runs them all and compares their outputs.

Exact computations in three languages. The first program exists in Python, Julia and C, all in exact integer or rational arithmetic, and the three versions print the same 218 lines; the shell script compares them line by line and against a stored copy. The lines fall into six groups.

- F Fermat varieties X_m^n for $n = 2$, $m \le 12$ and for $n = 4$, $m \le 8$: the characters of Proposition 3.6, the primitive Betti number against $((m-1)^{n+2} + (-1)^n(m-1))/m$, the middle Hodge number, the Hodge characters and those of pair type; for surfaces, $h^{1,1} = (2m^3 - 6m^2 + 7m)/3$, the Picard number ρ, and Aoki and Shioda's formula $\rho = 3(m-1)(m-2) + 1$ for $(m, 6) = 1$ [AS83]. The values $\rho = 20, 37, 86$ for $m = 4, 5, 6$ appear among them.
- J Fermat curves of degree $m \le 30$: every orbit of characters under $(\mathbb{Z}/m)^\times$ has an odd type function, as in the proof of Lemma 3.3.
- S The switch lemma (Lemma 5.1) for all pairs of lists with $p \le 4$ rows and $s \le 3$ columns: the column correction reaches the target by two-row switches, each preserving the column sums, in at most $s\lfloor p/2 \rfloor$ steps.
- G The signs $(-1)^{p(p-1)/2}$ of the permutations that regroup a product of $2p$ classes of degree one between interleaved and blocked order, for $p \le 10$.
- W The dimensions $3n$ and n^2 of the Prym locus and the Weil family, and the count of Remark 6.13: the pairs (m, k) with $3(m+k) + 3k \ge (m+k)^2$, which are $k \le 2$ for $m = 2$, $k = 0$ for $m = 3$, and none for $4 \le m < 60$, $k < 200$; and the count of Proposition 6.15: for $2 \le n \le 12$ and every genus q, the Riemann–Hurwitz relation, the dimension $q + 2n - 1$ of the family, and its unique maximum $3n$ at $q = n + 1$, $r = 0$; and for Proposition 6.16, the pairs (a, b) with $0 \le a \le 2$, $1 \le b \le 2$ and $a + b \equiv 0 \pmod 3$, which are $(1, 2)$ and $(2, 1)$, the dimensions g, $g-1$, $g-1$ of V_0, V_1, V_2 adding up to the genus $3g - 2$ of C, and the dimensions $2n^2$ of E and $3n$ of $H^0(X, K_X^{\otimes 2})$.
- L The rules of Table 1: the closed sets of Lemma 7.1; the valuation in which (CM) holds and (HC) fails; the five minimal sets of Theorem D, found by enumerating all sets of statements; and the fact that (L) and (M) form the only minimal set avoiding (F2), (F3′), (IP), (CM) and (SR).

Lean. A first Lean file is checked by the kernel of Lean 4 with the core library only, without `sorry` and without `native_decide`; every theorem in it depends only on the standard axioms. It states the rules of Table 1 as an inductive derivability relation and proves: that the four sets of Lemma 7.1 and the set $\{(\mathrm{CM}), (\mathrm{Fer})\}$ are closed; Theorem D itself, for an arbitrary set of

statements; Corollary 7.2(i); and the derivations of (HC) from each of the five minimal sets. It also proves, for all sizes at once, the counting identity behind the switch lemma, the equivalence $3n < n^2 \iff n \geq 4$, the inequality of Remark 6.13 for every $m \geq 4$ and $k \geq 0$, and the count of Proposition 6.15 for all n, q and r, and the eigenspace bookkeeping of Proposition 6.16. By evaluation in the kernel, it computes the Picard numbers 20, 37 and 86 of the Fermat quartic, quintic and sextic surfaces from the characters, and it checks that the sets O_{21}, O_{33} and O_{39} of Section 3.7 are balanced, contain no pair, generate, and have no direct or lowering move, and that the three identities in the proof of Lemma 3.26 hold with every part balanced.

Lean with Mathlib. A second Lean project, which uses Mathlib, holds a formalization of Theorem 3.9, about 3600 lines in 15 files. It follows the proof of Section 3.6 step by step: the counting function, Dirichlet characters and their conductors, Fourier inversion on $(\mathbb{Z}/e)^\times$, the subgroups K_p and their orders, split functions and Lemmas 3.15 to 3.17, Proposition 3.18, Lemmas 3.19 and 3.20, and Step 7. Two of its theorems state Theorem 3.9 for every m prime to 6, and a third derives the conclusion of Theorem B for an abstract predicate "the line $V(\alpha)$ is algebraic" from the two geometric inputs, algebraicity for characters that contain a pair and for 5-standard characters. Theorem 3.27, for m divisible by 3, is not formalized there; it is checked by the searches below for $m \leq 105$.

The nonvanishing of $B_{1,\chi}$ for odd primitive χ is proved there as well, from the nonvanishing of $L(1, \bar\chi)$ in Mathlib, along the classical route of Step 2. Expanding $\bar\chi(n)$ by the Gauss sum $\tau(\chi)$ gives

$$\tau(\chi) \sum_{n \geq 1} \frac{\bar\chi(n)}{n} = -\sum_{s=1}^{e-1} \chi(s) \log(1 - e^{2\pi i s/e}) = -\frac{\pi i}{e} \sum_{s=1}^{e-1} \chi(s)\, s,$$

where the logarithmic series converges on the unit circle away from 1 by Dirichlet's test and Abel's limit theorem, the real parts cancel because χ is odd, and $\arg(1 - e^{i\theta}) = \theta/2 - \pi/2$ for $0 < \theta < 2\pi$. A summation by parts, uniform in $\sigma \in [1, 2]$, shows that the value at $s = 1$ of the analytic continuation of $L(s, \bar\chi)$ is the sum of the conditionally convergent series on the left. The geometric inputs and the description of the Hodge classes by characters remain hypotheses of the formal statements, since Mathlib has no Hodge theory. With them, the main theorems, among them the classification and the conclusion of Theorem B with $B_{1,\chi} \neq 0$ discharged, depend only on the standard axioms, and no file uses `sorry`.

Fermat fourfolds by exhaustive search. A second program exists in Python, Julia and C. For every m prime to 6 up to a bound it lists the balanced multisets of four and of six nonzero residues modulo m and checks Theorem 3.9: every quadruple contains a pair, and every sextuple without a pair is 5-standard. The three versions print the same lines for $m \leq 55$, and the C version is run up to $m = 125$ and compared with a stored copy of its output; for $m = 125$, for instance, it finds 41688 balanced sextuples, of which 24 contain no pair, all of them 5-standard.

A third program, in the same three languages, treats the odd m divisible by 3. It lists the balanced sextuples, and for each one that contains no pair and generates $\mathbb{Z}/m$ it finds a direct or a lowering move (Definition 3.23), or checks that tA is one of O_{21}, O_{33}, O_{39} for a unit t, as Theorem 3.27 asserts. It also checks the identities of Lemma 3.26 and that the three exceptional sets have no move. The three versions print the same lines for $m \leq 63$, and the C version is run up to $m = 105$ and compared with a stored copy of its output; for $m = 105$ it finds 27690 balanced sextuples, of which 78 contain no pair and 54 of these generate $\mathbb{Z}/105$, each with a direct move. These searches check the theorems only for the degrees listed; the proofs cover every m.

Macaulay2. Two scripts use Macaulay2. The first computes the primitive Hodge numbers of the Fermat surfaces and fourfolds from the Jacobian ring, $h^{n-q,q}_{\mathrm{prim}} = \dim R_{(q+1)m-(n+2)}$ with $R = \mathbb{Q}[x_0, \dots, x_{n+1}]/(x_0^{m-1}, \dots, x_{n+1}^{m-1})$ [Gri69], and checks that they agree with the character counts of item F. The second takes random smooth plane curves of degree $4 \le d \le 8$ modulo the prime 32003 and checks Max Noether's theorem in the form used in Section 6.6: the multiplication map $\mathrm{Sym}^2 H^0(K) \to H^0(2K)$ has rank $3g-3$, so $\dim I_2 = (g-2)(g-3)/2$. A smooth curve of full rank modulo p is smooth and of full rank over $\mathbb{Q}$, so these are checks over $\mathbb{Q}$.

Degree 114**.** A program in Python, Julia and C checks the finite statements of Section 4: Lemma 4.6, by listing $\langle t\alpha_1\rangle$ and $\langle t\alpha_2\rangle$ for the 36 units t; that ν_{19} is 1 on β and vanishes on every pair and on the 98 standard multisets of level 114; and the weights and degrees of the two families. The Python and Julia versions also check Proposition 4.19 in exact rational arithmetic at three values of λ, and a computer algebra system confirms the closed formula for I_{20}. The lattice computation described in Appendix B is done with exact integer matrices. In the Lean project with Mathlib, a further file proves Lemma 4.6(a),(b), the vanishing of ν_{19} on the pairs and on the standard multisets, the identities in the proof of Proposition 4.19 (that $\sum_k u_k = 0$, the two residues, the closed formula and its value at $\lambda = 2$) and the weight and degree conditions of both families; its theorems depend only on the standard axioms. The certificate of Appendix B is run at 1000 and at 600 bits.

Degree 44**.** A second program in Python, Julia and C checks Lemma 4.22, by listing $\langle t\alpha\rangle$ and $\langle tb\rangle$ for the 20 units t; that ν_{11} is 1 on β and vanishes on every pair and on the 24 standard multisets of level 44; and the weights and degrees of the family of Proposition 4.23; the three versions print the same 14 lines. They also enumerate the balanced multisets of four and six nonzero elements of $\mathbb{Z}/44$, 293 and 3164 of them, and find $\nu_{11} = 0$ on each. The Python version repeats, with exact integer matrices, the lattice computation of Appendix B at level 44. In the Lean project with Mathlib, a further file proves Lemma 4.22, the vanishing of ν_{11} on the pairs and on the standard multisets, and the weight and degree conditions of the family, with the standard axioms only. The certificate for Proposition 4.23 is run at 1000 and at 600 bits, and a contour integral of $\eta_{v^\dagger}$ over a circle enclosing all zeros of the blocks, computed with 40 digits by the trapezoidal rule, agrees with the enclosed value to 15 digits.

Degree 52**.** A third program in Python, Julia and C checks Lemma 4.25, by listing $\langle t\alpha\rangle$ and $\langle t\alpha'\rangle$ for the 24 units t; that ν_{13} is 1 on β and vanishes on every pair and on the 28 standard multisets of level 52; that ν_{13} vanishes on the 399 and 4884 balanced multisets of four and six nonzero elements of $\mathbb{Z}/52$; and the weights and degrees of the families of Propositions 4.26 and 4.28. The three versions print the same 20 lines. The Python version repeats the lattice computation of Appendix B at level 52. In the Lean project with Mathlib, a further file proves Lemma 4.25, the vanishing of ν_{13} on the pairs and on the standard multisets, and the weight, degree and chart conditions of both families, with the standard axioms only. One program runs both certificates, at 1000 and at 600 bits. Independently of it, for each family the integral of η_v over a circle enclosing all zeros of the blocks and the sum of the integrals over small circles around the zeros of u_i alone, computed with 40 digits by the trapezoidal rule, agree with each other and with the enclosed value to 15 digits; this also checks that η_v has no other poles in $\mathbb{C}$.

Degree 51**.** A fourth program in Python, Julia and C checks Lemma 4.30, by listing $\langle t\alpha\rangle$ and $\langle t\alpha'\rangle$ for the 32 units t modulo 102; that ν_{17} is 1 on β and vanishes on every pair and on the 88 standard multisets of level 102; that ν_{17} vanishes on the 1456 and 32490 balanced multisets of four and six nonzero elements of $\mathbb{Z}/102$, which it enumerates by matching halves; and the

weights and degrees of the families of Propositions 4.31 and 4.33. The three versions print the same 19 lines. The Python version repeats the lattice computation of Appendix B at level 102. In the Lean project with Mathlib, a further file proves Lemma 4.30, the vanishing of ν_{17} on the pairs and on the standard multisets, and the weight, degree and chart conditions of both families, with the standard axioms only. The certificates run at 1000 and at 600 bits, and the contour integrals described for degree 52 agree with the enclosed values to 15 digits.

References. A script looks up every entry of the bibliography online, through its DOI where it has one and through Crossref, DataCite, arXiv or the publisher's page otherwise, and compares title, year, volume and pages; its report is stored next to it.

Except for the certificates of Appendix B, none of these checks touches the geometry the paper relies on, and none of them bears on Question 6.14.

Appendix B. The certificates

This appendix proves Propositions 4.20, 4.23, 4.26, 4.28, 4.31 and 4.33. Each proof is a finite computation in interval arithmetic: every number below is an exact rational ball, and every operation returns a ball that contains the exact result. We used Arb [Joh17] at a working precision of 1000 bits. The programs and the data are archived with the repository of this paper. We describe the computation for Proposition 4.20 in detail; the others have the same structure and are summarized at the end.

The Krawczyk test. Write $x \in \mathbb{C}^{24}$ for the coordinates (f, h, g, a, b, c) of Section 4.5, in this order, and $E(x, \kappa) \in \mathbb{C}^{24}$ for the vector of coefficients $E_0, \dots, E_{23}$. Fix $\kappa_0 = (-17 + 3i)/10$ and a point $\tilde{x} \in \mathbb{C}^{24}$ whose real and imaginary parts are decimal numbers with 120 significant digits, and let $X \subseteq \mathbb{C}^{24}$ be the box of points whose real and imaginary parts differ from those of $\tilde{x}$ by at most 10^{-95}. Let Y be a complex matrix close to $J(\tilde{x}, \kappa_0)^{-1}$ (with exact rational entries), and let

$$K(X) = \tilde{x} - Y\,E(\tilde{x}, \kappa_0) + \big(I - Y\,J(X, \kappa_0)\big)(X - \tilde{x}),$$

evaluated in interval arithmetic, where $J(X, \kappa_0)$ is an interval matrix containing the Jacobian at every point of X. Regarding $\mathbb{C}^{24}$ as $\mathbb{R}^{48}$, the holomorphic map $E(\cdot, \kappa_0)$ becomes a smooth map of $\mathbb{R}^{48}$, the complex interval computations enclose the corresponding real ones, and Krawczyk's theorem applies: if $K(X)$ lies in the interior of X, then $E(\cdot, \kappa_0)$ has exactly one zero x^* in X, and every matrix in $J(X, \kappa_0)$, in particular $J(x^*, \kappa_0)$, is invertible [Kra69, Moo77], [Neu90, §5.1].

The computation gives $K(X)$ inside the interior of X, with every coordinate of $K(X)$ of radius below 10^{-180}. Let $p^* = (x^*, \kappa_0)$. This proves Proposition 4.20(i).

Nondegeneracy. On the box X, the balls enclosing a, b, c and $1 + a + b$, and the resultants of the six pairs among F, H, G, K, computed as determinants of Sylvester matrices, do not contain 0. Their absolute values exceed 0.46, 0.50, 0.070, 1.69, and, for the resultants of (F, G), (F, H), (F, K), (K, G), (H, G) and (H, K), $9.7 \cdot 10^{-6}$, 0.47, 0.88, 0.14, 0.19 and 1.0. This proves Proposition 4.20(ii).

The residue. The tangent vector v^* has κ-component 1, and its other components are $\dot{x} = -J^{-1}\,\partial E/\partial\kappa$, which we enclose by solving the linear system over the balls of X. With the derivatives of F, H, G and of a, b, c given by $\dot{x}$, and $\dot{K} = -s(s-1)$, we enclose the coefficients of the polynomials $\mathcal{D}_{v^*}$ and $\mathcal{M}$ of Section 4.5, reduce both modulo F^5, and solve the 35×35

linear system for the coefficients of R_{v^*}; Arb's solver certifies that the system is invertible. The coefficient of s^{34} is enclosed in

$$r_{34} \in [75.2385455180 \pm 1.1 \cdot 10^{-11}] + [-19.2657017116 \pm 3.1 \cdot 10^{-11}]\, i,$$

which does not contain 0; its absolute value exceeds 77.66. By the discussion after Corollary 4.21, $I_{03}(p^*, v^*) = r_{34} \neq 0$, which proves Proposition 4.20(iii).

Independent checks. The point $\tilde{x}$ was found by Newton's method from a homotopy, and the same computation repeated at 600 bits gives the same conclusions. The equality $B_{114} = S_{114} + \mathbb{Z}[\beta]$ of Corollary 4.5 is also checked without Theorem 4.4: the integer matrix whose rows are the classes of the 113 pairs, the 98 standard multisets of level 114, the element (57) and $[\beta]$ has rank 95 and is saturated, as its Smith normal form shows, while the lattice of $x = \sum_y c_y(y) \in R_{114}$ with $\sum_y c_y \overline{ty} = 57 \sum_y c_y$ for every unit t has rank $114 - 1 - 18 = 95$; since (57) has odd length and $2\,(57)$ lies in D_{114}, this gives $B_{114} = S_{114} + \mathbb{Z}[\beta]$. The same computation at level 57 gives $[B_{57} : S_{57}] = 2$.

The family of degree 44**.** For Proposition 4.23 the unknowns are $x = (p, q, r_1, c_0, c_1, c_3) \in \mathbb{C}^{12}$ and the equations are $E_0, \ldots, E_{11}$ of Section 4.6. We fix $\kappa_0 = (4 - 3i)/10$, a point $\tilde{x}$ with 120 significant digits, and the box X of radius 10^{-95} around it. The Krawczyk image $K(X)$ lies in the interior of X, with coordinates of radius below 10^{-185}; this gives the zero x^*, the point $p^\dagger = (x^*, \kappa_0)$ and part (i). On X the balls enclosing c_0, c_1, c_3, p_0, q_0 and the resultants of (P, Q), (P, S) and (Q, S) do not contain 0; their absolute values exceed 0.52, 0.80, 0.30, 1.57, 0.89, 1.05, 4.42 and 2.66, and $|\kappa_0| = 1/2$. This is part (ii).

For part (iii) we enclose $\dot{x} = -J'^{-1}\partial E/\partial\kappa$ on X, form the polynomial $\mathcal{N}_{v^\dagger}$ of Section 4.6 and the denominator $c_0c_1c_3s^3P^2Q^3S$, and expand their quotient at $s = \infty$ by division of power series in s^{-1}, whose leading coefficient $c_0c_1c_3$ does not vanish. The coefficient of s^{-1} is enclosed in a disc of radius $3 \cdot 10^{-71}$ around

$$I_{02}(p^\dagger, v^\dagger) \approx -0.177444420578390 + 0.065416149270899\, i,$$

whose absolute value exceeds 0.189. By the discussion after Corollary 4.24 this proves (iii). As a check on the regularity used there, every coefficient of the remainder of $\mathcal{N}_{v^\dagger}$ modulo P^2S is enclosed in a ball of numbers of absolute value below 10^{-70}. The same computation at 600 bits gives the same conclusions, and the lattice computation described above, at level 44 with the 43 pairs, the 24 standard multisets, the element (22) and $[\beta]$, gives rank 33, the rank of the lattice of balanced classes, and a saturated matrix; so $B_{44} = S_{44} + \mathbb{Z}[\beta]$ and $[B_{44} : S_{44}] = 2$ without Theorem 4.4.

The families of degree 52**.** For Proposition 4.26 the unknowns are $x = (g, f, c_0, c_1, c_3, c_4) \in \mathbb{C}^{15}$ and the equations are $E_0, \ldots, E_{14}$ of Section 4.7. We fix $\kappa_0 = -(3 + 2i)/10$, a point $\tilde{x}$ with 120 significant digits, and the box X of radius 10^{-95} around it. The Krawczyk image $K(X)$ lies in the interior of X, with coordinates of radius below 10^{-184}; this gives the zero x^*, the point $p^A = (x^*, \kappa_0)$ and part (i). On X the balls enclosing c_0, c_1, c_3, c_4, g_0, f_0 and the resultants of (G, F), (G, H) and (F, H) do not contain 0; their absolute values exceed 3.16, 3.04, 0.139, 1.24, 0.69, 0.121, 56.4, 5.05 and 0.644. This is part (ii).

For part (iii) we enclose $\dot{x} = -J_A^{-1}\partial E/\partial\kappa$ on X, form the polynomial $\mathcal{N}_{v^A}$ and the denominator $c_0c_1c_3c_4s^2F^3GH^3$ of Section 4.7, and expand their quotient at $s = \infty$ by division of power series in s^{-1}, whose leading coefficient $c_0c_1c_3c_4$ does not vanish. The coefficient of s^{-1} is enclosed in a disc of radius $5 \cdot 10^{-68}$ around

$$I_{02}(p^A, v^A) \approx -5.38867785237614 + 19.2154179849628\, i,$$

whose absolute value exceeds 19.95.

For Proposition 4.28 the unknowns are $x = (q_0, q_1, c_0, c_1, c_3, c_4) \in \mathbb{C}^6$, and we fix $\kappa_0 = (3+6i)/10$. The Krawczyk image of the box of radius 10^{-95} around a point with 120 significant digits lies in its interior, with coordinates of radius below 10^{-187}. On the box, c_0, c_1, c_3, c_4, and the resultants of (s, L), (s, Q), $(s-t, L)$, $(s-t, Q)$ and (L, Q), which are κ_0, q_0, $1+\kappa_0$, $1+q_1+q_0$ and $\kappa_0^2 - q_1\kappa_0 + q_0$, have absolute values above 0.58, 0.29, 1.31, 0.61, 0.67, 1.81, 1.43, 1.35 and 1.50. The coefficient of s^{-1} in the expansion of $\mathcal{N}_{v^B}/(c_0c_1c_3c_4s^3(s-1)^3L^3Q)$ is enclosed in a disc of radius $4 \cdot 10^{-87}$ around

$$I_{12}(p^B, v^B) \approx -5.53944549467493 - 5.38246966308485\, i,$$

whose absolute value exceeds 7.72. Both computations give the same conclusions at 600 bits. The lattice computation, at level 52 with the 51 pairs, the 28 standard multisets, the element (26) and $[\beta]$, gives rank 39, the rank of the lattice of balanced classes, and a saturated matrix; so $B_{52} = S_{52} + \mathbb{Z}[\beta]$ and $[B_{52} : S_{52}] = 2$ without Theorem 4.4.

The families of degree 102**.** For Proposition 4.31 the unknowns are the coordinates other than κ, $x = (q_0, q_1, f_1, c_1, c_2, c_3, c_4) \in \mathbb{C}^7$, and the equations are $E_0, \dots, E_6$ of Section 4.8. We fix $\kappa_0 = -(637+53i)/500$, a point $\tilde{x}$ with 320 significant digits, and the box X of radius 10^{-95} around it. The Krawczyk image $K(X)$ lies in the interior of X, with coordinates of radius below 10^{-187}; this gives the zero x^*, the point $p_6 = (x^*, \kappa_0)$ and part (i). On X the balls enclosing c_1, c_2, c_3, c_4, q_0 and the resultant of (Q, F) do not contain 0; their absolute values exceed 1.02, 1.04, 1.73, 1.35, 0.344 and 1.32. This is part (ii). For part (iii) we enclose $\dot{x} = -J_6^{-1}\partial E/\partial\kappa$ on X, form $\mathcal{N}_{v_6}$ and the denominator $c_1c_2c_3c_4s^2Q^2F^3$ of Section 4.8, and expand their quotient at $s = \infty$. The coefficient of s^{-1} is enclosed in a disc of radius $5 \cdot 10^{-90}$ around

$$I_{04}(p_6, v_6) \approx 1.13299117817861 + 1.06868517120741\, i,$$

whose absolute value exceeds 1.55.

For Proposition 4.33 the unknowns are $x = (q_1, f, g, c_1, c_2, c_3, c_4) \in \mathbb{C}^{19}$, and we fix $\kappa_0 = -(147+117i)/500$. The Krawczyk image of the box of radius 10^{-95} around a point with 320 significant digits lies in its interior, with coordinates of radius below 10^{-183}. On the box, c_1, c_2, c_3, c_4, and the resultants of (Q, F), (Q, s), (Q, G), (F, s) and (F, G), the second and fourth of which are κ_0 and f_0 up to sign, have absolute values above 1.61, 2.13, 0.914, 1.62, 0.548, 0.375, 0.716, 0.685 and 0.078; the resultant of (s, G) is 1. The coefficient of s^{-1} in the expansion of $\mathcal{N}_{v_{18}}/(c_1c_2c_3c_4s^4Q^5FG^5)$ is enclosed in a disc of radius $9 \cdot 10^{-77}$ around

$$I_{23}(p_{18}, v_{18}) \approx -2.46442837913866 - 3.88281035492264\, i,$$

whose absolute value exceeds 4.59. Both computations give the same conclusions at 600 bits. The lattice computation, at level 102 with the 101 pairs, the 88 standard multisets, the element (51) and $[\beta]$, gives rank 85, the rank of the lattice of balanced classes, and a saturated matrix; so $B_{102} = S_{102} + \mathbb{Z}[\beta]$ and $[B_{102} : S_{102}] = 2$ without Theorem 4.4.

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